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Classical metrics from quantum amplitudes

The Reissner-Nordström-Tangherlini metric from
the graviton emission of charged scalars

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Contents

Introduction	1
1 Quantization of the gravitational field	5
1.1 Free spin-2 particle	6
1.2 Lagrangian of pure gravitation	9
1.2.1 Cosmological term	11
1.3 Gravitation and massive scalar particles	12
1.4 Gravitation and photons	16
1.5 Divergences and Effective Field Theories	17
1.6 Classical limit	19
2 Metrics from Scattering Amplitudes	23
2.1 General l -loop Case	23
2.2 From Amplitudes to Metric Components	27
2.3 Quantum computation massive case	28
2.3.1 Tree-level	28
2.3.2 One-loop	29
2.4 Quantum Computations photons case	32
2.4.1 One-Loop	32
2.4.2 Two-loop	35
3 Non-minimal couplings	43
3.1 Reissner-Nordström-Tangherlini Metric Renormalization	46
3.1.1 Renormalised metric in four dimensions ($d = 3$)	46
3.1.2 Renormalised metric in five dimensions ($d = 4$)	48
4 Metrics in de Donder gauge	53
4.1 Schwarzschild-Tangherlini metric	55
4.2 Reissner-Nordström-Tangherlini metric	56
4.3 Numerical Computations of Metrics in de Donder gauge	57
Conclusions	63
A Faddeev-Popov ghosts	65
B Sum over the permutations	67

C Master Integral	69
C.1 Master Integral Fourier transform	69
C.2 Master Integral identities	71
D Three Graviton vertex	75
E Feynman Rules	79
Bibliography	81

Introduction

Building a quantum field theory of gravitation is notoriously difficult because of the non-linearity of the field equations and the presence of a dimensionful coupling constant κ . General relativity can be seen as the $\hbar \rightarrow 0$ limit of a path integral quantum theory with an action that minimally contains the Einstein-Hilbert action [1]. The gravitational force results from the exchange of the massless spin-2 graviton field. The gravitons interact with themselves as well as the matter fields [2–6]. The self-interaction of the graviton field yields to the non-linearity in the field equations and makes gravity a non-renormalizable theory. In an old way of thinking, only renormalizable theories are considered workable quantum theories. In the original works of DeWitt [3–5], 't Hooft [7], Veltman [2] and others [8–10] in 1970s, the authors after providing a quantum theory of gravity focused the attention of the reader on the divergences arising from the loop diagrams. The attempt to develop a fully renormalizable theory at all order failed.

However quantum predictions can be made in non-renormalizable theories. The effective field theory formalism encompasses this problem and shows that a quantized theory of gravity is well defined order by order in a derivative expansion. The first paper about this approach appeared twenty years later by Donoghue [11]. The key point of effective field theories is the separation of quantum effects due to low energy particles from the unknown physics at higher energies. This summarises the problem of the new degrees of freedom and interactions expected from a quantum theory of gravity beyond the Planck scale. Higher energy effects are represented by the most general effective lagrangians consistent with the symmetry of general relativity (local invariance under coordinate transformation) which contains infinite terms. Since the gravitational interactions are easily organised in an energy expansion (1.47), general relativity fits well in the framework of EFT. It is possible to make quantum predictions well below the high energy scales using the low energy degrees of freedom. The study of quantum effects at low energies or large scales is dominated by the propagation of massless particles (including gravitons) and so within these limits, we can obtain predictions on quantum gravity. In order to obtain the renormalized components at fixed dimension and loop order, we insert in the action all the non-minimal couplings permitted by the symmetry of the theory, then it is possible to perform a redefinition of the coupling constants in order to absorb the divergences of the theory (which also are consistent with the symmetry).

It has been firstly noticed by Iwasaki in 1971 [12] that cancellations of $\hbar$ factors lead to the definition of classical general relativity perturbatively through the loop expansion. In particular from the analysis of the $\hbar$ factors, one can generate classical results from one-loop processes in the presence of massless particles (associated with

a specific non-analytic term in momentum space). These non-analytic terms produce the long-distance classical contributions from the loops. It is possible to reconstruct these contributions with tree-like structures from loops of triangle graphs [13] with the unitary sewing [14, 15]. In [16] the authors conclude that the loop expansion does not coincide with the $\hbar$ expansion in the presence of coupling with massless fields. Therefore the usual QFT path integral limit for $\hbar \rightarrow 0$ does not correspond to classical physics in the sense of loop expansion, in this case quantum effects of gravity appears also from loop contributions. We can therefore extract contributions to the classical limit from loop computations as done in [13, 17].

In the literature the one-loop computation has been already done in four dimensions for the Reissner-Nordström and Kerr-Newman metrics (2002) [18] and up to three-loop for the Schwarzschild-Tangherlini metric (2020) [17]. The Reissner-Nordström-Tangherlini metric in general dimension has been computed at one-loop order by Fragomeno (2021) [19] in her master thesis project. In this work, we compute for the first time in the literature the renormalized contributions to the classical limit of the Reissner-Nordström-Tangherlini metric up to two-loop order in any dimension. We will show that the diagrams of a scalar line emitting a graviton contributing to the classical limit of the metric are all tree-like diagrams [16]. We will find that the static limit will project all the tree-structure legs attached to the scalar field to a point and so all the diagrams contribute as sunset diagrams [17] (diagrams in which we have an in and an out line with loops starting all from the same vertex and ending all in another).

It is interesting to obtain these metrics from a QFT formalism because they describe different astrophysical objects among which black holes. These modern methods can be used to compute different processes such as binary mergers and try to compute the gravitational wave emitted [20].

The thesis is organised as follows. In the first chapter (1) we will provide a path to the quantisation of the gravitational field starting from the analysis of a massless spin-2 field and link the result to the free gravitational lagrangian in the weak-field limit. Then we will add the coupling with massive and electromagnetic fields. Some Feynman rules of the theory will be provided and then we will define EFT and the description of gravity as an EFT. In Chapter 2 we will be initially provided the general tools to solve the diagrams involving a scalar line and an outgoing graviton leg. The general l -loop diagram will be linked to the l -loop massless sunset diagram. After the general l -loop treatment it will be given the computations for both scalar and photons processes up to two-loop order. The result obtained in this chapter is the metric contributions to each order in any dimension. These components have ultraviolet divergences that will be deleted by a smart choice of the coupling constants in Chapter 3. In this chapter, we will introduce a set of non-minimal couplings built from the action of powers of covariant derivatives on a single power of the Riemann tensor and its projections. These couplings will have a divergence of the same type of the loop divergences of the diagrams due to the symmetry proprieties. A smart choice of the coupling constants will permit us to cancel these divergences and obtain a renormalized metric at every fixed order. In Chapter 4 we will obtain the same metric from classical computations, derived from a change of coordinate of the metric in the de Donder gauge. This shows that the

procedure of including non-minimal couplings that delete loop divergences of the theory is equivalent to a change of coordinate and therefore it seems not to affect the space-time geometry.

In order to give a better fluidity to the reading, all the stand-alone demonstrations, as well as long computations, have been included in the appendices.

We will set $\hbar = c = 1$ except when explicitly stated, and use mostly negative signature metric $(+, -, -, \dots)$.

Chapter 1

Quantization of the gravitational field

In this chapter we will follow one of the many paths for the quantization of the gravitational field. In order to do that we will study the massive spin-2 particle and find an analogy with free gravity in the weak-field limit. After that, we will start to couple gravity with scalar particles and photons. This will lead us to the definition of an effective field theory and the theoretical aspects that will be useful for the computations in the next chapters.

General relativity is the theory of gravitation and space-time. It has been formulated by Einstein [21]. This theory is based on the Principle of Equivalence of inertial and gravitational mass and the Principle of General Covariance [22, 23]. The first principle claims the equivalence of the inertial and gravitational mass, and therefore the equivalence of a uniform accelerated mass and a free-falling equivalent mass. The second principle states the invariance under a general coordinate transformation of the physical equations.

This theory links the distribution of mass to the changes in the space-time curvature, and this can be related to the motion of a free particle. The study of the curvature of a manifold can be done by introducing the metric tensor $g_{\mu\nu}$. This tensor describes the geometry of a variety and is possible to build other tensors from it. First we can define the Christoffel symbols or Affine Connections $\Gamma_{\mu\nu}^{\lambda}$ (that are not tensors) which can be obtained by the equation

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2}g^{\lambda\alpha}(g_{\alpha\nu,\mu} + g_{\mu\alpha,\nu} - g_{\mu\nu,\alpha}) . \quad (1.1)$$

These quantities are useful for deriving the derivative of a vector field (or a tensor) on a manifold, named Covariant Derivative and defined by

$$V^{\mu}_{;\nu} = V^{\mu}_{,\nu} + V^{\alpha}\Gamma_{\nu\alpha}^{\mu} .$$

The tensor that describes the curvature of a manifold is the Riemann tensor

$$R^{\rho}_{\sigma\mu\nu} = \Gamma^{\rho}_{\nu\sigma,\mu} - \Gamma^{\rho}_{\mu\sigma,\nu} + \Gamma^{\rho}_{\mu\alpha}\Gamma^{\alpha}_{\nu\sigma} - \Gamma^{\rho}_{\nu\alpha}\Gamma^{\alpha}_{\mu\sigma} ,$$

which is antisymmetric in the first and in the second pair of indices, and is symmetric under the exchange of the two pair of indices.

By contracting the Riemann tensor with the metric we can construct the Ricci tensor

$$R_{\mu\nu} \equiv g^{\alpha\beta} R_{\alpha\mu\beta\nu} = R^\alpha{}_{\mu\alpha\nu} ,$$

and the scalar (or Ricci) curvature

$$R \equiv g^{\alpha\beta} R_{\alpha\beta} = R^\alpha{}_\alpha . \quad (1.2)$$

As the Riemann tensor both the Ricci tensor and the curvature are linear in the second derivatives of $g_{\mu\nu}$ and non-linear in the first derivatives. In the LIF (Locally Inertial Frame) $\Gamma^\lambda_{\mu\nu} = 0$ and the non-linear part of the Riemann tensor vanishes.

The link between the distribution of matter, given by the stress-energy tensor $T_{\mu\nu}$, and the metric is given by the Einstein's field equations

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -\frac{\kappa^2}{4} T_{\mu\nu} , \quad (1.3)$$

where $G_{\mu\nu}$ is the Einstein's tensor and $\kappa^2 \equiv 32\pi G_N$.

Einstein's field equations are ten partial differential equations for the ten independent components of $g_{\mu\nu}$, but these equations are not independent. Because of the Bianchi identities¹, $G^{\mu\nu}{}_{;\nu} = 0$ and this reduces the independent components to six. The unknown components of the metric are non-physical and due to the invariance of the theory under coordinate transformation, more precisely the metric is defined up to a coordinate transformation that involves four scalar functions. This gauge freedom is particularly useful in finding solutions to Einstein's equation.

1.1 Free spin-2 particle

The starting point in the quantization of the gravitational field is to study it as a free spin-2 field in four dimensions. The reasons are that it can not be described by a vector field (such as the electromagnetic field) otherwise it would have a repulsive force, and neither by a scalar field because it couples with the energy-momentum tensor.

A spin-2 particle is described by a symmetric tensor with two indices. We have to compute the free propagator $f_{\alpha\beta\mu\nu}(k)$. Considering a source emitting such field on mass shell, there is no way that this field decays in a vector or scalar field, so the field propagates itself free. The contribution of this diagram is then

$$j_{\mu\nu} \text{ (blob) } \text{ (wavy line) } V_{\alpha\beta} \quad j_{\mu\nu} \frac{f_{\alpha\beta\mu\nu}(k)}{k^2 + m^2 - i\epsilon} V_{\alpha\beta} . \quad (1.4)$$

Since 1.4 vanish $f_{\mu\nu\alpha\alpha}(k) = 0$ if $k^2 = m^2$, so $k^\beta f_{\mu\nu\alpha\beta}(k) = 0$. Next we have to impose the propriety of definite positiveness of the imaginary part of the self interaction diagram because this is equal to the probability of emitting a particle. Summarising the proprieties that define $f_{\mu\nu\alpha\beta}(k)$

$$k^\beta f_{\mu\nu\alpha\beta}(k) = 0 \quad \text{and} \quad \text{Im}\{ \text{ (blob) (wavy line) (blob) } \} > 0 . \quad (1.5)$$

¹A set of differential equations satisfied by the Riemann tensor, $R_{\alpha\beta\mu\nu;\lambda} + R_{\alpha\beta\lambda\mu;\nu} + R_{\alpha\beta\nu\lambda;\mu} = 0$.

The only tensor that satisfies these conditions is

$$f_{\mu\nu\alpha\beta}(k) = \frac{1}{2} \left(\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta} \right) + \quad (1.6)$$

$$+ \frac{1}{2} \left(\eta_{\mu\alpha} \frac{k_\nu k_\beta}{m^2} + \eta_{\nu\beta} \frac{k_\mu k_\alpha}{m^2} + \eta_{\mu\beta} \frac{k_\nu k_\alpha}{m^2} + \eta_{\nu\alpha} \frac{k_\mu k_\beta}{m^2} \right) \quad (1.7)$$

$$+ \frac{2}{3} \left(\frac{1}{2} \eta_{\mu\nu} - \frac{k_\mu k_\nu}{m^2} \right) \left(\frac{1}{2} \eta_{\alpha\beta} - \frac{k_\alpha k_\beta}{m^2} \right), \quad (1.8)$$

where we omitted the presence of a multiplicative constant. Such a constant would have turned out to be one [2].

In the rest frame, where $k = (m, 0)$, the expression for the propagator simplifies noticing that the contribution from the momentum parts arises only if all the indices are temporal and in this case it vanishes. The result under this condition is

$$f_{\mu\nu\alpha\beta}(k) = \begin{cases} \frac{1}{2}(\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha}) - \frac{1}{2}\eta_{\mu\nu}\eta_{\alpha\beta} & \text{for } \mu, \nu, \alpha, \beta \neq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (1.9)$$

We can define a basis of symmetric polarisation tensors $e_{\mu\nu}^\lambda(k)$ that generates the propagator $f_{\mu\nu\alpha\beta}(k) = \sum_\lambda e_{\mu\nu}^\lambda(k) e_{\alpha\beta}^\lambda(k)$. From 1.5 we obtain the proprieties of the polarisation tensors:

$$e_{\mu\mu} = 0 \quad \text{and} \quad k_\mu e_{\mu\nu} = 0 .$$

The only tensors satisfying these proprieties are:

$$\sqrt{\frac{2}{3}} \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \sqrt{\frac{1}{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \sqrt{\frac{1}{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (1.10)$$

$$\sqrt{\frac{1}{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \sqrt{\frac{1}{2}} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (1.11)$$

The Lagrangian corresponding to the propagator 1.6 has the form

$$\mathcal{L} = \mathcal{L}_{sym} + \frac{1}{4}m^2(h_{\mu\mu}h_{\nu\nu} - h_{\mu\nu}h_{\mu\nu}), \quad (1.12)$$

with

$$\begin{aligned} \mathcal{L}_{sym} \equiv & -\frac{1}{2}\partial_\lambda h_{\lambda\mu}\partial_\mu h_{\nu\nu} + \frac{1}{2}\partial_\lambda h_{\lambda\mu}\partial_\nu h_{\nu\mu} \\ & -\frac{1}{4}\partial_\lambda h_{\mu\nu}\partial_\lambda h_{\mu\nu} + \frac{1}{4}\partial_\lambda h_{\mu\mu}\partial_\lambda h_{\nu\nu}, \end{aligned} \quad (1.13)$$

obtained inverting the propagator. We are interested in the limit $m \rightarrow 0$, but in the propagator this is not possible because of the divergent terms. So we can not define a Lagrangian and we can neither take the limit of 1.12 for $m \rightarrow 0$.

The problem is similar to the one of QED in which the massless propagator gives negative probabilities. These are avoided by introducing the gauge invariance of the theory. We will require the invariance under the gauge transformation

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \eta_\nu + \partial_\nu \eta_\mu, \quad (1.14)$$

where η_μ are four arbitrary functions of the coordinates. We omit terms proportional to $\eta_{\mu\nu}$ because they contribute as a scale transformation which gives no contribution to the physics of the problem. The lagrangian $\mathcal{L}_{sym}$ is invariant under this gauge transformation.

We can therefore add a convenient gauge fixing term that kills the zero-mode of the lagrangian. This mode would make the propagator ill defined [2, 6]. In the harmonic gauge the gauge fixing term is $-\frac{1}{2}C_\mu^2$, with $C_\mu \equiv \partial_\nu h_{\mu\nu} - \frac{1}{2}\partial_\mu h$. The effect of such terms in the lagrangian is

$$\mathcal{L} = \mathcal{L}_{sym} - \frac{1}{2}C_\mu^2 = -\frac{1}{4}\partial_\lambda h_{\mu\nu}\partial_\lambda h_{\mu\nu} + \frac{1}{8}\partial_\lambda h_{\mu\mu}\partial_\lambda h_{\nu\nu} . \quad (1.15)$$

It can be written in a more convenient form introducing the tensor $V_{\alpha\beta\mu\nu}$ as follow

$$\mathcal{L} = -\frac{1}{2}\partial_\lambda h_{\alpha\beta}V_{\alpha\beta\mu\nu}\partial_\lambda h_{\mu\nu} \quad \text{with} \quad V_{\alpha\beta\mu\nu} \equiv \frac{1}{2}\eta_{\alpha\mu}\eta_{\beta\nu} - \frac{1}{4}\eta_{\alpha\beta}\eta_{\mu\nu} . \quad (1.16)$$

The vertex above is now relatively easy to invert. Performing integration by part to all the terms we can write

$$\mathcal{L} = -\frac{1}{2}h_{\mu\nu}\left(I^{\mu\nu\alpha\beta} - \frac{1}{2}\eta^{\mu\nu}\eta^{\alpha\beta}\right)\square h_{\alpha\beta} \equiv -\frac{1}{2}h_{\mu\nu}P_{(4)}^{\mu\nu\alpha\beta}\square h_{\alpha\beta} , \quad (1.17)$$

where we defined

$$I^{\mu\nu\alpha\beta} \equiv \eta^{\mu\alpha}\eta^{\nu\beta} + \eta^{\mu\beta}\eta^{\nu\alpha} \quad (1.18)$$

and

$$P_{(4)}^{\mu\nu\alpha\beta} \equiv I^{\mu\nu\alpha\beta} - \frac{1}{2}\eta^{\mu\nu}\eta^{\alpha\beta} . \quad (1.19)$$

In the momentum space, we obtain a propagator of the form

$$\mathcal{P}_{\alpha\beta\mu\nu}^{(4)}(q) = \frac{1}{q^2 - i\epsilon} \left(\eta_{\alpha\mu}\eta_{\beta\nu} + \eta_{\alpha\nu}\eta_{\beta\mu} - \eta_{\alpha\beta}\eta_{\mu\nu} \right) = \frac{1}{q^2 - i\epsilon} P_{\alpha\beta\mu\nu}^{(4)} . \quad (1.20)$$

Another way to evaluate the propagator is by renaming the ten independent components of $h_{\mu\nu}$ to ten new components ψ_i , changing the indices via the correspondence:

$\mu\nu$	11	22	33	44	12	13	14	24	34	34
i	1	2	3	4	5	6	7	8	9	10

(1.21)

The Lagrangian 1.15 in the new components becomes

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2}\sum_{i=5}^{10}\partial_\mu\psi^i\partial_\mu\psi^i - \frac{1}{4}\sum_{i=1}^4\partial_\mu\psi^i\partial_\mu\psi^i + \frac{1}{8}\left(\partial_\mu\sum_{i=1}^4\psi^i\right)^2 \\ &\equiv -\frac{1}{2}\partial_\mu\psi^iV^{ij}\partial_\mu\psi^j \end{aligned}$$

with V^{ij} defined by

$$\begin{aligned} V^{ij} &= \delta^{ij} && \text{if } i, j \geq 5, \\ &= \frac{1}{4} \begin{pmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix} && \text{if } i \geq 1, 4 \geq j, \\ &= 0 && \text{otherwise,} \end{aligned}$$

which can easily be inverted obtaining

$$\begin{aligned}
 (V^{-1})^{ij} &= \delta^{ij} && \text{if } i, j \geq 5, \\
 &= \begin{pmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix} && \text{if } i \geq 1, 4 \geq j, \\
 &= 0 && \text{otherwise.}
 \end{aligned}$$

Performing the indices change defined by the table 1.21 backward we have the free spin-2 particle propagator (in agreement with the one written above)

$$\mathcal{P}_{\alpha\beta\mu\nu}^{(4)}(q) = \frac{1}{q^2 - i\epsilon} \frac{1}{2} \left(\eta_{\alpha\mu} \eta_{\beta\nu} + \eta_{\alpha\nu} \eta_{\beta\mu} - \eta_{\alpha\beta} \eta_{\mu\nu} \right). \quad (1.22)$$

It is possible to perform the same computations in generic dimension D . The propagator has the expression

$$\mathcal{P}_{\alpha\beta\mu\nu}(q) = \frac{1}{q^2 - i\epsilon} \frac{1}{2} \left(\eta_{\alpha\mu} \eta_{\beta\nu} + \eta_{\alpha\nu} \eta_{\beta\mu} - \frac{2}{D-2} \eta_{\alpha\beta} \eta_{\mu\nu} \right) = \frac{1}{q^2 - i\epsilon} P_{\alpha\beta\mu\nu}, \quad (1.23)$$

with $P_{\alpha\beta\mu\nu}$ defined by

$$P_{\alpha\beta\mu\nu} \equiv \frac{1}{2} \left(\eta_{\mu\alpha} \eta_{\nu\beta} + \eta_{\mu\beta} \eta_{\nu\alpha} - \frac{2}{D-2} \eta_{\mu\nu} \eta_{\alpha\beta} \right). \quad (1.24)$$

1.2 Lagrangian of pure gravitation

In this section, the analogy between the free spin-2 quantized field and the free gravitational field will be shown, and we will provide a lagrangian of such field.

Since the only tensor that can be constructed from $h_{\mu\nu}$ is the Riemann tensor, the Lagrangian of pure gravitation must be a polynomial in the Riemann tensor. The first terms are written below

$$\mathcal{L} = \sqrt{-g} \left(c_0 + c_1 R + c_2 R^2 + c_3 R_{\mu\nu} R^{\mu\nu} + c_4 R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta} + \dots \right), \quad (1.25)$$

with c_i arbitrary constants and $\sqrt{-g} \equiv \sqrt{-\det g}$ that makes covariant the infinitesimal integration volume. The term c_0 is the cosmological term, and it will be ignored in the following. The terms with $i \geq 2$ are higher derivative terms because the Riemann tensor contains second order derivatives. These terms, of the form $\partial^2 h \partial^2 h$, will produce propagators behaving as q^{-4} which are good for convergence of the diagrams but contain non-physical states that lead to negative probabilities and acasual behaviours. We must therefore not allow such terms for the moment, but these will be useful for the formulation of gravity as an effective field theory. This assumption is reasonable also for the following reason. Considering only the terms in $c_1 R$ and $c_2 R^2$ and dropping the indices to focus only on the derivatives, the equation of motion becomes $\partial^2 h + \frac{2c_2}{c_1} \partial^2 h \partial^2 h$, which correspond to a propagator in the momentum space of the form

$$\mathcal{P}(q) = \frac{1}{q^2 + \frac{2c_2}{c_1} q^4} = \frac{1}{q^2} - \frac{1}{q^2 + \frac{c_1}{2c_2}},$$

which leads to a potential proportional to

$$V(r) \propto \frac{1}{r} - \frac{e^{-\frac{r}{\sqrt{c_1/2c_2}}}}{r} .$$

The constant c_1 is fixed by the Einstein field equations to be $c_1 = \frac{1}{16\pi G_N}$ and imagining a $c_2 \sim 1$ the denominator of the exponent in the potential is $\sqrt{\frac{c_1}{2c_2}} \sim 10^{-35}\text{m}$. An upper bound for c_2 has been found out by Stelle [24] who gives the experimental bound $c_2 < 10^{74}$. Therefore at low energies there is no effect on any observable physics of the R^2 terms and even more so the higher order ones.

The only terms that we consider in the lagrangian form the so called Einstein-Hilbert action, defined by

$$S_{EH} \equiv \frac{2}{\kappa^2} \int d^D x \sqrt{-g(x)} R(x) , \quad (1.26)$$

with $\kappa \equiv \sqrt{32\pi G_N}$ fixed by the Einstein field equation through the principle of least action.

We will show that the lagrangian implicitly defined above is the generalisation of 1.13 beyond the weak field limit. To do so we need to compute g and R under this hypothesis. Defining the metric $g_{\mu\nu}$ as the flat metric plus a small deviation $h_{\mu\nu}$, that is

$$g_{\mu\nu} \simeq \eta_{\mu\nu} + h_{\mu\nu} , \quad (1.27)$$

is possible to express the lagrangian as a power expression of the $h_{\mu\nu}$. The inverse of the metric reads:

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + h^{\mu\alpha} h_{\alpha}{}^{\nu} + O(h^3) ,$$

obtained from the condition $g^{\mu\alpha} g_{\alpha\nu} = \delta^{\mu}{}_{\nu}$. The indices of $h_{\mu\nu}$ can be raised or lowered with the flat metric.

The Christoffel symbols (1.1) for this metric become

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} \eta^{\lambda\alpha} (\partial_{\nu} h_{\mu\alpha} + \partial_{\mu} h_{\alpha\nu} - \partial_{\alpha} h_{\mu\nu}) + O(h^2) . \quad (1.28)$$

The $\sqrt{-g}$ part becomes (using $\sqrt{\det -g} = e^{\frac{1}{2} \text{tr} \ln(\eta+h)}$)

$$\sqrt{-g} = 1 + \frac{1}{2} h_{\mu}{}^{\mu} + \frac{1}{8} h_{\mu}{}^{\mu} h_{\nu}{}^{\nu} - \frac{1}{4} h_{\mu\nu} h^{\mu\nu} + O(h^3) .$$

The Ricci tensor and the scalar curvature take the expressions

$$\begin{aligned} R_{\mu\nu} &\equiv \partial_{\lambda} \Gamma_{\mu\nu}^{\lambda} - \partial_{\nu} \Gamma_{\mu\lambda}^{\lambda} + \Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\mu\nu}^{\sigma} - \Gamma_{\sigma\nu}^{\lambda} \Gamma_{\mu\lambda}^{\sigma} \\ &= \frac{1}{2} (\square h_{\mu\nu} - \partial_{\alpha} \partial_{\mu} h^{\alpha}{}_{\nu} - \partial_{\alpha} \partial_{\nu} h^{\alpha}{}_{\mu} + \partial_{\mu} \partial_{\nu} h^{\alpha}{}_{\alpha}) + O(h^2) \\ &\rightarrow R = \square h^{\mu}{}_{\mu} - \partial_{\alpha} \partial_{\mu} h^{\alpha\mu} + O(h^2) . \end{aligned}$$

Combining these equations one has a contribution to the lagrangian of the form (including the action multiplicative constant κ)

$$\frac{1}{\kappa^2} \sqrt{-g} R = \frac{1}{\kappa^2} \left(-\frac{1}{2} \partial_{\lambda} h_{\lambda\mu} \partial_{\mu} h_{\nu\nu} + \frac{1}{2} \partial_{\lambda} h_{\lambda\mu} \partial_{\nu} h_{\nu\mu} - \frac{1}{4} \partial_{\lambda} h_{\mu\nu} \partial_{\lambda} h_{\mu\nu} + \frac{1}{4} \partial_{\lambda} h_{\mu\mu} \partial_{\lambda} h_{\nu\nu} \right) + O(h^3) . \quad (1.29)$$

That is equivalent to the expression for the lagrangian of the free spin-2 field $h_{\mu\nu}$ (1.13) if we change the definition of the variation in the weak field limit (1.27) by a factor of κ , i.e.

$$\begin{aligned} g_{\mu\nu} &= \eta_{\mu\nu} + \kappa h_{\mu\nu} \\ g^{\mu\nu} &= \eta^{\mu\nu} - \kappa h^{\mu\nu} + \kappa^2 h^\mu{}_\alpha h^{\alpha\nu} + \dots \quad \sqrt{-g} = 1 + \frac{\kappa}{2} h - \frac{\kappa^2}{4} h^\alpha{}_\beta h^\beta{}_\alpha + \frac{\kappa^2}{8} h^2 + \dots \end{aligned} \quad (1.30)$$

So the propagator of the free gravitational field is equivalent to the massless spin-2 free particle in equations 1.22 and 1.23.

It is also possible to generalise the gauge invariance of the lagrangian defined in equation 1.14, now it is the invariance of the theory under a general coordinate translation of a constant vector a^μ , $x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$. As done for gauge theories we can promote this global invariance to a local invariance for the translation of a vector $a(x)^\mu$ [25]. This is the general invariance for a Lorentz transformation $x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu(x) x^\nu$ with $\Lambda^\mu{}_\nu(x) \equiv \frac{\partial x'^\mu}{\partial x^\nu}$ (from now on we will omit the dependence on coordinates of the Lorentz matrix, except when it is necessary, for a easier notation), under which the differential dx^μ transforms as $dx'^\mu = \Lambda^\mu{}_\nu dx^\nu$ and then the new metric becomes

$$g'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}(x) \equiv \bar{\Lambda}_\mu{}^\alpha \bar{\Lambda}_\nu{}^\beta g_{\alpha\beta}(x) ,$$

where we defined the inverse matrix $\bar{\Lambda}_\mu{}^\nu \equiv \frac{\partial x^\mu}{\partial x'^\nu}$. With such transformations it is easy to verify that the proper time interval $d\tau'^2 = g'_{\mu\nu}(x') dx'^\mu dx'^\nu = g_{\mu\nu}(x) dx^\mu dx^\nu = d\tau^2$ is invariant.

With a gauge invariance, it is tricky to use the path integrals method because of the over counting of the gauge copies. We need to introduce a gauge fixing term and the associated Faddeev-Popov [10] ghosts field to fix the gauge orbit appropriately, this will be done in more detail later.

1.2.1 Cosmological term

In the lagrangian of pure gravitation (1.25) we must add a term of the form $c_0 \sqrt{g}$ because the volume element $dx \sqrt{g}$ in the action is invariant under general coordinate transformations, and therefore is an acceptable contribution to the action.

This contribution does not contain derivatives of the field perturbations and was initially omitted in the treatment. The effects of such a term are analogous to a mass term that maintains the gauge invariance of the theory. The difference with a normal mass term is that the cosmological one does not increase the number of polarization states of the gravitational field. If we included this term in the expansion of the pure gravitational lagrangian, we would have obtained a quadratic term of the form $-\frac{1}{2} h_{\alpha\beta} V_{\alpha\beta\mu\nu} (\Box - c_0 \kappa^2) h_{\mu\nu}$, which is equal to the lagrangian 1.16 substituting the momenta $q^2 \rightarrow q^2 + c_0 \kappa^2$. This substitution has to be performed also in the propagator. It is possible to show that such a propagator would give negative probabilities if we study the exchange of a graviton between two sources [2]. This suggests that the inclusion of such term in the quantization of the field has to be managed more accurately or has to include other terms (for example tadpole

diagrams).

The effect of such a term on the equation of motion 1.3 would be simply

$$G_{\mu\nu} + \frac{c_0}{\kappa^2} g_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \frac{c_0}{\kappa^2} g_{\mu\nu} = -\frac{\kappa^2}{4} T_{\mu\nu} .$$

1.3 Gravitation and massive scalar particles

In this section we will analyse the interaction of the scalar and gravitational field. Here and in the following we take the cosmological constant to be zero. Considering a real scalar field $\phi(x)$ of mass m in a flat background the lagrangian density has the form

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 . \quad (1.31)$$

The translational invariance implies by Noether's theorem the existence of a conserved current

$$T_{\mu\nu} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi - \eta_{\mu\nu} \mathcal{L} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \left(\eta_{\mu\nu} \partial^\alpha \phi \partial_\alpha \phi - m^2 \phi^2 \right) . \quad (1.32)$$

By the definition of a conserved current using the equations of motion $\partial_\nu T^{\mu\nu} = 0$. To introduce the scalar particle in a curved background we just need to write the matter action in an invariant form in the sense of General Relativity (promoting $\eta_{\mu\nu}$ to $g_{\mu\nu}$ and inserting the invariant volume element in the integral). The matter action then reads

$$S_m = \int d^4x \sqrt{-g} \left(\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \right) . \quad (1.33)$$

As a check of consistence we can compute the conserved current associated with this action using the variational principle and see if it matches the promotion to General Relativity of 1.32. Using the general propriety for the variation of a matrix M

$$\begin{aligned} \delta \det M &= \det(M + \delta M) - \det M = e^{\text{Tr} \ln(M + \delta M)} - e^{\text{Tr} \ln M} \\ &= e^{\text{Tr}(\ln M + M^{-1} \delta M)} - e^{\text{Tr} \ln M} \quad \downarrow \ln(x + \delta x) \simeq \ln x + x^{-1} \delta x \\ &= e^{\text{Tr} \ln M} \text{Tr}(M^{-1} \delta M) \quad \downarrow e^{(x + \delta x)} \simeq e^x (1 + \delta x) \\ &= \det M \text{Tr}(M^{-1} \delta M) , \end{aligned}$$

and therefore the variation with respect to $g^{\mu\nu}$ yields

$$\frac{\delta \sqrt{-\det g}}{\delta g^{\mu\nu}} = -\frac{\sqrt{-g}}{2} g_{\mu\nu} . \quad (1.34)$$

Using this propriety it is straightforward to obtain the variation of the action 1.33

$$\frac{\delta S_m}{\delta g^{\mu\nu}} = \frac{1}{2} \sqrt{-g} \left[\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - m^2 \phi^2) \right] .$$

We see that

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}} \quad (1.35)$$

is obtained from 1.32 promoting the Lorentz indices to the world ones and replacing the metric as previously described.

The full invariant free action for the matter field coupling with gravity is given by:

$$S = S_{EH} + S_m . \quad (1.36)$$

As done for the interaction of the Electromagnetic field and the scalar field in QED the minimal-coupling term (dependent on the charge) of the lagrangian can be built from the conserved current as $eA_\mu J^\mu$.

The lagrangian density in the weak field limit has the form

$$\sqrt{-g}\mathcal{L} = \sqrt{-g}\left(\frac{R}{\kappa^2} - \mathcal{L}_{GF}\right) - \frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu} . \quad (1.37)$$

The expansion of only the gravitational part of the lagrangian has been already computed in 1.29. As done for the free spin-2 particle we need a gauge fixing term to be able to invert the differential operator that acts on the fields and have the possibility to extract a propagator. The gauge fixing term in the harmonic gauge is the same that we used in the free case of equation 1.15 and can be expressed more compactly by introducing the quantity

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h . \quad (1.38)$$

The expression for the gauge fixing term is therefore

$$\mathcal{L}_{GF} = \frac{1}{2}C_\mu^2 = \frac{1}{2}\partial^\lambda \bar{h}_{\mu\lambda} \partial_\lambda \bar{h}^{\mu\lambda} . \quad (1.39)$$

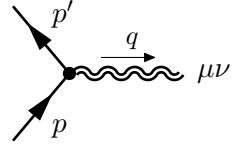
This term removes the problems in the lagrangian and integrating by parts it is possible to express it in the form:

$$\sqrt{-g}\mathcal{L} = \frac{1}{2}h_{\mu\nu}\square\left(I^{\mu\nu\alpha\beta} - \frac{1}{2}\eta^{\mu\nu}\eta^{\alpha\beta}\right)h_{\alpha\beta} - \frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu} . \quad (1.40)$$

This expression gives us the first important Feynmann rules of the theory of gravity coupled with massive spinless particles. At the lowest order we obtain the graviton propagator of the form

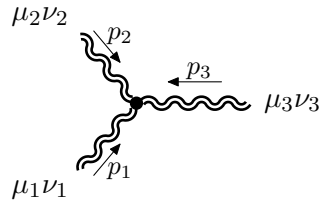
$$\begin{array}{c} \mu\nu \quad \overset{q}{\text{~~~~~}} \quad \alpha\beta \end{array} = \mathcal{P}_{\alpha\beta\mu\nu}^{(4)}(q) = \frac{iP_{(4)}^{\mu\nu\alpha\beta}}{q^2} .$$

The vertex including the matter can be obtained from the interaction term. Taking the expression of the interaction term and of $T^{\mu\nu}$ for a scalar field ϕ (1.32) the corresponding vertex is given by



$$\begin{aligned}
&= \tau_{\mu\nu}(p, p', m) = \\
&= -\frac{i\kappa}{2} \left[(p_\mu p'_\nu + p_\nu p'_\mu) - \eta_{\mu\nu}(p \cdot p' - m^2) \right].
\end{aligned} \tag{1.41}$$

Expanding the gravitational lagrangian up to $O(h^4)$ gives terms with a product of three gravitational fields $h_{\mu\nu}$ or derivatives, which correspond to a vertex with three entries. Expanding up to every order leads always to a vertex with a number of gravitons equal to the order of expansion, so at each loop order new vertexes appear in the theory. For the three gravitons vertex the expression has been calculated for the first time by DeWitt [5] in 1967 and has the form

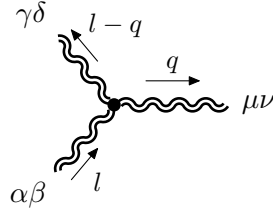


$$\begin{aligned}
&= \tau^{\mu_1 \nu_1 \mu_2 \nu_2 \mu_3 \nu_3}(p_1, p_2, p_3) = \\
&= \frac{i\kappa}{2} \text{Sym} \left[-\frac{1}{4} P_3(p \cdot p' \eta^{\mu_1 \nu_1} \eta^{\mu_2 \nu_2} \eta^{\mu_3 \nu_3}) - \frac{1}{4} P_6(p^{\mu_2} p'^{\nu_2} \eta^{\mu_1 \nu_1} \eta^{\mu_3 \nu_3}) + \frac{1}{4} P_3(p \cdot p' \eta^{\mu_1 \mu_2} \eta^{\nu_1 \nu_2} \eta^{\mu_3 \nu_3}) \right. \\
&\quad + \frac{1}{2} P_6(p \cdot p' \eta^{\mu_1 \nu_1} \eta^{\mu_2 \mu_3} \eta^{\nu_2 \nu_3}) + P_3(p^{\mu_2} p'^{\nu_3} \eta^{\mu_1 \nu_1} \eta^{\nu_2 \mu_3}) - \frac{1}{2} P_3(p^{\nu_2} p'^{\mu_1} \eta^{\nu_1 \mu_2} \eta^{\mu_3 \nu_3}) \\
&\quad + \frac{1}{2} P_3(p^{\mu_3} p'^{\nu_3} \eta^{\mu_1 \mu_2} \eta^{\nu_1 \nu_2}) + \frac{1}{2} P_6(p^{\mu_3} p'^{\nu_3} \eta^{\mu_1 \mu_2} \eta^{\nu_1 \nu_2}) + P_6(p^{\mu_2} p'^{\nu_3} \eta^{\nu_2 \mu_1} \eta^{\nu_1 \mu_2}) \\
&\quad \left. + P_3(p^{\mu_2} p'^{\mu_1} \eta^{\nu_2 \mu_3} \eta^{\nu_3 \nu_1}) - P_3(p \cdot p' \eta^{\nu_1 \mu_2} \eta^{\nu_2 \mu_3} \eta^{\nu_3 \mu_1}) \right].
\end{aligned}$$

The "Sym" indicates that a symmetrization has to be performed on each index pair. The "P" indicates that a summation has to be performed on all the distinct permutations of the momentum-index triplets (the pedices give the number of terms)² In the literature another expression is present of the three graviton vertex, given in 1995 by Donoghue [11]:

²For example:

$$\text{Sym}[P_3(p^{\mu_2} p'^{\mu_1} \eta^{\nu_2 \mu_3} \eta^{\nu_3 \nu_1})] = p_1^{\mu_2} p_2^{\mu_1} \eta^{\nu_2 \mu_3} \eta^{\nu_3 \nu_1} + p_1^{\mu_3} p_3^{\mu_1} \eta^{\nu_3 \mu_2} \eta^{\nu_2 \nu_1} + p_2^{\mu_3} p_3^{\mu_2} \eta^{\nu_3 \mu_1} \eta^{\nu_1 \nu_2}.$$



$$= \tau_{\alpha\beta\gamma\delta}^{\mu\nu}(l, q) =$$

$$\begin{aligned}
= & \frac{i\kappa}{2} \left\{ P_{\alpha\beta\gamma\delta}^{(4)} \left[l^\mu l^\nu + (l - q)^\mu (l - q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] \right. \\
& + 2q_\lambda q_\sigma \left[I_{\alpha\beta}^{\lambda\sigma} I_{\gamma\delta}^{\mu\nu} + I_{\gamma\delta}^{\lambda\sigma} I_{\alpha\beta}^{\mu\nu} - I_{\alpha\beta}^{\lambda\mu} I_{\gamma\delta}^{\sigma\nu} - I_{\alpha\beta}^{\sigma\nu} I_{\gamma\delta}^{\lambda\mu} \right] \\
& + q_\lambda q^\mu \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\lambda\nu} + \eta_{\gamma\delta} I_{\alpha\beta}^{\lambda\nu} \right) + q_\lambda q^\nu \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\lambda\mu} + \eta_{\gamma\delta} I_{\alpha\beta}^{\lambda\mu} \right) \\
& - q^2 \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\mu\nu} + \eta_{\gamma\delta} I_{\alpha\beta}^{\mu\nu} \right) - \eta^{\mu\nu} q^\lambda q^\sigma \left(\eta_{\alpha\beta} I_{\gamma\delta\lambda\sigma} + \eta_{\gamma\delta} I_{\alpha\beta\lambda\sigma} \right) \\
& + 2q^\lambda \left[I_{\alpha\beta}^{\sigma\lambda} I_{\gamma\delta\sigma\nu} (l - q)^\mu + I_{\alpha\beta}^{\lambda\sigma} I_{\gamma\delta\sigma}^\mu (l - q)^\nu - I_{\gamma\delta}^{\lambda\sigma} I_{\alpha\beta\sigma}^\nu l^\mu - I_{\gamma\delta}^{\lambda\sigma} I_{\alpha\beta\sigma}^\mu l^\nu \right] \\
& + q^2 \left(I_{\alpha\beta}^{\sigma\mu} I_{\gamma\delta\sigma}^\nu + I_{\alpha\beta\sigma}^\nu I_{\gamma\delta}^{\sigma\mu} \right) + \eta^{\mu\nu} q^\lambda q_\sigma \left(I_{\gamma\delta}^{\rho\sigma} I_{\alpha\beta\lambda\rho} + I_{\alpha\beta}^{\rho\sigma} I_{\gamma\delta\lambda\rho} \right) \\
& + (l^2 + (l - q)^2) \left(I_{\alpha\beta}^{\sigma\mu} I_{\gamma\delta\sigma}^\nu + I_{\alpha\beta}^{\sigma\nu} I_{\gamma\delta\sigma}^\mu - \frac{1}{2} \eta^{\mu\nu} P_{\alpha\beta\gamma\delta}^{(4)} \right) \\
& \left. - l^2 \eta_{\alpha\beta} I_{\gamma\delta}^{\mu\nu} - (l - q)^2 \eta_{\alpha\beta\gamma\delta} I_{\alpha\beta}^{\mu\nu} \right\} . \tag{1.42}
\end{aligned}$$

This vertex is in general dimension and the appearance of the projector $P_{\alpha\beta\gamma\delta}^{(4)}$, which seems to be a fix in the dimensions, is just a coincidence of the lagrangian expression. This vertex has been computed with the background field method [11] and this is the reason why it is different from the one of DeWitt. We think that is wrong to use this vertex if it appears as an internal one because the emitted graviton leg is treated differently with respect to the others (due to the background field method expansion). If in a diagram an internal 3graviton vertex appears, the DeWitt's one should be used or a symmetrization has to be performed in order to use the Donoghue's vertex.

The gauge fixing (1.39) method applied to the lagrangian 1.40 has to be implemented also in the path integral formalism and so we have the same behaviours described in Appendix A. In particular there is the arising of ghost particles.

Since in gravity the gauge fixing condition has a vector form, the ghosts have to be fermionic vectors. The gauge invariance of the quantum metric is given by

$$h'_{\mu\nu} = h_{\mu\nu} + \partial_\mu \eta_\nu + \partial_\nu \eta_\mu \tag{1.43}$$

in the de Donder gauge, where η_μ is a function of the coordinates for the reasons given in the previous chapter.

We need to compute the Faddeev-Popov determinant which depends on the derivative of the gauge fixing function with respect to the group parameter (A.2). The gauge fixing term becomes

$$C'_\mu \equiv \partial^\alpha h'_{\alpha\mu} - \frac{1}{2} \partial_\mu h' = C_\mu + \square \eta_\mu ,$$

and so A.2 with this notation and this gauge becomes

$$\Delta(h) = \det \left(\frac{\delta C'_\mu}{\delta \eta_\nu} \right) = \det(\square \eta_{\mu\nu}) = \int \delta c_\alpha \delta \bar{c}_\beta e^{i \int d^4 x \sqrt{-g} \bar{c}^\mu (\square \eta_{\mu\nu}) c^\nu} .$$

As anticipated we see that the ghost particles are fermionic vectors due to the tensorial form of the gauge transformation.

Expanding the determinant of the metric we can compute the Feynmann rules for the ghosts field. At the lowest order we have the free ghost propagator of the form

$$\mu \cdots \blacktriangleright^q \cdots \nu = \frac{\eta_{\mu\nu}}{q^2}.$$

Expanding to second order we have the 2ghosts-1graviton vertex given by

$$\begin{array}{c} \alpha \\ \vdots \\ p' \\ \vdots \\ \bullet \\ \vdots \\ p \\ \vdots \\ \beta \end{array} \begin{array}{c} \xrightarrow{q} \\ \text{wavy line} \\ \mu\nu \end{array} = -\frac{i\kappa}{2} \left[\eta_{\alpha\beta} p_\mu p'_\nu + \eta_{\alpha\beta} p_\nu p'_\mu - \eta_{\alpha\mu} q_\nu p'_\beta - \eta_{\alpha\nu} q_\mu p'_\beta \right].$$

The structure of all vertices is of an incoming ghost particle and the corresponding outgoing anti-particle (or the opposite). The diagrams that we can build from these components with no external ghost particles are all with internal loops of the artificial particle and the anti-particle and the contribution given to the amplitudes is none. That is the reason why these particles will not be taken into account when doing future computations.

1.4 Gravitation and photons

In this section, we will provide all the Feynman rules for the theory of gravity coupled to scalar and photons fields.

As done for the massive scalar field the first step to quantize the lagrangian of the electromagnetic and complex scalar fields is to write it in a covariant form. A general covariant version of QED lagrangian is given by [15]

$$\mathcal{L} = \sqrt{-g} \left\{ -\frac{1}{4} (g^{\alpha\mu} g^{\beta\nu} F_{\alpha\nu} F_{\mu\beta}) + [(\partial_\mu + ieA_\mu) \phi]^* (g^{\mu\nu}) [(\partial_\nu + ieA_\nu) \phi] - m^2 |\phi|^2 \right\}, \quad (1.44)$$

where $F_{\mu\nu} \equiv D_\mu A_\nu - D_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the covariant electromagnetic tensor. To obtain the Feynmann rules of the theory we need to perform the weak field limit (1.30) and expand the lagrangian in powers of the quantized fields $h_{\mu\nu}$, ϕ and A_μ . In order to do that we ignore for the moment the part of the lagrangian with only scalar and gravitational fields and we will expand the rest up to the first order in κ . The part in which we are interested in is

$$\begin{aligned} & \sqrt{-g} \left\{ -\frac{1}{4} (g^{\alpha\mu} g^{\beta\nu} F_{\alpha\nu} F_{\mu\beta}) + g^{\mu\nu} [e^2 A_\mu A_\nu \phi^2 + ie(\phi \partial_\mu \phi^* A_\nu - \phi^* \partial_\nu \phi A_\mu)] \right\} \simeq \left. \begin{array}{l} g^{\alpha\mu} g^{\beta\nu} \simeq \eta^{\alpha\mu} \eta^{\beta\nu} - \\ - \kappa(\eta^{\alpha\mu} h^{\beta\nu} + \\ + \eta^{\beta\nu} h^{\alpha\mu}) \end{array} \right\} \\ & \simeq \sqrt{-g} \left\{ -\frac{1}{4} (F_{\mu\nu})^2 + \eta^{\mu\nu} [e^2 A_\mu A_\nu \phi^2 + ie(\phi \partial_\mu \phi^* A_\nu - \phi^* \partial_\nu \phi A_\mu)] + \right. \\ & \quad \left. + \kappa \left[\frac{1}{2} F^\mu{}_\nu F_{\mu\beta} h^{\beta\nu} - h^{\mu\nu} (e^2 A_\mu A_\nu \phi^2 + ie(\phi \partial_\mu \phi^* A_\nu - \phi^* \partial_\nu \phi A_\mu)) \right] \right\} \simeq \left. \right\} \sqrt{-g} \simeq 1 + \frac{\kappa}{2} h \\ & \simeq -\frac{1}{4} (F_{\mu\nu})^2 + e^2 (A_\mu)^2 \phi^2 + ie(\phi \partial_\mu \phi^* A_\mu - \phi^* \partial_\mu \phi A_\mu) \\ & \quad + \kappa \left[\frac{1}{2} F^\mu{}_\nu F_{\mu\beta} h^{\beta\nu} - h^{\mu\nu} (e^2 A_\mu A_\nu \phi^2 + ie(\phi \partial_\mu \phi^* A_\nu - \phi^* \partial_\nu \phi A_\mu)) - \frac{1}{8} h (F_{\mu\nu})^2 + \frac{\epsilon^2}{2} h (A_\mu)^2 \phi^2 \right]. \end{aligned}$$

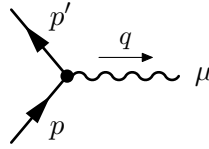
Substituting

$$\begin{aligned} F^\mu{}_\nu F_{\mu\beta} &= \partial_\mu A_\nu \partial^\mu A_\beta + \partial^\nu A_\mu \partial_\beta A_\mu - \partial^\mu A_\nu \partial_\beta A_\mu - \partial^\nu A_\mu \partial_\mu A_\beta \\ (F_{\mu\nu})^2 &= 2(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A_\mu) , \end{aligned}$$

we obtain the expansion of the lagrangian (without the fully scalar term) up to first order in κ

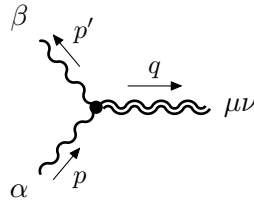
$$\begin{aligned} \mathcal{L} \simeq & -\frac{1}{4}(F_{\mu\nu})^2 + e^2(A_\mu)^2\phi^2 + ie(\phi\partial_\mu\phi^*A_\mu - \phi^*\partial_\mu\phi A_\mu) \\ & + \kappa \left\{ \frac{1}{2}h^{\beta\nu}(\partial_\mu A_\nu \partial^\mu A_\beta + \partial^\nu A_\mu \partial_\beta A_\mu - \partial^\mu A_\nu \partial_\beta A_\mu - \partial^\nu A_\mu \partial_\mu A_\beta) - \right. \\ & - \frac{1}{4}h(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A_\mu) - \\ & \left. - h^{\mu\nu} \left(e^2 A_\mu A_\nu \phi^2 + ie(\phi\partial_\mu\phi^*A_\mu - \phi^*\partial_\mu\phi A_\mu) + \frac{e^2}{2}h(A_\mu)^2\phi^2 \right) \right\} . \end{aligned}$$

From the above expression, it is possible to extract all the propagators and the lowest order vertices of the theory. The two scalars one photon vertex is visible in the first row of $\mathcal{L}$ and is



$$= \tau^\mu(p, p') = -ie(p + p')^\mu .$$

The two photons one graviton vertex comes from the order κ terms and takes the expression



$$= \tau_{\mu\nu, \alpha, \beta}(p, p') =$$

$$\begin{aligned} = & i\kappa \left\{ P_{\mu\nu\alpha\beta}^{(4)}(p \cdot p') + \frac{1}{2} \left[\frac{1}{2}\eta_{\mu\nu}(p_\alpha p'_\beta + p_\beta p'_\alpha) + \eta_{\alpha\beta}(p_\mu p'_\nu + p_\nu p'_\mu) \right. \right. \\ & \left. \left. - \frac{1}{2} \left(\eta_{\beta\nu}(p_\mu p'_\alpha + p_\alpha p'_\mu) + \eta_{\alpha\nu}(p_\mu p'_\beta + p_\beta p'_\mu) + \eta_{\alpha\mu}(p_\nu p'_\beta + p_\beta p'_\nu) + \eta_{\beta\mu}(p_\nu p'_\alpha + p_\alpha p'_\nu) \right) \right] \right\} . \end{aligned} \quad (1.45)$$

As mentioned in the scalar section the presence of the $\sqrt{-g}$ in front of the lagrangian leads to an infinite expansion with an infinite number of vertices at all orders. We showed the technique to compute these rules but at higher orders the computations become too long and difficult to be reported in this study. It is important to know that extra vertices exist and the expressions for those needed in the computations performed in the next chapters are reported in Appendix D.

1.5 Divergences and Effective Field Theories

Once the propagators and the vertices of the theory have been computed we can proceed, as done for QED, with the evaluation of one-loop corrections. Unfortunately

the theory will manifest problems in these computations. It is useful to estimate the degree of divergence $\mathcal{D}$ of a Feynman diagram with V vertices and I internal lines, which is the total power in momentum. Every internal line is proportional to q^{-2} (1.23) and the vertex function like q^2 (E.4). Accounting also the integration over the loop momentum in D dimensions is given by

$$\mathcal{D} = DL + 2V - 2I ,$$

substituting the topological relation $L = 1 + I - V$ we have

$$\mathcal{D} = 2 + (D - 2)L .$$

For $D > 2$ the degree of divergence increases with the loop order and therefore the theory is not perturbatively renormalizable.

The full computations can be found in [2, 5] and lead to the one-loop counterterm for pure gravity of the form

$$\Delta\mathcal{L} = \frac{\sqrt{-g}}{D-4} \left(\alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} \right) , \quad (1.46)$$

which is divergent in $D = 4$.

There is the possibility to treat gravity as an effective theory, a theory that is valid and complete at a certain energy scale. Effective field theory techniques have been developed for dividing the scales of a problem and are used to make expectations at the lowest energy scales (or the largest distance scales). In our case the lightest scale is the photon and graviton scales (zero energy), so we will then separate the effects of massive degrees of freedom from the massless ones. Any effective theory is built up from terms with a higher and higher number of derivatives coupling to the fields obeying the gauge symmetries. Any effective field theory can be seen as an expansion in the *light* field below a certain scale. At energies above the scale transition, the presence of massive *heavy* fields will be manifested. We must differentiate the effects of these two fields. The *heavy* fields can be seen as a local effect on the lagrangian, as a shift in the coefficients of the general lagrangian. It can be seen from the power expression of the propagator $(q^2 - m^2)^{-1} \simeq -m^{-2} - q^2 m^{-2} - q^4 m^{-4}$ which in real space is a delta (local effect) plus higher derivatives terms. The *light* fields instead have also nonlocal effects on the low-energy lagrangian. The massless propagator is of the form q^{-2} and can't be expanded in $q^2 = 0$. Therefore the massless particles can not be integrated out of the theory but must be included explicitly.

As previously stated we need to build up the lagrangian of the effective theory from the invariants of the theory. For the pure gravity case, this is simply because all the possible invariants of the theory are constructed from the contractions of the Riemann tensor. The effective pure gravitational lagrangian is

$$\mathcal{L} = \sqrt{-g} \left(c_0 + c_1 R + c_2 R^2 + c_3 R_{\mu\nu} R^{\mu\nu} + c_4 R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta} + \dots \right) . \quad (1.47)$$

The coefficients represent the presence of effective coupling in the action dependent on the energy scale. As we see the first extra terms fix the divergences of the one-loop lagrangian (1.46) in the redefinition of the coupling constants. This happens at all orders with the higher terms and in principle, we would need an infinite number of

terms to develop an exact theory at all energy scales.

For the interaction lagrangian we need to be careful with the derivatives with respect to the *light* fields ($\tilde{\partial}$) and the ones with respect to *heavy* fields (∂). The $\tilde{\partial}$ terms generate powers of momentum and ∂ terms powers of interacting mass that is usually much bigger than the momentum terms. For the scalar graviton theory the first terms to be included are:

$$\begin{aligned}\sqrt{-g}R(\partial_\mu\phi)^2 & (\tilde{\partial}\tilde{\partial}\partial\partial) \\ \sqrt{-g}R^{\mu\nu}\partial_\mu\phi\partial_\nu\phi & (\tilde{\partial}\tilde{\partial}\partial\partial) \\ \sqrt{-g}m^2R\phi^2 & (\tilde{\partial}\tilde{\partial}) ,\end{aligned}$$

and are the one-loop divergences of the minimal theory. For the graviton photon case these divergences are of the type:

$$\begin{aligned}\sqrt{-g}(T_{\mu\nu})^2 & (\tilde{\partial}\tilde{\partial}\tilde{\partial}\tilde{\partial}) \\ \sqrt{-g}T_{\mu\nu}R^{\mu\nu} & (\tilde{\partial}\tilde{\partial}\tilde{\partial}\tilde{\partial}) ,\end{aligned}$$

where $T_{\mu\nu}$ is the Belinfante–Rosenfeld stress–energy tensor

$$T_{\mu\nu} = F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}g_{\mu\nu}F^{\alpha\beta}F_{\alpha\beta} . \quad (1.48)$$

Therefore the pieces we need to include in the minimal lagrangian to make it an effective field theory are:

$$\mathcal{L}_{scalar} = \sqrt{-g} \left[c_1 R^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + c_2 R (\partial_\mu \phi)^2 + c_3 m^2 R \phi^2 + \dots \right] \quad (1.49)$$

$$\mathcal{L}_{photon} = \sqrt{-g} \left[d_1 (T_{\mu\nu})^2 + d_2 R_{\mu\nu} T^{\mu\nu} + \dots \right] , \quad (1.50)$$

where the dots indicate more terms with higher derivative couplings and mixed contribution with both photons, gravitons and scalar couplings.

We will see that only the scalar couplings need to be added in order to renormalize both scalar and photonic contributions since the divergences are of the same order and both respect the symmetry of the problem.

1.6 Classical limit

In this section we will explain how classical contributions arise from loop computations. Following the argument given in [16] it is possible to show that the expansion in $\hbar$ not always coincides with loop expansion, and therefore classical terms (not dependent on $\hbar$) can arise from loop terms.

The usual analogy between the loop and $\hbar$ expansions comes from the following reasoning. In a Feynman diagram, every vertex will contribute with a factor $\frac{1}{\hbar}$ because it comes from the expansion of the factor $\exp \frac{i}{\hbar} \int d^4x \mathcal{L}_{int}$ in the partition function. Instead, every propagator will contribute with a factor $\hbar$, due to the commutation relations. Therefore in every diagram with V vertices and I internal lines we will have a contribution in terms of $\hbar$ of the type $\hbar^{I-V+1}$. Using the topology relation for a L -loop diagram $L = I - V + 1$, we see that for this diagram the contribution is $\hbar^L$, and that is where the analogy between $\hbar$ expansion and loop

expansion arises.

We will prove that this argument is not always true. This comes from the fact that also the mass is dependent on $\hbar$ because of the Klein-Gordon equation $((\square + \frac{m^2}{\hbar^2})\phi(x) = 0)$. This changes the propagator dependence and therefore changes the diagram contribution. In particular, the right propagator with all the $\hbar$ explicit is

$$\langle 0|T(\phi(x)\phi(y)|0\rangle = \int \frac{d^4q}{(2\pi)^4} \frac{i\hbar e^{iq(x-y)}}{q^2 - \frac{m^2}{\hbar^2} + i\epsilon}.$$

In the following, we will show the example of the classical contribution found in QED from loop computations, given in [16].

In order to do that we have to define the form factors. Taking the transition density

$$\langle p_2|T_{\mu\nu}(x)|p_1\rangle,$$

with the usual normalization $\langle p_2|p_1\rangle = 2E_1(2\pi)^3\delta^3(\vec{q})$. If we require that for a scalar field $T_{\mu\nu}$ is a second rank conserved tensor we have the general expansion

$$\langle p_2|T_{\mu\nu}(0)|p_1\rangle = \frac{1}{\sqrt{4E_1E_2}} \left[2P_\mu P_\nu F_1(q^2) + (q_\mu q_\nu - \eta_{\mu\nu} q^2) F_2(q^2) \right],$$

where $P_\mu \equiv \frac{1}{2}(p_1 + p_2)_\mu$ and $q_\mu \equiv (p_1 - p_2)_\mu$ the momentum transferred.

The functions $F(q^2)$ are called form factors and are a momentum space description of the structure of a particle. At the lowest order conservation of momentum requires $F_1(q^2) = 1$, as can be seen from

$$\langle p_2|\hat{P}_\mu|p_1\rangle = \langle p_2|\int d^3x T_{\mu 0}(x)\Big|_{q^2=0}|p_1\rangle = P_\mu \langle p_2|p_1\rangle,$$

and there is no constraint on $F_2(q^2)$. In QED at the lowest order in the fine-structure constant we have $F_1(q^2) = 1$ and $F_2(q^2) = -1/2$. These results are modified at higher orders, in particular at one-loop corrections of order e^2 arises. These can be evaluated from all the possible one loop diagrams of the emission of a photon. The forms factors at one loop have the expressions [16]:

$$F_1(q^2) = 1 + \frac{e^2}{16\pi^2} \frac{q^2}{m^2} \left(\frac{3}{4} \frac{m\pi^2}{\sqrt{-q^2}} - \frac{8}{3} + 2 \log \frac{-q^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots$$

$$F_2(q^2) = -\frac{1}{2} + \frac{e^2}{16\pi^2} \left(\frac{m\pi^2}{2\sqrt{-q^2}} - \Omega - \frac{26}{9} + \frac{4}{3} \log \frac{-q^2}{m^2} \right) + \dots$$

Inserting the $\hbar$ hidden in the mass $(\frac{m}{\hbar})$, in the electric charge $(\frac{e^2}{\hbar})$ and in the momentum $(q = \hbar k)$, we have for $F_1(q^2)$

$$F_1(k^2) = 1 + \frac{e^2}{16\pi^2\hbar} \frac{\hbar^2 k^2}{m^2} \left(\frac{3}{4} \frac{m\pi^2}{\sqrt{-\hbar^2 k^2}} - \frac{8}{3} + 2 \log \frac{-\hbar^2 k^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots$$

$$= 1 + \frac{3e^2\sqrt{-k^2}}{64m} + \frac{\hbar e^2 k^2}{16\pi^2 m^2} \left(-\frac{8}{3} + 2 \log \frac{-\hbar^2 k^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots$$

The second term comes from a one-loop contribution but it is a classical term since it is of order zero in $\hbar$.

We see that the classical contributions arise from the non-analytic term with the square root of the momentum q , which is generated by the long-range propagation of massless particles.

This term is given by the diagram in which the EMT couples with the photon lines in a tree-like structure. Analysing the result classically it is possible to see this effect in the following way. The EMT couples with the electric field surrounding the particle and the particle itself. At tree-level the EMT accounts only for the interaction with the particle. But the particle has a classical field attached to it and the EMT couples classically also with this field. The one-loop tree-like diagram represents this coupling, which is due only to the classical parts of all the ingredients despite the loop computations. Also in the case in which the coupling constant is dimensionful (as gravity), the key feature of the classical contributions is given by these square root non-analytic terms.

The analysis done in this chapter is the starting point for the scattering computations of Chapter 2. All the Feynman rules previously given are summarised in Appendix E for faster consultation. All these rules will be the bricks needed for building the diagrams. The arguments about the classical contributions will be fundamental in extracting the contributions from the diagrams. The EFT treatment of gravity will allow us to obtain the renormalized metric through the non-minimal couplings constants redefinition, as done in Chapter 3.

Chapter 2

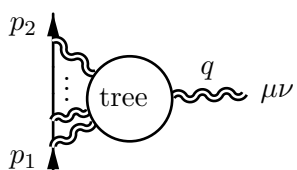
Metrics from Scattering Amplitudes

We will first give the general method to obtain a metric from diagrams containing a scalar line emitting a graviton. Then we will do the explicit computations for the massive and charged terms up to two-loop order (included). The computations for the massive part have already been given at three-loop order [17]. For the charged case the one-loop case has been computed in four dimensions [18] and general dimension [19]. The computations for the charged part at two-loop order are given here for the first time in literature.

2.1 General l -loop Case

We will describe the general method to obtain the l -loop corrections to the Schwarzschild-Tangerlini metric.

The action for a scalar interacting with the gravitational field is given by 1.36. In analogy with the Compton scattering, we will study all the diagrams with a scalar line connected with a tree diagram that emits a graviton. All the diagrams are of the type

$$\mathcal{M}(p_1, q) \equiv \text{diagram} \quad \text{with} \quad \begin{cases} q \equiv p_1 - p_2 \\ p_1^2 = p_2^2 = m^2 \end{cases}.$$


It is possible to expand the tree blob in different contributions ordered by the number of loops inside it. Therefore we can expand $\mathcal{M}$ in the number of loop, namely the last contribution to the l -loop expansion is $\mathcal{M}^{(l)}$. It is possible also to expand the EMT, which is related to the diagram $\mathcal{M}$, as $\langle T^{\mu\nu}(q^2) \rangle = \sum_l \langle T_{(l)}^{\mu\nu}(q^2) \rangle$. At each loop order we have the identity

$$i\mathcal{M}^{(l)}(p_1, q) = -\frac{i\sqrt{32\pi G_N}}{2} \langle T_{(l)}^{\mu\nu}(q^2) \rangle \epsilon_{\mu\nu} ,$$

where $\epsilon_{\mu\nu}$ is the polarisation tensor.

As done for the EMT we can expand the metric in loop contributions as $g_{\mu\nu} = 1 + \sum_{l=1}^{\infty} h_{\mu\nu}^{(l)}$. With this expansion the analogy between the loop expansion and the post-Minkowskian expansion of the metric is visible. The link between the EMT and the gravitational field is given by the Einstein equations. In the de Donder gauge these are

$$\square \bar{h}_{\mu\nu}^{(l)} \equiv \square (h_{\mu\nu}^{(l)} - \eta_{\mu\nu} h^{(l)}) = -\frac{\kappa^2}{2} T_{\mu\nu}^{(l)}. \quad (2.1)$$

Taking the trace of both sides it is possible to find h , then substituting and expressing the result in the real space

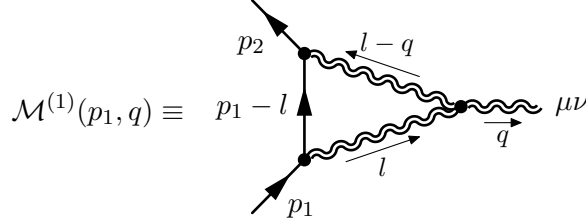
$$h_{\mu\nu}^{(l+1)}(\vec{x}) = -\frac{\kappa^2}{2} \int \frac{d^d q}{(2\pi)^d} e^{i\vec{x} \cdot \vec{q}} \frac{1}{\vec{q}^2} \left[\langle T_{\mu\nu}^{(l)}(q^2) \rangle - \frac{1}{d-1} \eta_{\mu\nu} \langle T^{(l)}(q^2) \rangle \right]. \quad (2.2)$$

For the reasons given above we will compute $\mathcal{M}$ under these conditions

$$\text{Static Limit} \quad q_\mu \simeq (0, \vec{q}) \quad \text{and} \quad \vec{q}^2 \ll m^2 \quad (2.3)$$

$$\text{Classical Limit} \quad \hbar \rightarrow 0 \quad \text{with} \quad \frac{q}{\hbar} \quad \text{fixed.} \quad (2.4)$$

As already mentioned $\mathcal{M}$ is the sum of all the diagrams with a different number of loops with two scalar and one graviton external lines. Following the reasoning given in [13] we will show that only some parts of these diagrams will contribute in these limits. At the first order in the number of loops the only diagram that contributes is



The contributions to the integral over the loop momenta l are proportional to

$$\int d^D l \frac{1}{l^2 + i\epsilon} \frac{1}{(l - q)^2 + i\epsilon} \frac{1}{(p_1 - l)^2 - m^2 + i\epsilon}.$$

From the static limit condition we can derive $(p - l)^2 - m^2 = l^2 - 2l_\mu p^\mu \simeq l_0^2 - 2ml_0$. We can then integrate the l_0 component using the Cauchy Theorem. The integrand term has a singularity in $l_0 = 2m - \frac{i\epsilon}{2m}$ and one in $l_0 = \frac{i\epsilon}{2m}$. Closing the contour of integration in the lower half-plane we surround only the first singularity that contributes with $\text{Res}(2m) = \frac{1}{2m}$. Omitting the multiplicative constants we are left with an integral over the spatial components of l that is equivalent to the integral over two static sources of massless fields located at different positions.

Applying the classical limit on the $\mathcal{M}$ diagrams is then equivalent to cutting the scalar line and inserting as many static sources as the vertices on this line connected to the gravitons of the tree structure [13, 16]. In the following, we will show that this leads to corrections to the post-Newtonian expansion of the Schwarzschild-Tangerlini

$$\int_{|\vec{l}| \ll m} d^d l \frac{1}{l^2} \frac{1}{(\vec{l} - \vec{q})^2} \quad \Longleftrightarrow \quad \text{Diagram: A vertex connected to two external lines (one solid, one wavy) and one internal loop line (wavy).}$$

metric.

For the general l -loop case it is convenient to split the diagram into the scalar plus gravitons line and the tree-like structure. Calling the scalar line as

$$L_{\mu_1 \nu_1 \dots \mu_{l+1} \nu_{l+1}}(p_1, p_2, l_1, \dots, l_{l+1}) \equiv \text{Diagram: A vertical chain of vertices connected by wavy lines. The top vertex has incoming momentum } p_2 \text{ and outgoing momentum } l_{l+1} \text{ and index } \mu_{l+1} \nu_{l+1}. \text{ The bottom vertex has incoming momentum } p_1 \text{ and outgoing momentum } l_1 \text{ and index } \mu_1 \nu_1. \text{ Intermediate vertices have indices } \mu_2 \nu_2, \dots. \text{ The chain is labeled } l_2, \dots. \text{ } =$$

$$= (i)^{l+1} \frac{\tau_{\mu_1 \nu_1}(p_1, p_1 - l_1) \dots \tau_{\mu_{l+1} \nu_{l+1}}(p_1 - \sum_{j=1}^l l_j, p_2)}{\prod_{i=1}^l [(p_1 - \sum_{j=1}^i l_j)^2 - m^2 + i\epsilon]}, \quad (2.5)$$

where $\tau_{\mu\nu}$ is given in E.2. Then we define the tree-structure part of the diagram as

$$M_{\mu_1 \nu_1 \dots \mu_{l+1} \nu_{l+1}} \equiv \text{Diagram: A circle labeled 'tree' with multiple wavy lines entering and exiting. The top line is labeled } \mu_{l+1} \nu_{l+1}, \text{ the bottom line } \mu_1 \nu_1, \text{ and the right line } \mu \nu. \text{ Other lines are labeled } \mu_{\nu 2}, \dots. \text{ The circle is labeled } q.$$

The sum over all possible diagrams created from the contraction of these two elements ($\mathcal{M}^{(l)}$) is now

$$\mathcal{M}^{(l)}(p_1, q) = \frac{i}{\sqrt{4E_1 E_2}} \int \frac{\prod_{n=1}^{l+1} d^D l_n}{(2\pi)^D} \left(\sum_{\sigma \in \mathcal{G}} L_{\mu_1 \nu_1 \dots \mu_{l+1} \nu_{l+1}}(p_1, p_2, l_{\sigma(1)}, \dots, l_{\sigma(l+1)}) \right) \times$$

$$\times i^{l+1} \prod_{i=1}^{l+1} \frac{P^{\mu_i \nu_i \alpha_i \beta_i}}{l_i^2 + i\epsilon} M_{\alpha_1 \beta_1 \dots \alpha_{l+1} \beta_{l+1}}, \quad (2.6)$$

where $\sum_{\sigma \in \mathcal{G}}$ is the sum over the group $\mathcal{G}$ of all possible permutations of $l+1$ elements.

The static limit makes some simplifications in the expression of the scalar line, as showed in the one-loop case. Using this condition the scalar line 2.5 becomes

$$L_{\mu_1\nu_1\dots\mu_{l+1}\nu_{l+1}}(p_1, p_2, l_1, \dots, l_{l+1}) = (i)^{l+1} \frac{\tau_{\mu_1\nu_1}(p_1, p_1 - l_1) \dots \tau_{\mu_{l+1}\nu_{l+1}}(p_1 - \sum_{j=1}^l l_j, p_2)}{\prod_{i=1}^l \left[(p_1 - \sum_{j=1}^i l_j)^2 - m^2 + i\epsilon \right]} \left(\tau_{\mu\nu}(p_1, p_1 - l) \simeq -i\kappa m^2 \delta_{\mu\nu} \right)$$

$$\simeq (i)^{2l} \left(\frac{\kappa}{2} m^2 \right)^{l+1} \frac{\prod_{i=1}^{l+1} \delta_{\mu_i}^0 \delta_{\nu_i}^0}{\prod_{i=1}^{l+1} \left[(p_1 - \sum_{j=1}^i l_j)^2 - m^2 + i\epsilon \right]} .$$

As done for the one-loop case the denominator contribution given by the scalar line can be expressed in terms of the time component of the loop-momenta. Thanks to the static condition, the denominator becomes

$$\begin{aligned} (p_1 - L)^2 - m^2 + i\epsilon &= L^2 - 2p_{1\mu}L^\mu + i\epsilon \simeq L_0^2 - \vec{L}^2 - 2mL_0 + i\epsilon \\ &= \left(L_0 - m - \sqrt{\vec{L}^2 + m^2 - i\epsilon} \right) \left(L_0 - m - \sqrt{\vec{L}^2 - m^2 - i\epsilon} \right) \\ &\simeq \left(L_0 - 2m - \frac{\vec{L}^2}{2m} + i\epsilon \right) \left(L_0 + \frac{\vec{L}^2}{2m} - i\epsilon \right) \simeq -2m(L_0 - i\epsilon) . \end{aligned} \quad (2.7)$$

Substituting inside the scalar line

$$L_{\mu_1\nu_1\dots\mu_{l+1}\nu_{l+1}}(p_1, p_2, l_1, \dots, l_{l+1}) \simeq (i)^{2l} \left(\frac{\kappa}{2} m^2 \right)^{l+1} \prod_i^{l+1} \frac{\frac{-1}{2m} \delta_{\mu_i}^0 \delta_{\nu_i}^0}{\sum_j^i l_j^0 - i\epsilon} .$$

Using the identity B.1 given in Appendix B

$$\sum_{\sigma \in \mathcal{G}} \prod_{i=1}^{l+1} \frac{1}{\sum_{j=1}^i x_{\sigma(j)}} = \prod_{i=1}^{l+1} \frac{1}{x_i} ,$$

and the Cauchy Theorem to integrate the time component of each loop momentum (obtaining a contribution of $(2\pi i)^{l+1}$)

$$\mathcal{M}^{(l)}(p_1, q) = (-i)^l \left(\frac{\kappa}{4} m \right)^{l+1} \int \frac{\prod_{n=1}^{l+1} d^d l_n}{(2\pi)^d} \prod_{i=1}^{l+1} \frac{P^{00\alpha_i\beta_i}}{l_i^2 + i\epsilon} M_{\alpha_1\beta_1\dots\alpha_{l+1}\beta_{l+1}} \Big|_{l_i^0=0} .$$

This diagram is equivalent to the one in which the massive line is collapsed to a point.

In the static limit the contraction of P and M will give contributions $\propto \sqrt{G_N}^l q^2$ [17] and higher powers of $\vec{q}^2$ will contribute to higher powers in $\hbar$ (it will be computed up to two loop order in the next sections), and therefore all these diagram can be expressed in terms of the massless sunset master integral with l -loop defined by

$$J_{(l)}(q^2) \equiv \int \frac{\prod_{n=1}^l d^d l_n}{(2\pi)^d} \frac{\vec{q}^2}{\prod_{i=1}^l \vec{l}_i^2 (\vec{l}_1 + \dots + \vec{l}_l + \vec{q})} \quad (2.8)$$

and

$$\mathcal{M}^{(l)}(p_1, q) \propto (-i)^l \left(\frac{\kappa}{4} m \right)^{l+1} \sqrt{G_N}^l J_{(l)}(q^2) .$$

The master integrals develop ultraviolet poles inducing divergences in the stress-energy tensor. It will be shown that at every fixed loop order it is possible to tune the coefficients of the higher-derivative terms added to the lagrangian in such a way that these divergences disappear. This renormalization process is shown in Chapter 3.

2.2 From Amplitudes to Metric Components

Finally we have all the ingredients to link the scattering computations to metrics. As discussed in the previous section the dependence of the diagrams $\mathcal{M}$ in q is all inside the master integral $J^{(l)}(q^2)$, which provides ultraviolet divergences to the theory. Therefore the general EMT that is conserved can be written in the following form

$$\langle T_{\mu\nu}^{(l)} \rangle = (\frac{\kappa^2}{16})^l m \left[c_1^{(l)}(d) \delta_\mu^0 \delta_\nu^0 + c_2^{(l)}(d) \left(\frac{q_\mu q_\nu}{q^2} - \eta_{\mu\nu} \right) \right] J^{(l)}(q^2). \quad (2.9)$$

Substituting inside 2.2 we can evaluate the metric correction to the l -loop order as

$$\begin{aligned} h_{\mu\nu}^{(l+1)}(\vec{q}) &= -\frac{16(\frac{\kappa^2}{16})^{l+1}}{q^2} \left[c_1^{(l)} \delta_\mu^0 \delta_\nu^0 + c_2^{(l)} \left(\frac{q_\mu q_\nu}{q^2} - \eta_{\mu\nu} \right) - \frac{\eta_{\mu\nu}}{d-1} \left(c_1^{(l)} + c_2^{(l)} - (d+1)c_2^{(l)} \right) \right] J^{(l)}(q^2) \\ &= -\frac{16(\frac{\kappa^2}{16})^{l+1}}{q^2} \left[c_1^{(l)} \left(\delta_\mu^0 \delta_\nu^0 - \frac{\eta_{\mu\nu}}{d-1} \right) + c_2^{(l)} \left(\frac{q_\mu q_\nu}{q^2} - \frac{\eta_{\mu\nu}}{d-1} \right) \right] J^{(l)}(q^2). \end{aligned}$$

We need to pass to real space applying the Fourier transformation to $h_{\mu\nu}^{(l+1)}(\vec{q})$ as $h_{\mu\nu}^{(l+1)}(\vec{x}) = \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q} \cdot \vec{x}} h_{\mu\nu}^{(l+1)}(\vec{q})$, and so substituting $h_{\mu\nu}^{(l+1)}(\vec{q})$ we have the two contributes

$$\begin{aligned} h_{\mu\nu}^{(l+1)}(\vec{x}) &= -16(\frac{\kappa^2}{16})^{l+1} \left\{ \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q} \cdot \vec{x}} \frac{J^{(l)}(q^2)}{q^2} \left[c_1^{(l)} \left(\delta_\mu^0 \delta_\nu^0 - \frac{\eta_{\mu\nu}}{d-1} \right) - \frac{c_2^{(l)}}{d-1} \eta_{\mu\nu} \right] + \right. \\ &\quad \left. + \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q} \cdot \vec{x}} \frac{q_\mu q_\nu}{q^4} J^{(l)}(q^2) c_2^{(l)} \right\}. \end{aligned}$$

Using the identities (Appendix C.1)

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q} \cdot \vec{x}} \frac{J^{(l)}(q^2)}{q^2} &= \left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{4\pi^{d/2} r^{d-2}} \right)^{l+1} \equiv \left(\frac{4\rho(r, d)}{\kappa^2} \right)^{l+1} \\ \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q} \cdot \vec{x}} \frac{q_\mu q_\nu}{q^4} J^{(l)}(q^2) &= \left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{4\pi^{d/2} r^{d-2}} \right)^{l+1} \frac{1}{2-l(d-2)} \left(-\delta_{ij} + (l+1)(d-2) \frac{x_i x_j}{r^2} \right), \end{aligned}$$

where we have defined the radial parameter:

$$\rho(r, d) \equiv \frac{\Gamma(d/2) G_N m \pi}{\pi^{d/2} r^{d-2}},$$

the metric becomes

$$h_{\mu\nu}^{(l+1)}(\vec{x}) = -16 \left(\frac{\rho(r, d)}{4} \right)^{l+1} \left[c_1^{(l)} \left(\delta_\mu^0 \delta_\nu^0 - \frac{\eta_{\mu\nu}}{d-1} \right) - \frac{c_2^{(l)}}{d-1} \eta_{\mu\nu} + c_2^{(l)} \frac{-\delta_{ij} + (l+1)(d-2) \frac{x_i x_j}{r^2}}{2-l(d-2)} \right].$$

To express the metric in the form of 4.5 we need to use the equations 4.5 in the hypothesis that the metric components are written as $h_i(r) = h_i^0 + \sum_{l \geq 0} h_i^l(r)$, with $h_i^0 = (-1, 1, 0)$. To do so we compute the 00 and ij components of the metric:

$$\begin{aligned} h_{00}^{l+1} &= -16 \left(\frac{\rho(r, d)}{4} \right)^{l+1} \frac{1}{d-1} \left[(d-2)c_1^{(l)} - c_2^{(l)} \right] \\ h_{ij}^{l+1} &= -16 \left(\frac{\rho(r, d)}{4} \right)^{l+1} \left[\left(c_1^{(l)} - c_2^{(l)} - c_2^{(l)} \frac{d-1}{2-l(d-2)} \right) \frac{\delta_{ij}}{d-1} + \frac{c_2^{(l)}(l+2)(d-2)}{2-l(d-2)} \frac{x_i x_j}{r^2} \right]. \end{aligned}$$

Comparing with 4.5 we have

$$\begin{cases} h_0^{(l+1)}(r, d) &= -\frac{16}{d-1} \left((d-2)c_1^{(l)}(d) + c_2^{(l)}(d) \right) \left(\frac{\rho(r, d)}{4} \right)^{l+1} \\ h_1^{(l+1)}(r, d) &= \frac{16}{d-1} \left(c_1^{(l)}(d) - \left(1 + \frac{d-1}{2-l(d-2)} \right) c_2^{(l)}(d) \right) \left(\frac{\rho(r, d)}{4} \right)^{l+1} \\ h_2^{(l+1)}(r, d) &= 16 \frac{(d-2)(l+1)}{2-l(d-2)} c_2^{(l)}(d) \left(\frac{\rho(r, d)}{4} \right)^{l+1} . \end{cases} \quad (2.10)$$

The components of the metric are divergent for $l = 1 \vee d = 4$ and $l = 2 \vee d = 3$.

2.3 Quantum computation massive case

In this chapter we will compute the contribution to the metric components at each loop order, obtaining the stress-energy tensor from the Feynman Rules (E) and then from this is possible to extract the metric through the constants $c_1^{(l)}(d)$ and $c_2^{(l)}(d)$.

2.3.1 Tree-level

At tree-level the only possible diagram contributing to the amplitude $\mathcal{M}^0$ is

$$\mathcal{M}^0(p_1, q) = \begin{array}{c} \text{Diagram showing a vertex where two incoming lines (momenta } p \text{ and } p') \text{ meet a wavy line (momentum } q \text{) labeled } \mu\nu. \end{array}$$

The vertex 2scalars-1graviton is given in E.2. In the static limit it takes the expression

$$\tau_{\mu\nu}(p_1, p_2) = \frac{i\kappa}{2} \left[(p_\mu p'_\nu + p_\nu p'_\mu) - \eta_{\mu\nu}(p \cdot p' - m^2) \right] \simeq i\kappa m^2 \delta_\mu^0 \delta_\nu^0 . \quad (2.11)$$

Substituting inside the expression of $\mathcal{M}^0$ obtained by the Feynmann rules gives

$$\begin{aligned} \mathcal{M}^0(p_1, q) &= \frac{i}{\sqrt{4E_1 E_2}} \frac{i\kappa}{2} \epsilon^{\mu\nu} \left[(p_\mu p'_\nu + p_\nu p'_\mu) - \eta_{\mu\nu}(p \cdot p' - m^2) \right] \simeq \\ &\simeq -\frac{\kappa}{2} m \epsilon^{\mu\nu} \delta_\mu^0 \delta_\nu^0 . \end{aligned}$$

Comparing with the expression of $T_{\mu\nu}^{(0)}$ given in 2.9 and 2.1 it is possible to find the coefficients $c_1^{(0)}$ and $c_2^{(0)}$. these are

$$\begin{cases} c_1^{(0)} = 1 \\ c_2^{(0)} = 0 . \end{cases}$$

Using this result it is possible to compute the first post-Minkowskian correction to the metric trough the equation 2.10. This is found to be:

$$\begin{cases} h_0^{(1)}(r, d) = -4 \frac{d-2}{d-1} \rho(r, d) \\ h_1^{(1)}(r, d) = \frac{d}{d-1} \rho(r, d) \\ h_2^{(1)}(r, d) = 0 . \end{cases}$$

At tree level the metric is not divergent in $d > 1$ so these results don't need the renormalization process and are already comparable with the classical computations. In particular these values match with the pure massive terms in 4.22 for the general dimension case and their values for fixed d are in agree for $d = 3$ (4.23) and $d = 4$ (4.24).

2.3.2 One-loop

For the one-loop case, the only diagram contributing to the process is the one already introduced in the general l -loop computations. This diagram is given by

$$\mathcal{M}^{(1)}(p_1, q) \equiv \begin{array}{c} \text{Diagram: A loop of two wavy lines. The left vertex has incoming momenta } p_1 \text{ and } p_2, \text{ and an outgoing momentum } p_1 + l. \text{ The right vertex has an outgoing momentum } \mu\nu. \text{ The loop momenta are } l \text{ and } l + q. \end{array} = -\frac{\kappa}{2} \epsilon_{\mu\nu} T_1^{\mu\nu}(q^2),$$

with the stress-energy tensor defined by the Feynman rules as

$$\begin{aligned} T_1^{\mu\nu}(q^2) &= \frac{i}{\sqrt{4E_1 E_2}} \int \frac{d^D l}{(2\pi)^D} \frac{\tau^{\rho\sigma}(p_1, p_1 + l) P_{\rho\sigma cd} \tau_{cdab}^{\mu\nu}(l, q) P_{ab\alpha\beta} \tau^{\alpha\beta}(p_2, p_1 + l)}{(l^2 - i\epsilon)((l + q)^2 - i\epsilon)((p_1 + l)^2 - m^2 - i\epsilon)} \\ &= \frac{i}{\sqrt{4E_1 E_2}} \int \frac{d^D l}{(2\pi)^D} \frac{\tau^{\rho\sigma}(p_1, p_1 + l) \tilde{\tau}_{\rho\sigma\alpha\beta}^{\mu\nu}(l, q) \tau^{\alpha\beta}(p_2, p_1 + l)}{(l^2 - i\epsilon)((l + q)^2 - i\epsilon)((p_1 + l)^2 - m^2 - i\epsilon)}, \end{aligned}$$

where we have defined the new 3gravitons vertex defined by

$$\tilde{\tau}_{\rho\sigma\alpha\beta}^{\mu\nu}(l, q) = P_{\alpha\beta ab} P_{\rho\sigma cd} \tau_{abcd}^{\mu\nu}(l, q).$$

This will be useful for the future computations because every vertex appears in the diagrams with the attached legs and therefore the it has to be contract with the projectors. In the Appendix D we computed the full expression of this vertex in general dimension and found useful proprieties that simplifies the following treatment. As said before (2.11) in the static limit the scalar vertices become deltas that projects the lower indices of the graviton vertex to the zero components. Than $T_1^{\mu\nu}$ becomes

$$T_1^{\mu\nu}(q^2) = -\frac{i\kappa^2}{2} m^3 \int \frac{d^D l}{(2\pi)^D} \frac{\tilde{\tau}_{0000}^{\mu\nu}(l, q)}{(l^2 - i\epsilon)((l + q)^2 - i\epsilon)((p_1 + l)^2 - m^2 - i\epsilon)}.$$

The static condition simplifies the scalar part of the denominator (2.7) and then the integration over the time component of the loop momentum expresses the integral as the master integral $J_1(q^2)$. We can integrate the zero component of the loop momentum with the Cauchy Theorem. In order to simplify the computation we do the following trick. The integrand is of the form

$$\frac{1}{(l^2 - i\epsilon)((l + q)^2 - i\epsilon)((p_1 + l)^2 - m^2 - i\epsilon)}.$$

We can write the contribution from the scalar line in two different ways (obtained by the exchange of the internal graviton momenta) as

$$\frac{1}{(l^2 - i\epsilon)((l+q)^2 - i\epsilon)} \frac{1}{2} \left[\frac{1}{(p_1 + l)^2 - m^2 - i\epsilon} + \frac{1}{(p_2 - l)^2 - m^2 - i\epsilon} \right] .$$

The static condition (2.7) simplifies the scalar part of the denominator and it becomes

$$\frac{1}{(l^2 - i\epsilon)((l+q)^2 - i\epsilon)} \frac{1}{2} \left[\frac{1}{2ml_0 - i\epsilon} - \frac{1}{2ml_0 + i\epsilon} \right] . \quad (2.12)$$

We have four poles coming from the first fraction in $l_0 = \pm|\vec{l}| \mp i\epsilon$ and $l_0 = \pm|\vec{l} + \vec{q}| \mp i\epsilon$ and two poles from the terms inside the bracket in $2ml_0 = \pm i\epsilon$. In the hypothesis of closing both the integral in the upper half plane the contributions of the first fraction cancel each other and the only contributing residue is at $2ml_0 = i\epsilon$. We could see this also from the following argument. If we sum the two terms in the bracket of 2.12 we see that his term contributes as $O(\epsilon)$ so in order to have a non-vanish result the only contributing residue is the one inside the brackets [13].

So we obtain

$$T_1^{\mu\nu}(q^2) = \frac{\kappa^2}{4} m^2 \int \frac{d^d l}{(2\pi)^d} \frac{\tilde{\tau}_{0000}^{\mu\nu}(l, q)}{(l^2 - i\epsilon)((l+q)^2 - i\epsilon)} \Big|_{l_0=0} , \quad (2.13)$$

with (D.3)

$$\begin{aligned} \tilde{\tau}_{0000}^{\mu\nu}(l, q) = & \frac{i\kappa}{2} \frac{1}{d-1} \left[(d-2) \left(l^\mu l^\nu + (l+q)^\mu (l+q)^\nu + q^\mu q^\nu + \frac{3}{2} \eta^{\mu\nu} \vec{q}^2 \right) - \right. \\ & \left. - 2(d-2) \left(\vec{l}^2 + (\vec{l} + \vec{q})^2 \right) \left(\delta_0^\mu \delta_0^\nu - \frac{\eta^{\mu\nu}}{4} \right) - 2(d-3) \vec{q}^2 \delta_0^\mu \delta_0^\nu \right] . \end{aligned}$$

In order to obtain the coefficients $c_1^{(1)}$ and $c_2^{(1)}$ we will perform the following strategy. It will only be necessary to compute T^{00} and $\delta_{ij} T^{ij}$ that are easier than $T^{\mu\nu}$ (being the components $T^{0i} = 0$), we can then obtain the coefficients from these values thanks to the equation 2.9. The 2.9 for $l = 1$ takes the expression

$$\langle T_{\mu\nu}^{(1)}(q^2) \rangle = \frac{\kappa}{4} m^2 \left[c_1^{(1)}(d) \delta_\mu^0 \delta_\nu^0 + c_2^{(1)}(d) \left(\frac{q_\mu q_\nu}{q^2} - \eta_{\mu\nu} \right) \right] J_1(q^2) ,$$

with

$$J_1(q^2) \equiv \int \frac{d^d l}{(2\pi)^d} \frac{\vec{q}^2}{\vec{l}^2 (\vec{l} - \vec{q})^2} = \frac{\Gamma(2 - \frac{d}{2}) \Gamma(\frac{d}{2} - 1)^2}{(4\pi)^{\frac{d}{2}} \Gamma(d-2)} (\vec{q}^2)^{\frac{d}{2}} - 1 .$$

Computing the T^{00} and $\delta_{ij} T^{ij}$ it is possible to obtain two independent equations in order to compute the $c_i^{(1)}$. The equations are:

$$\begin{cases} T^{00} = \frac{\kappa}{4} m^2 \left(c_1^{(1)}(d) - c_2^{(1)}(d) \right) J_1(q^2) \\ \delta_{ij} T^{ij} = \frac{\kappa}{4} m^2 (d-1) c_2^{(1)}(d) J_1(q^2) . \end{cases} \quad (2.14)$$

For the T^{00} component only the component with all indices set to zero of $\tilde{\tau}$ is needed. In the hypothesis that all the momenta's time component is set to zero

(for q is required by the static limit and for l by the complex integration) it has the expression D.3 [17]

$$\tilde{\tau}_{0000}^{00}(l, q) = \frac{i\kappa}{2} \frac{1}{d-1} \left[\frac{3}{2}(d-2)\vec{q}^2 - \frac{3}{2}(d-2)(\vec{l}^2 + (\vec{l} + \vec{q})^2) - 2(d-3)\vec{q}^2 \right] .$$

We have to insert this inside 2.13. The terms proportional to $\vec{q}^2$ give immediately contributions proportional to the master integral $J_1(q^2)$. The term with $\vec{l}^2 + (\vec{l} + \vec{q})^2$ has to be rearranged. We will give different methods to extract this contribution during the future treatment. One of these is the following.

If the expression of the numerator permits this is possible to split it in two pieces and make a change of variables to obtain the same integral, in this case it is possible to that as

$$\int d^d l \frac{\vec{l}^2 + (\vec{l} + \vec{q})^2}{\vec{l}^2 (\vec{l} + \vec{q})^2} = \int d^d l \frac{1}{\vec{l}^2} + \int d^d l \frac{1}{(\vec{l} + \vec{q})^2} = 2 \int d^d l \frac{1}{\vec{l}^2} .$$

This term is divergent but does not depend on $\vec{q}$, these terms can be ignored as they will be include in the counterterm part with the same degree of divergence.

So the vertex contributes as

$$\tilde{\tau}_{0000}^{00} = -\frac{i\kappa}{4} \frac{d-6}{d-1} \vec{q}^2$$

and substituting inside 2.13 we have the finale expression of the zero-zero component of the stress-energy tensor

$$T_1^{00}(q^2) = \frac{\kappa^2}{4} m \left(-2 \frac{d-6}{d-2} \right) J_1(q^2) .$$

For the $\delta_{ij} T_1^{ij}$ we have to compute $\tilde{\tau}_{0000}^{ij}$. Considering that the terms with $\vec{l}^2 + (\vec{l} + \vec{q})^2$ can be neglected, it becomes

$$\tilde{\tau}_{0000}^{ij}(l, q) = \frac{i\kappa}{2} \frac{d-2}{d-1} \left[l^i l^j + (l+q)^i (l+q)^j + q^i q^j - \frac{3}{2} \delta^{ij} \vec{q}^2 \right] .$$

Contracting with the delta and again neglecting the terms that don't depend on q ,

$$\delta_{ij} \tilde{\tau}_{0000}^{ij}(l, q) = \frac{i\kappa}{4} \frac{d-2}{d-1} (3d-2) \vec{q}^2 ,$$

and therefore

$$\delta_{ij} T_1^{ij} = \frac{\kappa^2}{4} \left[-2 \frac{d-2}{d-1} (3d-2) \right] J_1(q^2) .$$

Using the equations 2.14 we can find the values of $c_1^{(1)}$ and $c_2^{(1)}$. From the second equation we have $c_2^{(1)}(d)$ and then substituting inside the first we obtain the other coefficient. The values found are

$$\begin{cases} c_1^{(1)}(d) = -\frac{2}{(d-1)^2} (4d^2 - 15d + 10) \\ c_2^{(1)}(d) = -2 \frac{(d-2)(3d-2)}{d-1} . \end{cases}$$

From these values, we can compute the metric components with the equation 2.10 as done for the tree-level case. We have

$$\begin{cases} h_0^{(2)}(r, d) = 8 \frac{(d-2)^2}{(d-1)^2} \rho(r, d)^2 \\ h_1^{(2)}(r, d) = -4 \frac{2d^2 - 9d + 14}{(d-4)(d-1)^2} \rho(r, d)^2 \\ h_2^{(2)}(r, d) = -4 \frac{(d-2)^2(3d-2)}{(d-4)(d-1)^2} \rho(r, d)^2 . \end{cases} \quad (2.15)$$

The last two components are divergent in $d = 4$ as predicted in the general treatment. These match the values of 4.22 for general dimension and of 4.23 for $d = 3$. In $d = 4$ we need to add certain counterterms to heal the divergences (Section 3).

2.4 Quantum Computations photons case

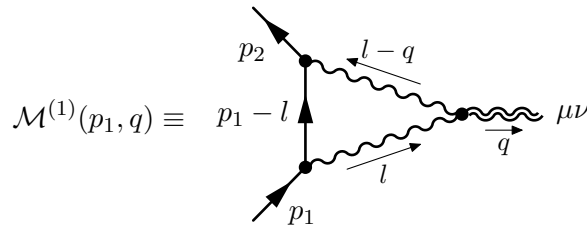
The treatment done for the scalar case can be extended to the case in which the theory is coupled also to an electromagnetic field. The only changes in the quantity computed above are the new Feynman rules (Section 1.4) that arise from the presence of the photons. These influence the expression of the EMT, changing his coefficients c_i , but we can still use the equation 2.9 with a different multiplicative constant (due to the new coupling constant α). Also, the expression of the metric components 2.10 is still valid because obtained by the tensor proprieties of the EMT which are not dependent on the new multiplicative constant.

We have then all the ingredients to compute the metric components, and the standard path done in the scalar case will be performed. For each loop order (up to the second) we compute the diagram's contribution given by the Feynman rules of the theory and from this, we will find the coefficients c_i of 2.9 and compute the metric components thanks to 2.10. The components computed can be compared in general dimension directly with expression 4.22 but for certain fixed dimensions, a renormalization process has to be performed in order to absorb the divergences of the metric.

For the tree-level case, there are no photons effects on the metric, since the external lines have to be two scalars and a graviton and the only possible diagram is the one treated in the Section 2.3.1. Therefore this study will start with the one-loop case.

2.4.1 One-Loop

The first contribution to this process given by photons coupling with gravitational and scalar fields comes from the one-loop diagram. This coupling provides a 2scalar-1photon vertex and a 2photons-1graviton vertex which can be combined to build the following diagram



With the stress-energy tensor defined with the Feynman rules (Appendix D) as

$$T_1^{\mu\nu}(q^2) = \frac{-i}{\sqrt{4E_1E_2}} \int \frac{d^D l}{(2\pi)^D} \frac{\tau^\alpha(p_1, p_1 - l) \tau^{\mu\nu}_{\alpha, \beta}(l, l - q) \tau^\beta(p_2, p_1 - l)}{(l^2 + i\epsilon)((l - q)^2 + i\epsilon)((p_1 - l)^2 - m^2 + i\epsilon)} .$$

Using the static limit condition of the new scalar-photons vertex and the usual expansion of the denominator

$$\begin{aligned} \tau^\alpha \tau^\beta &\simeq -4m^2 e^2 \delta_0^\alpha \delta_0^\beta \\ (l - p_1)^2 - m^2 &\simeq -2ml_0 , \end{aligned}$$

the stress energy tensor can be written as

$$T_1^{\mu\nu}(q^2) = 2ie^2 m \int \frac{d^D l}{(2\pi)^D} \frac{\tau^{\mu\nu}_{0,0}(l, l - q)}{(l^2 + i\epsilon)((l - q)^2 + i\epsilon)(-2m(l_0 + i\epsilon))} .$$

The last part of the denominator can be integrated out with the time-component of the loop momentum. Using Cauchy Theorem with contributions coming only from the pole in $l_0 = 0$ (for the reason explained before (2.7)), the integral becomes

$$T_1^{\mu\nu}(q^2) = e^2 \int \frac{d^d l}{(2\pi)^d} \frac{\tau^{\mu\nu}_{0,0}(l, l - q)}{(l^2 + i\epsilon)((l - q)^2 + i\epsilon)} \Big|_{l_0=0} .$$

The expression of the 2photons-1graviton vertex is given in E.3 and setting the lower indices to zero we obtain

$$\tau^{\mu\nu}_{0,0}(l, l - q) = i\kappa \left[P^{(4)\mu\nu}_{00} l \cdot (l - q) + l^\mu l^\nu - \frac{1}{2}(q^\mu l^\nu + q^\nu l^\mu) \right] .$$

It is possible to use the same procedure done for the chargeless case, in which we computed the T^{00} component and $\delta_{ij} T^{ij}$ from which we obtained the coefficients of the EMT thanks to 2.14. The only difference is the new multiplicative constant but the tensorial form of the identities has not been changed:

$$\begin{cases} T^{00} = \alpha\kappa\pi \left(c_1^{(1)}(d) - c_2^{(1)}(d) \right) J_1(q^2) \\ \delta_{ij} T^{ij} = \alpha\kappa\pi (d - 1) c_2^{(1)}(d) J_1(q^2) . \end{cases} \quad (2.16)$$

For the 00-component

$$\begin{aligned} T^{00} &= i\kappa e^2 \int \frac{d^d l}{(2\pi)^d} \frac{-\frac{1}{2} l \cdot (l - q)}{(l^2 + i\epsilon)((l - q)^2 + i\epsilon)} \Big|_{l_0=0} = \int \frac{d^d l}{(2\pi)^d} \frac{-\frac{1}{2} l \cdot (l - q)}{(l^2 + i\epsilon)((l - q)^2 + i\epsilon)} \Big|_{l_0=0} \\ &= \frac{i\kappa e^2}{4} \int \frac{d^d l}{(2\pi)^d} \frac{q^2 - l^2 - (l - q)^2}{(l^2 + i\epsilon)((l - q)^2 + i\epsilon)} \Big|_{l_0=0} . \end{aligned}$$

The only term contributing to T^{00} is the one with q^2 because the others are divergent integrals that do not depend on the transferred momentum q and can be neglected

as done for the chargeless case. The contribution to T^{00} is then proportional to the master integral as predicted in the general case and has the expression

$$T^{00} = \frac{i\kappa e^2}{4} J_1(q^2) .$$

For the $\delta_{ij}T^{ij}$ we have

$$\begin{aligned} \delta_{ij}T^{ij} &= i\kappa e^2 \int \frac{d^d l}{(2\pi)^d} \frac{1}{l^2(l-q)^2} \left[-\frac{d}{2} l \cdot (l-q) + l^2 + q \cdot l \right] \\ &= \frac{i\kappa e^2}{4} (d-2) J_1(q^2) , \end{aligned}$$

where we used the same procedure of T^{00} . Using the definition $\alpha \equiv e^2/4\pi$ we have

$$\begin{cases} T^{00} = i\kappa\alpha\pi J_1(q^2) \\ \delta_{ij}T^{ij} = i\kappa\alpha\pi(d-2)J_1(q^2) \end{cases}$$

and comparing with 2.16 we can solve the two equations to find the coefficients c_i (which are defined up to a common multiplicative constant). The result obtained is

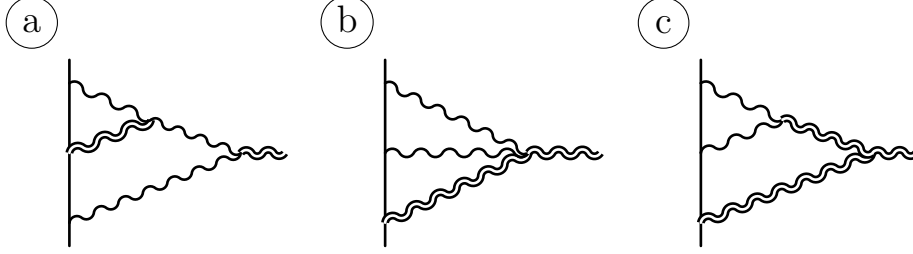
$$\begin{cases} c_1^{(1)} = -\frac{2d-3}{d-1} \\ c_2^{(1)} = -\frac{d-2}{d-1} . \end{cases}$$

From these values, we can compute the metric components with the equation 2.10. We have

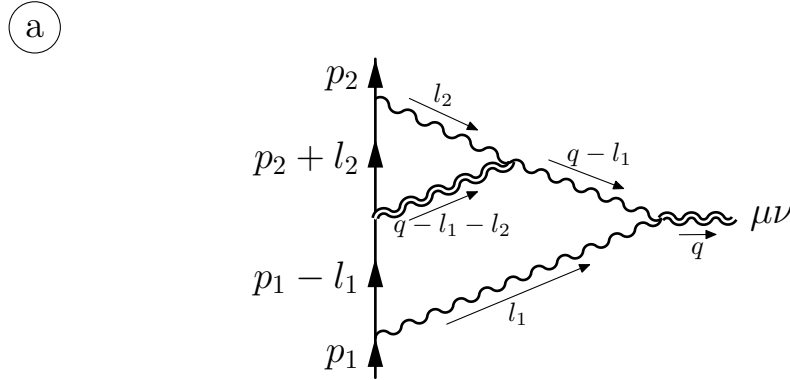
$$\begin{cases} h_0^{(2)}(r, d) = 2\frac{d-2}{d-1}\alpha G\tilde{\rho}(r, d)^2 \\ h_1^{(2)}(r, d) = -2\frac{d-3}{(d-4)(d-1)}\alpha G\tilde{\rho}(r, d)^2 \\ h_2^{(2)}(r, d) = 2\frac{(d-2)^2}{(d-4)(d-1)}\alpha G\tilde{\rho}(r, d)^2 . \end{cases} \quad (2.17)$$

The last two components are divergent in $d = 4$. These match the values of 4.22 for general dimension and of 4.23 for $d = 3$. In $d = 4$ and we need to add certain counterterms to heal the divergences and then will be possible to compare the result with the 4.24.

2.4.2 Two-loop



In the two-loop computations we have the contribution of three different diagrams. we will compute all the terms and then perform a weighted sum to obtain the final contribution to the EMT. The first diagram ((a)) is the one build only from the 2photons-1graviton E.3 vertex. The second diagram ((b)) is due by the 2photon-2graviton vertex E.5 given by the expansion of the free electro-gravitational lagrangian to the second order. As already said at each order the expansion of the lagrangian gives new vertex to the theory and these has to be considered to build the possible diagram of the process. The third and last diagram ((c)) is the more difficult to compute and is construct from the 2photons-1graviton vertex and from the 3graviton vertex E.4.



The stress-energy tensor defined with the Feynman rules (Appendix D) of this diagram is

$$T_{(a)}^{\mu\nu}(q^2) = \frac{1}{\sqrt{4E_1E_2}} \int \frac{d^D l_1}{(2\pi)^D} \frac{d^D l_2}{(2\pi)^D} \frac{i}{(p_1 - l_1)^2 - m^2} \frac{i}{(p_2 + l_2)^2 - m^2} \left(\frac{-i\tau^\alpha}{l_1^2} \right) \left(\frac{-i\tau^\beta}{l_2^2} \right) \times \\ \times \frac{i\tau^{ab}P_{abcd}}{(l_1 + l_2 - q)^2} \tau_{\beta,\gamma}^{cd}(p, p') \frac{-i\tau^{\mu\nu}_{\gamma,\alpha}(k, k')}{(l_1 - q)^2} ,$$

where we called the photons momenta of the first vertex p and p' and the ones of the second vertex k and k' .

Using the approximations:

$$\begin{aligned}\tau^\alpha \tau^\beta &\simeq -4m^2 e^2 \delta_0^\alpha \delta_0^\beta \\ \tau^{ab} &\simeq -i\kappa m^2 \delta_0^\alpha \delta_0^\beta \\ (l_i \pm p_i)^2 - m^2 &\simeq \pm 2ml_{i0} ,\end{aligned}\tag{2.18}$$

the tensor becomes

$$\begin{aligned}T_{(a)}^{\mu\nu}(q^2) &\simeq 2m^3 e^2 \kappa \int \frac{d^D l_1}{(2\pi)^D} \frac{d^D l_2}{(2\pi)^D} \frac{1}{(-2m(l_{10} + i\epsilon))2m(l_{20} + i\epsilon)} \times \\ &\times \frac{\tau^{\mu\nu}_{\gamma,0}(k, k') P_{00cd} \tau^{cd}_{0,\gamma}(p, p')}{l_1^2 l_2^2 (l_1 + l_2 - q)^2 (l_1 - q)^2} .\end{aligned}$$

We can perform the integral over the time-component of both momenta with the Cauchy Theorem. For the same reason of the one-loop case (2.7) the only contributing poles are the ones in $l_{0i} = 0$ and this time the multiplicative factor to be included is $\frac{1}{3}$. Therefore considering also the 2π factors at denominator we have

$$T_{(a)}^{\mu\nu}(q^2) = -\frac{1}{6} m e^2 \kappa \int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{\tau^{\mu\nu}_{\gamma,0}(k, k') \overbrace{P_{00cd} \tau^{cd}_{0,\gamma}(p, p')}^{\text{Diagram}}}{l_1^2 l_2^2 (l_1 + l_2 - q)^2 (l_1 - q)^2} \Big|_{l_{10}=l_{20}=0} .$$

From the vertex E.2 (symmetric in the lower indices) we compute the component

$$\tau^{\mu\nu}_{0,\alpha}(p, p') = i\kappa \left\{ P_{(4)0\alpha}^{\mu\nu} p \cdot p' + \frac{1}{2} \delta_\alpha^0 (p^\mu p'^\nu + p^\nu p'^\mu) - \frac{1}{4} (\delta_0^\nu p^\mu p'_\alpha + \delta_0^\nu p_\alpha p'^\mu + \delta_0^\mu p_\alpha p'^\nu + \delta_0^\mu p'^\nu p'_\alpha) \right\} .\tag{2.19}$$

Then the expression Diagram is

$$\text{Diagram} \equiv P_{00cd} \tau^{cd}_{0,\gamma}(p, p') = \left(\delta_c^0 \delta_d^0 - \frac{1}{d-1} \eta_{cd} \right) \tau^{cd}_{0,\gamma}(p, p') = i\kappa (p \cdot p') \frac{d-2}{d-1} \delta_\gamma^0 .$$

The tensorial form of Diagram is δ_γ^0 , it is then easy to obtain the full numerator of the integrand setting to zero the α index of the vertex component 2.19 (being the vertex symmetric in the exchange of the photons indices).

The numerator becomes

$$\tau^{\mu\nu}_{\gamma,0}(k, k') P_{00cd} \tau^{cd}_{0,\gamma}(p, p') = -\kappa^2 \frac{d-2}{d-1} (p \cdot p') \left[(k \cdot k') \left(\delta_0^\mu \delta_0^\nu - \frac{1}{2} \right) + \frac{1}{2} (k^\mu k'^\nu + k^\nu k'^\mu) \right] .$$

Now it is possible to insert the values of the momenta remembering that the vertex is defined with the first momentum incoming and the second outgoing. From the diagram we have

$$\begin{cases} p = l_2 & k = q - l_1 \\ p' = q - l_1 & k' = -l_1 . \end{cases}$$

As done for the one-loop case we compute the 00-component and the spatial trace of $T_{(a)}^{\mu\nu}$ in order to explicit the coefficient using 2.16.

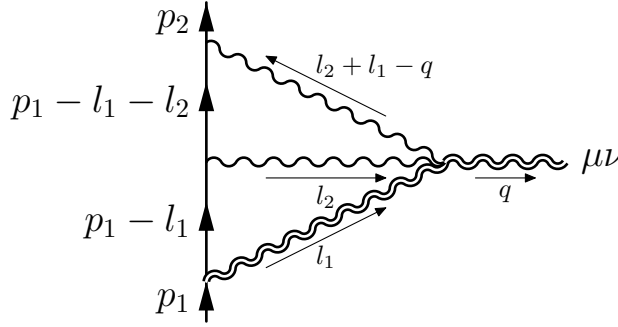
For the 00-component, remembering that all the momenta have no time component due to the complex integration

$$T_{(a)}^{00} = \frac{1}{12} e^2 \kappa^3 \frac{d-2}{d-1} \int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{l_2 \cdot (q - l_1)(q - l_1) \cdot (-l_1)}{l_1^2 l_2^2 (l_1 + l_2 - q)^2 (l_1 - q)^2} = \frac{m e^2 \kappa^3 (d-2)}{72(d-1)} J_2(q^2) .$$

For the spatial trace we have the same integral with different coefficients given by the tensorial part of the numerator

$$\delta_{ij} T_{(a)}^{ij} = \frac{m e^2 \kappa^3 (d-2)^2}{72(d-1)} J_2(q^2) .$$

(b)



The contribution of this diagram is given by

$$T_{(b)}^{\mu\nu}(q^2) = \frac{1}{2m} \int \frac{d^D l_1}{(2\pi)^D} \frac{d^D l_2}{(2\pi)^D} \frac{i}{(p_1 - l_1)^2 - m^2} \frac{i}{(p_1 - l_1 - l_2)^2 - m^2} \left(\frac{-i\tau^\alpha}{l_2^2} \right) \left(\frac{-i\tau^\beta}{(l_1 + l_2 - q)^2} \right) \times \\ \times \frac{i\tau^{ab} P_{ab}^{cd}}{l_1^2} \tau^{\mu\nu}_{cd,\alpha,\beta}(p, p') ,$$

where we used the 2photons-2graviton vertex E.5 with photons momenta named p and p' .

Using the approximations 2.18 given by the static limit condition we have

$$T_{(b)}^{\mu\nu}(q^2) \simeq 2im^3 e^2 \kappa \int \frac{d^D l_1}{(2\pi)^D} \frac{d^D l_2}{(2\pi)^D} \frac{1}{(l_{10}^2 - 2ml_{10} + i\epsilon)((l_1 + l_2)_0^2 + 2m(l_1 + l_2)_0 + i\epsilon)} \times \\ \times \frac{P_{00}^{cd} \tau^{\mu\nu}_{cd,0,0}(p, p')}{l_1^2 l_2^2 (l_1 + l_2 - q)^2} \\ = -\frac{1}{6} i m e^2 \kappa \int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{\overbrace{P_{00}^{\rho\sigma} \tau^{\mu\nu}_{\rho\sigma,0,0}(p, p')}^{\text{Diagram (b)}}}{l_1^2 l_2^2 (l_1 + l_2 - q)^2} \Big|_{l_{10}=l_{20}=0} .$$

For this computation the only components that contributes of the vertex E.5 are

$$\begin{aligned} \tau^{\mu\nu}{}_{\rho\sigma,0,0}(p, p') = & -i\frac{\kappa^2}{4} \left\{ -p \cdot p' \left(\eta^\mu{}_\rho \eta^\nu{}_\sigma + \eta^\nu{}_\rho \eta^\mu{}_\sigma - \eta^{\mu\nu} \eta_{\rho\sigma} \right) \right. \\ & + \eta^\mu{}_\rho (p^\nu p'_\sigma + p_\sigma p'^\nu + 2p \cdot p' \delta_0^\nu \delta_\sigma^0) + \eta^\mu{}_\sigma (p^\nu p'_\rho + p_\rho p'^\nu + 2p \cdot p' \delta_0^\nu \delta_\rho^0) \\ & + \eta^\nu{}_\rho (p^\mu p'_\sigma + p_\sigma p'^\mu + 2p \cdot p' \delta_0^\mu \delta_\sigma^0) + \eta^\nu{}_\sigma (p^\mu p'_\rho + p_\rho p'^\mu + 2p \cdot p' \delta_0^\mu \delta_\rho^0) \\ & - \eta^{\mu\nu} (p_\rho p'_\sigma + p_\sigma p'_\rho + 2p \cdot p' \delta_\rho^0 \delta_\sigma^0) - \eta_{\rho\sigma} (p^\mu p'^\nu + p^\nu p'^\mu + 2p \cdot p' \delta_0^\mu \delta_0^\nu) \\ & + \left(\delta_\rho^0 p^\mu - \delta_0^\mu p_\rho \right) \left(\delta_\sigma^0 p'^\nu - \delta_0^\nu p'_\sigma \right) + \left(\delta_\sigma^0 p^\nu - \delta_0^\nu p_\sigma \right) \left(\delta_\rho^0 p'^\mu - \delta_0^\mu p'_\rho \right) \\ & \left. + \left(\delta_\sigma^0 p^\mu - \delta_0^\mu p_\sigma \right) \left(\delta_\rho^0 p'^\nu - \delta_0^\nu p'_\rho \right) + \left(\delta_\rho^0 p^\nu - \delta_0^\nu p_\rho \right) \left(\delta_\sigma^0 p'^\mu - \delta_0^\mu p'_\sigma \right) \right\} . \end{aligned} \quad (2.20)$$

Expanding the numerator of the integrand we have

$$\bigcirc \equiv P_{00}{}^{\rho\sigma} \tau^{\mu\nu}{}_{\rho\sigma,0,0}(p, p') = \tau^{\mu\nu}{}_{00,0,0}(p, p') - \frac{1}{d-1} \eta^{\rho\sigma} \tau^{\mu\nu}{}_{\rho\sigma,0,0}(p, p') .$$

The first term of the sum is given directly by 2.20 and is

$$\tau^{\mu\nu}{}_{00,0,0}(p, p') = -\frac{i\kappa^2}{4} \{ p \cdot p' [4\delta_0^\nu \delta_0^\nu - \eta^{\mu\nu}] + p^\mu p'^\nu + p^\nu p'^\mu \} ,$$

the second one is given by the trace on the lower indices of 2.20, that is

$$\eta^{\rho\sigma} \tau^{\mu\nu}{}_{\rho\sigma,0,0}(p, p') = \frac{i\kappa^2}{4} (d-5) \{ p \cdot p' [2\delta_0^\nu \delta_0^\nu - \eta^{\mu\nu}] + p^\mu p'^\nu + p^\nu p'^\mu \} .$$

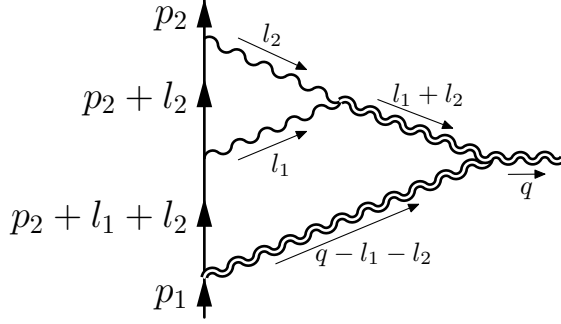
Then

$$\begin{aligned} \bigcirc & \equiv P_{00}{}^{\rho\sigma} \tau^{\mu\nu}{}_{\rho\sigma,0,0}(p, p') = \tau^{\mu\nu}{}_{00,0,0}(p, p') - \frac{1}{d-1} \eta^{\rho\sigma} \tau^{\mu\nu}{}_{\rho\sigma,0,0}(p, p') \\ & = -\frac{i\kappa^2}{4} \left\{ \frac{2(d-3)}{d-1} [p^\mu p'^\nu + p^\nu p'^\mu - p \cdot p' \eta^{\mu\nu}] + \frac{2(3d-7)}{d-1} p \cdot p' \delta_0^\mu \delta_0^\nu \right\} , \end{aligned}$$

substituting $p = l_2$ and $p' = l_1 + l_2 - q$ the contributions of this diagram to $T_{(b)}^{00}$ and $\delta_{ij} T_{(b)}^{ij}$ are

$$\begin{cases} T_{(b)}^{00} = -\frac{1}{6} m e^2 \kappa^3 \frac{d-2}{d-1} \int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{l_2 \cdot (l_1 + l_2 - q)}{l_1^2 l_2^2 (l_1 + l_2 - q)^2} = -\frac{1}{36} m e^2 \kappa^3 \frac{d-2}{d-1} J_2(q^2) \\ \delta_{ij} T_{(b)}^{ij} = -\frac{1}{12} m e^2 \kappa^3 \frac{(d-3)(d-2)}{d-1} \int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{l_2 \cdot (l_1 + l_2 - q)}{l_1^2 l_2^2 (l_1 + l_2 - q)^2} = -\frac{1}{72} m e^2 \kappa^3 \frac{(d-3)(d-2)}{d-1} J_2(q^2) . \end{cases}$$

(c)



The computations involved in this case are more difficult with respect to the other two cases because of the presence of the 3graviton vertex, which has a long expression and therefore makes difficult the compute of the amplitudes.

As done for the previous cases the EMT of this diagram has the expression

$$\begin{aligned}
 T_{(c)}^{\mu\nu}(q^2) &= \frac{1}{2m} \int \frac{d^D l_1}{(2\pi)^D} \frac{d^D l_2}{(2\pi)^D} \frac{i}{(p_2 + l_1 + l_2)^2 - m^2} \frac{i}{(p_2 + l_2)^2 - m^2} \left(\frac{-i\tau^\alpha}{l_2^2} \right) \left(\frac{-i\tau^\beta}{l_1^2} \right) \times \\
 &\quad \times \frac{i\tau^{ab} P_{abcd}}{(q - l_1 - l_2)^2} \tau^{\rho\sigma}_{\alpha,\beta}(p, p') \frac{iP_{\rho\sigma\gamma\delta}}{(l_1 + l_2)^2} \tau^{\mu\nu}_{cd,\gamma\delta}(k, k') \simeq \\
 &\simeq -2ime^2 m^3 \kappa \int \frac{d^D l_1}{(2\pi)^D} \frac{d^D l_2}{(2\pi)^D} \frac{1}{l_{20}^2 + 2ml_{20} + i\epsilon} \frac{1}{(l_1 + l_2)_0^2 + 2m(l_1 + l_2)_0 + i\epsilon} \times \\
 &\quad \times \frac{\tau^{\rho\sigma}_{0,0}(p, p') P_{\rho\sigma\gamma\delta} P_{00cd} \tau^{\mu\nu}_{cd,\gamma\delta}(k, k')}{l_1^2 l_2^2 (l_1 + l_2 - q)^2 (l_1 + l_2)^2} = \\
 &\quad \times \left(\text{3-graviton vertex symbol} \right) \\
 &= -\frac{me^2 \kappa}{6} \int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{\tau^{\rho\sigma}_{0,0}(p, p') P_{\rho\sigma\gamma\delta} P_{00cd} \tau^{\mu\nu}_{cd,\gamma\delta}(k, k')}{l_1^2 l_2^2 (l_1 + l_2 - q)^2 (l_1 + l_2)^2} \Big|_{l_{10}=l_{20}=0}.
 \end{aligned} \tag{2.21}$$

Is then possible to rewrite the numerator of the integrand using the vertex $\tilde{\tau}$ D.4 as

$$\left(\text{3-graviton vertex symbol} \right) \equiv \tau^{\rho\sigma}_{0,0}(p, p') P_{\rho\sigma\gamma\delta} P_{00cd} \tau^{\mu\nu}_{cd,\gamma\delta}(k, k') = \tau^{\rho\sigma}_{0,0}(p, p') \tilde{\tau}^{\mu\nu}_{00,\rho\sigma}(k, k').$$

As done for the previous cases we need to find the $T_{(c)}^{00}$ component and the spatial trace $\delta_{ij} T_{(c)}^{ij}$. The amplitude is proportional to the contraction of the photonic vertex with the graviton one. In this paragraph will be useful the components of $\tilde{\tau}$ computed in the Appendix D. As already computed in the one loop case the 2photons-1graviton vertex component with photonic's indices set to zero is

$$\tau^{\rho\sigma}_{0,0}(p, p') = i\kappa \left[\left(\delta_0^\rho \delta_0^\sigma - \frac{1}{2} \eta^{\rho\sigma} \right) p \cdot p' + \frac{1}{2} (p^\rho p'^\sigma + p'^\rho p^\sigma) \right].$$

For the $T_{(c)}^{00}$ component we have to fix in $\tilde{\tau}$ (D.4) the indices μ and ν to zero and contract with the vertex given above. The tensor $\tilde{\tau}^{00}_{00,\rho\sigma}$ has been computed from

the vertex and is D.6.

For $\mu = \nu = 0$ the numerator becomes

$$\bigcirc_{\otimes} = \frac{\kappa^2}{4} \frac{1}{d-1} \left\{ p \cdot p' \left[-dk'^2 + \frac{(3d-3)(d-2)}{d-1} k^2 + (d-2)(k-k')^2 \right] + (4d-8)(k' \cdot p)(k' \cdot p') \right\} .$$

Substituting the momenta from the diagram using the conventions of the vertexes and paying attention to the fact that the $\tilde{\tau}$ vertex is not symmetric in the lower indices

$$\begin{cases} p = l_2 & k = q - l_1 - l_2 \\ p' = -l_1 & k' = q . \end{cases} \quad (2.22)$$

Substituting inside $T_{(c)}^{\mu\nu}$ all the integrals less than one can be related easily to the master integral with the algebraic methods used in the easier cases and explained in the Appendix C.2. The only integral more difficult to solve is the one appearing from the last term of the numerator. It has the expression

$$\int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{q \cdot l_1 q \cdot l_2}{l_1^2 l_2^2 (l_1 + l_2)^2 (l_1 + l_2 - q)^2} , \quad (2.23)$$

and it is impossible to eliminate the dependence from q or rewrite the scalar product in a more convenient way with the previous methods. In order to compute the contributions proportional to the master integral we have to compute vectorial identities from the derivatives of the master integral and then use these to compute this integral [26]. We used a package of *Mathematica* named *LiteRed* [26, 27] in order to compute this integral. The result is given in C.5.

Using this identity the $T_{(c)}^{00}$ component of 2.21 is

$$T_{(c)}^{00}(q^2) = \frac{me^2 \kappa^3}{72} \frac{(3d^2 - 19d + 32)}{(d-4)(d-1)} J_2(q^2) .$$

For the spatial trace we need to contract with δ_{ij} the tensor $\tilde{\tau}^{ij}_{00,\rho\sigma}$ given in D.7 and then contract with the photonic vertex. We compute first the contraction with the delta in order to have a less long expression due to the symmetry of D.7.

$$\begin{aligned} \delta_{ij} \tilde{\tau}^{ij}_{00,\rho\sigma} = & \frac{i\kappa}{2} \left\{ \frac{d-2}{2} k^2 \delta_\rho^0 \delta_\sigma^0 - \frac{d-2}{2(d-1)} k^2 \eta_{\rho\sigma} + \frac{d-2}{2} (k-k')^2 \delta_\rho^0 \delta_\sigma^0 - \frac{d^2-7d+2}{2(d-1)^2} (k-k')^2 \eta_{\rho\sigma} \right. \\ & - \frac{2}{d-1} (k-k')^2 \delta_{ij} P_{\rho\sigma}^{ij} + \frac{3d-2}{2} k'^2 \delta_\rho^0 \delta_\sigma^0 - \frac{3d-10}{2(d-1)} k'^2 \eta_{\rho\sigma} + \frac{4}{d-1} k'_\lambda k'^\lambda \delta_{ij} P_{\rho\sigma}^{\lambda j} \\ & \left. - \frac{2}{d-2} k'^2 \delta_{ij} P_{\rho\sigma}^{ij} + \frac{2d}{d-1} k'_\alpha k'_\beta P_{\rho\sigma}^{\alpha\beta} \right\} . \end{aligned} \quad (2.24)$$

We have to contract this with the photonic vertex. We split the computations in three different contractions and then sum all together.

Defining

$$\tau^{\rho\sigma}_{0,0}(p, p') = \frac{i\kappa}{2} \left[\underbrace{p^\rho p'^\sigma + p'^\rho p^\sigma}_{(\alpha)} + \underbrace{2\delta_0^\rho \delta_0^\sigma p \cdot p' - \eta^{\rho\sigma} p \cdot p'}_{(\beta)} \right] ,$$

the contraction of 2.24 with the first term gives

$$(\alpha) = -\frac{\kappa^2}{4} \left\{ -\frac{d-2}{d-1} k^2 (p \cdot p') - \frac{d-6}{d-1} (k-k')^2 (p \cdot p') + \frac{3(d-2)}{d-1} k'^2 (p \cdot p') + \frac{4(d-2)}{d-1} (k' \cdot p)(k' \cdot p') \right\} ,$$

the contraction with the second

$$\textcircled{\beta} = -\frac{\kappa^2}{4}(p \cdot p') \left\{ \frac{(d-2)^2}{d-1} k^2 + \frac{(d-2)^2}{d-1} (k - k')^2 + \frac{(3d-2)(d-2)}{d-1} k'^2 \right\} ,$$

and the last contraction is

$$\textcircled{\gamma} = \frac{\kappa^2}{4}(p \cdot p') \left\{ -\frac{d-2}{d-1} k^2 + \frac{d+2}{d-1} (k - k')^2 + \frac{d-2}{d-1} k'^2 \right\} .$$

Summing all the terms we obtain

$$\begin{aligned} \tau^{\rho\sigma}_{0,0}(p, p') \delta_{ij} \tilde{\tau}_{00, \rho\sigma}^{ij}(k, k') &= \textcircled{\alpha} + \textcircled{\beta} + \textcircled{\gamma} = \\ &= -\frac{\kappa^2}{4} \frac{d-2}{d-1} \left\{ (d-2)k^2 + (d-4)(k - k')^2 + 3dk'^2 + 4(k' \cdot p)(k' \cdot p') \right\} . \end{aligned} \quad (2.25)$$

As discussed above the more tricky part of the integrand, derived from the substitution of 2.25 inside 2.21 with momenta 2.22, is the last term of 2.25. For this term, the usual algebraic methods performed above are no longer enough and the use of the package `LiteRed` gives the contribution to the master integral (using the script in the Appendix C.2).

The substitution inside 2.21 gives the spatial trace contribution of this diagram that is

$$\delta_{ij} T_{(c)}^{ij} = \frac{m e^2 \kappa^3}{72} \frac{(d-2)(5d^2 - 27d + 30)}{(d-4)(d-1)} J_2(q^2) .$$

Now that we computed all the contributions from all the different diagrams contributing to this process we need to sum them together with their multiplicity. This will be done in the next section.

Total contribution at two loops

The total contribution of the three diagrams computed above can be obtained by a weighted sum over the three contributions. In particular, we have to multiply each diagram by its multiplicity. Diagrams with the same contribution of $\textcircled{b}$ and $\textcircled{c}$ can be made from the different order of gravitons and photons attached to the scalar line. In particular there are three possible combinations. The $\textcircled{a}$ diagram has the same multiplicity by the sorting of the scalar line but for each of these diagrams, there is the possibility to exchange the two photon-graviton vertexes. Therefore this diagram will have a contribution that is double the others.

Therefore the total EMT of the two-loop computations is equal to

$$T^{\mu\nu} = 6T_{(a)}^{\mu\nu} + 3T_{(b)}^{\mu\nu} + 3T_{(c)}^{\mu\nu}$$

Summarising the different contribution to the 00-component and the spatial trace:

$$\begin{array}{|l}
T_{(a)}^{00} = \frac{me^2\kappa^3(d-2)}{72(d-1)}J_2(q^2) \\
\delta_{ij}T_{(a)}^{ij} = \frac{me^2\kappa^3(d-2)^2}{72(d-1)}J_2(q^2)
\end{array}
\quad
\begin{array}{|l}
T_{(b)}^{00} = -\frac{me^2\kappa^3(d-2)}{36(d-1)}J_2(q^2) \\
\delta_{ij}T_{(b)}^{ij} = -\frac{me^2\kappa^3(d-3)(d-2)}{72(d-1)}J_2(q^2)
\end{array}$$

$$\begin{array}{|l}
T_{(c)}^{00}(q^2) = \frac{me^2\kappa^3}{72} \frac{(3d^2-19d+32)}{(d-4)(d-1)} J_2(q^2) \\
\delta_{ij}T_{(c)}^{ij} = \frac{me^2\kappa^3}{72} \frac{(d-2)(5d^2-27d+30)}{(d-4)(d-1)} J_2(q^2) .
\end{array}$$

It is interesting how, for the 00-component, the contribution of the first two diagrams cancel each other out, making the last diagram the only contributing to this component.

Summing all the contributions we obtain

$$\begin{cases} T^{00} = \frac{me^2\kappa^2}{24} \frac{(3d^2-19d+32)}{(d-4)(d-1)} J_2(q^2) \\ \delta_{ij}T^{ij} = \frac{me^2\kappa^3}{24} \frac{2(d-2)(3d^2-16d+17)}{(d-4)(d-1)} J_2(q^2) . \end{cases}$$

Using the definition of the EMT and comparing with 2.16 it is possible to extract the coefficients

$$\begin{cases} c_1^{(2)} = \frac{(9d^3-66d^2+149d-100)}{8(d-4)(d-1)^2} \\ c_2^{(2)} = \frac{2(d-2)(8d^2-16d+17)}{3(d-4)(d-1)^2} . \end{cases}$$

From the coefficients we obtain the charged corrections at two-loop order to the metrics component by 4.6

$$\begin{cases} h_0^{(3)}(r) = -\frac{4(d-2)^2(3d-11)}{(d-4)(d-1)^2} m\alpha G_N^2 \tilde{\rho}(r)^3 \\ h_1^{(3)}(r) = \frac{8(3d^3-25d^2+69d-65)}{3(d-4)(d-3)(d-1)^2} m\alpha G_N^2 \tilde{\rho}(r)^3 \\ h_2^{(3)}(r) = -\frac{4(d-2)^2(3d^3-19d^2+33d-17)}{(d-4)(d-3)(d-1)^3} m\alpha G_N^2 \tilde{\rho}(r)^3 . \end{cases} \quad (2.26)$$

In $d = 3$ only $h_1^{(3)}$ and $h_2^{(3)}$ diverge and in $d = 4$ all the components diverge. To obtain the values of the metrics component in these dimensions we need to perform a renormalization process described in the following chapter. Instead, we can immediately compare with the charged part of the G_N^2 terms of the dimensionful result 4.22.

Chapter 3

Non-minimal couplings

Because of the dimensionful coupling constant κ and the nonlinear interactions to all orders in $h_{\mu\nu}$, gravity does not belong to the class of renormalizable theories. In a renormalizable theory it is possible to fix the divergences from loop diagrams with the redefinition of a finite number of parameters. In gravity, this is not possible because the number of parameters to be fixed increases with the number of loops. It is impossible in this case to have a theory working at each loop order but it is instead possible to build a renormalized theory at every fixed loop order, as it will be done in this chapter. This pattern is typical of effective field theories and it is not an obstacle in making quantum predictions.

The procedure carried out in this chapter is organised in these steps. First, the action is written in an energy expansion, with all the terms in agreement with the symmetry of the theory. At each order, the terms in the lagrangian up to that order determine the Feynman rules and the higher-order terms are treated as perturbations. Given the Feynman rules the quantum computations can be done. In these computations, ultraviolet divergences will appear (as already seen in the previous chapters), but since these are compatible with the symmetry of the lagrangian and are local can be absorbed in a redefinition of the parameters. The renormalized parameters must be determined experimentally and then the remaining relations between amplitudes are the predictions of the theory.

Given the free gravitational lagrangian 1.25 we could add infinite coupling terms made from the powers of the covariant derivative acting on a single power of Riemann tensor, Ricci tensor or the Ricci scalar. At a fixed loop order we can neglect the higher terms as done for these in the lagrangian and use the others to fix the divergences of the theory. In our computations we will need the counterterms up to the second order. To this order we need the first non-minimal couplings defined by the variation of the action:

$$\delta^{(1)}S = (G_N m)^{\frac{2}{d-2}} \int d^D x \sqrt{-g} \left\{ a_1^{(1)} R \partial_\mu \phi \partial^\mu \phi + a_2^{(1)} R^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right\} , \quad (3.1)$$

where $a_i^{(1)}$ are dimensionless parameters that can be redefined to fix the divergences and depend on the dimension. The absence of the contraction of two Riemann tensor is due to the Gauss-Bonnet theorem [25].

The power of the Newton constant is determined by dimension analysis, and since it has to be an integer this coupling will appear only in dimensions $D < 6$.

The second order corrections are given by the terms

$$\delta^{(2)}S = (G_N m)^{\frac{4}{d-2}} \int d^D x \sqrt{-g} \left\{ a_1^{(2)} \square R \partial_\mu \phi \partial^\mu \phi + a_2^{(2)} D_\mu D_\nu R \partial^\mu \phi \partial^\nu \phi + a_3^{(2)} \square R_{\mu\nu} \partial^\mu \phi \partial^\nu \phi \right\}. \quad (3.2)$$

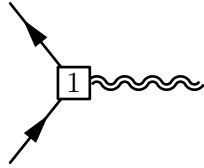
The terms with the Riemann tensor can be expressed in terms of these thanks to the Bianchi identities [17].

Higher orders non-minimal coupling will not contribute to the classical limit because they will contribute to higher orders in the transferred momentum q and therefore are sub-leading terms (it also can be seen in the following computations).

It will be shown that the only couplings that change the metric components are the first ones (3.1) and so only these will be the used in the renormalization process [17]. Will be also shown that the only term of this coupling that varies the metric components is the term $a_1^{(1)}$, the other ($a_2^{(1)}$) is needed for exemple in the renormalization of the EMT but this is not done in this work.

Tree-level insertions

The non-minimal couplings can be seen as the presence of new vertexes in the theory (represented with a boxed number). For the coupling 3.1 the insertion of this vertex in the tree-level diagram gives a variation in EMT of the type



$$\delta^{(1)}T_{\mu\nu}^{(0)} = -\bar{q}^2 (G_N m)^{\frac{2}{d-2}} m \left\{ 2a_1^{(1)} \left(\frac{q_\mu q_\nu}{q^2} - \eta_{\mu\nu} \right) - a_2^{(1)} \delta_\mu^0 \delta_\nu^0 \right\}. \quad (3.3)$$

From this variation it is possible to compute the corresponding variation in the metrics components using the 2.2 with $l = 0$

$$\delta^{(1)}h_{\mu\nu}^{(2)}(\vec{x}) = -16\pi (G_N m)^{\frac{d}{d-2}} \int \frac{d^d q}{(2\pi)^d} e^{i\vec{x}\cdot\vec{q}} \left\{ 2a_1^{(1)} \left(\frac{q_\mu q_\nu}{q^2} - \eta_{\mu\nu} \right) - a_2^{(1)} \delta_\mu^0 \delta_\nu^0 - \frac{a_2^{(1)} - 2da_1^{(1)}}{d-1} \eta_{\mu\nu} \right\}.$$

Using the proprieties of the Fourier transform (Appendix C.1) the only contributing term is the one with the transferred momentum components ($F_{ij}(0, r)$).

The metric becomes

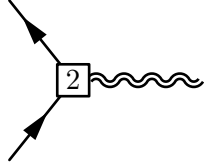
$$\delta^{(1)}h_{\mu\nu}^{(2)}(\vec{x}) = -16 \frac{1}{\pi^{\frac{d-2}{2}}} \left(\frac{(G_N m)^{\frac{1}{d-2}}}{r} \right)^d \Gamma\left(\frac{d}{2}\right) a_1^{(1)} \left(\delta_{ij} + (\alpha - d) \frac{x_i x_j}{\vec{x}^2} \right). \quad (3.4)$$

From this we can extract the metric components in Cartesian coordinates h_0 , h_1 and h_2 defined by 4.5. In particular it is possible to see that this counterterm doesn't vary the h_0 and that the variation of the other two components depends only in the $a_1^{(1)}$ parameter as anticipated. To compute the h_2 the basic Gamma propriety $\Gamma(d+1) = d\Gamma(d)$ is needed. The variation at one-loop order of the metric

components due to the introduction of a non-minimal coupling of the form 3.1 are then

$$\begin{cases} \delta^{(1)}h_0^{(2)}=0 \\ \delta^{(1)}h_1^{(2)}=\frac{16a_1^{(1)}\Gamma(\frac{d}{2})}{\pi^{\frac{d-2}{2}}}\left(\frac{(G_N m)^{\frac{1}{d-2}}}{r}\right)^d \\ \delta^{(1)}h_2^{(2)}=-\frac{32a_1^{(1)}\Gamma(\frac{d+2}{2})}{\pi^{\frac{d-2}{2}}}\left(\frac{(G_N m)^{\frac{1}{d-2}}}{r}\right)^d. \end{cases} \quad (3.5)$$

The second order minimal coupling contribution can be seen as the precence in the theory of a vertex Inserting this value inside 2.2 we have that the contribution to



$$\delta^{(2)}T_{\mu\nu}^{(0)} = -\vec{q}^4 (G_N m)^{\frac{4}{d-2}} m \left\{ 2 \left(a_1^{(2)} + \frac{1}{2} a_2^{(2)} \right) \left(\frac{q_\mu q_\nu}{q^2} - a_2^{(2)} \eta_{\mu\nu} \right) - a_2^{(1)} \delta_\mu^0 \delta_\nu^0 \right\}.$$

the metric variations is proportional to the Fourier transformation

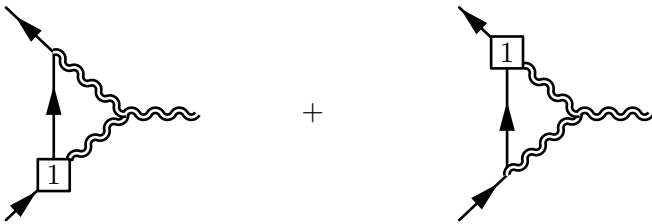
$$\int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} |\vec{q}|^2 e^{i\vec{x} \cdot \vec{q}}$$

which is zero. The higher minimal couplings will have a higher power of the transferred momentum and therefore all of them won't contribute to changes in the metric (in [17] it is shown for the third-order ones). So as anticipated the only contribution to the metric in the static limit is given by 3.1.

Then the metric at all orders is changed only by the addition in the theory of the vertex 3.3 corresponding to 3.1. We will perform the computations up to two-loop order. At one-loop order the variation are given by the already computed tree-level graph 3.3 and have the values 3.5.

One-loop insertion

For the second order corrections we have the contribution of the two one-loop diagrams:

$$\delta^{(1)}T_{\mu\nu}^{(1)} =$$


$$+$$

Which gives the contribution to the metric components of the form [17]

$$\begin{cases} \delta^{(1)} h_0^{(3)} = 64 a_1^{(1)} \frac{(d-2)\Gamma(\frac{d}{2})^2}{(d-1)\pi^{d-2}} \left(\frac{(G_N m)^{\frac{1}{d-2}}}{r} \right)^{2(d-1)} \\ \delta^{(1)} h_1^{(3)} = -64 a_1^{(1)} \frac{(d-2)\Gamma(\frac{d}{2})^2}{(d-1)\pi^{d-2}} \left(\frac{(G_N m)^{\frac{1}{d-2}}}{r} \right)^{2(d-1)} \\ \delta^{(1)} h_2^{(3)} = 128 a_1^{(1)} \frac{(d-2)\Gamma(\frac{d}{2})^2}{(d-1)\pi^{d-2}} \left(\frac{(G_N m)^{\frac{1}{d-2}}}{r} \right)^{2(d-1)} \end{cases} \quad (3.6)$$

3.1 Reissner-Nordström-Tangherlini Metric Renormalization

Through the scattering computations we computed the general dimension corrections to the Reissner-Nordström-Tangherlini metric in the de Donder gauge. This metric is in agree with 4.22 derived from the post-Minkowskian expansion of the metric in this gauge. The metric components are divergent in certain dimensions. In particular in $d = 3$ both h_1 and h_2 diverges at two loops, and in $d = 4$ all the components diverge at one and two-loop order. In the dimensions in which the metric doesn't diverge (for example $d = 5$) no renormalization procedure has to be carried on and the right, finite, metrics component are given directly by the diagram computations. The divergences derived from the massive and the charged part of the metric have the same leading of the divergences defined by the couplings 3.1 and 3.2 (which are only couplings between the scalar and the gravitational field). It is possible to obtain the right corrections to the metric using the variations 3.5 and 3.6 with the coupling constant $a_1^{(1)}$ function of both the couplings m and α .

3.1.1 Renormalised metric in four dimensions ($d = 3$)

In $d = 3$ the metric components will have ultraviolet divergences from two loop order and the addition of a non-minimal coupling in the action can absorb this divergence.

Two-loop renormalization $d = 3$

The two-loop contributions to the metric components is given by the previous quantum computations for the photon case (2.26) and for the scalar case are given in [17] (Eq. 3.44), and added together are

$$\begin{cases} h_0^{(3)}(r) = -\frac{8(3d-7)(d-2)^3}{(d-4)(d-1)^3} m^3 G_N^3 \tilde{\rho}(r)^3 - \frac{4(d-2)^2(3d-11)}{(d-4)(d-1)^2} m\alpha G_N^2 \tilde{\rho}(r)^3 \\ h_1^{(3)}(r) = \frac{8(7d^4-63d^3+214d^2-334d+212)}{3(d-4)(d-3)(d-1)^3} m^3 G_N^3 \tilde{\rho}(r)^3 + \frac{8(3d^3-25d^2+69d-65)}{3(d-4)(d-3)(d-1)^2} m\alpha G_N^2 \tilde{\rho}(r)^3 \\ h_2^{(3)}(r) = -\frac{8(d-2)^2(2d^3-13d^2+25d-10)}{(d-4)(d-3)(d-1)^3} m^3 G_N^3 \tilde{\rho}(r)^3 - \frac{4(d-2)^2(3d^3-19d^2+33d-17)}{(d-4)(d-3)(d-1)^3} m\alpha G_N^2 \tilde{\rho}(r)^3 \end{cases}, \quad (3.7)$$

with the radial parameter defined by

$$\tilde{\rho}(r) \equiv \frac{\Gamma\left(\frac{d-2}{2}\right)}{\pi^{\frac{d-2}{2}}} \frac{1}{r^{d-2}} \stackrel{d=3}{=} \Gamma\left(\frac{d-2}{2}\right) \frac{1}{\sqrt{\pi} r^{d-2}} .$$

In order to perform the dimensional regularisation method all the non divergent coefficients multiplying the metric components 3.7 can be evaluated in $d = 3$, making sure to leave the terms with that are complicated functions of d unchanged.

The result is

$$\begin{cases} h_0^{(3)}(r) = -\frac{2m\alpha G_N^2}{r^3} - \frac{2m^3 G_N^3}{r^3} \\ h_1^{(3)}(r) = \frac{1}{d-3} \left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{\sqrt{\pi}} \right)^3 \left(\frac{4}{3} m\alpha G_N^2 - \frac{2}{3} m^3 G_N^3 \right) r^{-3(d-2)} \\ h_2^{(3)}(r) = \frac{1}{d-3} \left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{\sqrt{\pi}} \right)^3 (-4m\alpha G_N^2 + 2m^3 G_N^3) r^{-3(d-2)} . \end{cases}$$

The first component is non divergent and can be already compared with the 4.23. The other two components are equal less then a multiplicative constant, in particular we can obtain $h_2^{(3)} = -3h_1^{(3)}$. We will so perform the renormalization procedure only on the $h_1^{(3)}$ component and then obtain the other with this relation.

Using the dimensional regularization method we study the system in dimension $d = 3 + \epsilon$ where ϵ is a small deviation and then obtain the values for $d = 3$ taking the limit $\epsilon \rightarrow 0$. This choice permits to explicit the divergences of the theory and extract from these the non-divergent terms.

In terms of ϵ the radial dependence of h_1 can be rewritten as

$$r^{-3(d-2)} = r^{-3(1+\epsilon)} = r^{-3} e^{\ln(r^{-3\epsilon})} = r^{-3} e^{-3\epsilon \ln r} \simeq r^{-3} (1 - 3\epsilon \ln r) ,$$

and the Gamma as

$$\left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{\sqrt{\pi}} \right)^3 = \left(\frac{\Gamma\left(\frac{1}{2} + \frac{\epsilon}{2}\right)}{\sqrt{\pi}} \right)^3 \simeq \left(\frac{\sqrt{\pi} - \frac{\epsilon\gamma}{2}}{\sqrt{\pi}} \right)^3 \simeq 1 - \frac{3\gamma\epsilon}{2\sqrt{\pi}} , \quad (3.8)$$

where γ is the Euler-Mascheroni constant $\gamma \equiv \sum_{n=1}^{\infty} \left[\frac{1}{n} - \ln \left(1 + \frac{1}{n} \right) \right] = 0.577216 \dots$. Substituting inside h_1 we obtain

$$\begin{aligned} h_1^{(3)}(r) &= \frac{1}{\epsilon} \left(\frac{4}{3} \frac{m\alpha G_N^2}{r^3} - \frac{2}{3} \frac{m^3 G_N^3}{r^3} \right) \left(1 - \frac{3\gamma}{2\sqrt{\pi}} \epsilon \right) (1 - 3\epsilon \ln r) \simeq \\ &\simeq \frac{1}{\epsilon} \left(\frac{4}{3} \frac{m\alpha G_N^2}{r^3} - \frac{2}{3} \frac{m^3 G_N^3}{r^3} \right) \left\{ 1 - 3\epsilon \left[\ln \left(\frac{r}{\sqrt{G_N m}} \right) + \ln \left(e^{\frac{\gamma}{2\sqrt{\pi}}} \right) \right] \right\} \equiv \\ &\equiv \frac{1}{\epsilon} \left(\frac{4}{3} \frac{m\alpha G_N^2}{r^3} - \frac{2}{3} \frac{m^3 G_N^3}{r^3} \right) (1 - 3\epsilon \ln cr) , \end{aligned}$$

where we performed a dimensional regularisation of the logarithm argument and then we defined the constant $c \equiv \frac{\gamma}{2\sqrt{G_N m \pi}}$. Only the first term is divergent in the limit $\epsilon \rightarrow 0$. Also the non-minimal coupling 3.1 provides a divergence of this type at this order (3.6) and we will use the freedom of the coupling constant to avoid this

divergence.

The variation to the metric from the presence of a non-minimal coupling of the type 3.1 is given by 3.6 and has to be expressed in terms of ϵ . Performing the same procedure described for the h_1 component on $\delta^{(1)}h_1^{(2)}$:

$$\begin{aligned}\delta^{(1)}h_1^{(3)}(r) &= -64a_1^{(1)} \frac{(d-2)\Gamma(\frac{d}{2})^2}{(d-1)\pi^{d-2}} \left(\frac{(G_N m)^{\frac{1}{d-2}}}{r} \right)^{2(d-1)} = \leftarrow (d=3) \\ &= -32a_1^{(1)} \frac{\Gamma(\frac{d}{2})^2}{\pi} (G_N m)^4 r^{-2(d-1)} \simeq -16a_1^{(1)} \left(\frac{G_N m}{r} \right)^4 + 32a_1^{(1)} \left(\frac{G_N m}{r} \right)^4 \epsilon \ln cr .\end{aligned}$$

In order to obtain the full renormalised metric, we have to add to the component h_1 also this contribution. It is possible to fix the coupling constant $a_1^{(1)}$ in order to delete the divergent term of h_1 . We want so to fix $a_1^{(1)}$ in order to vanish the sum

$$-16a_1^{(1)} \left(\frac{G_N m}{r} \right)^4 + \frac{1}{\epsilon} \left(\frac{4}{3} \frac{m\alpha G_N^2}{r^3} - \frac{2}{3} \frac{m^3 G_N^3}{r^3} \right) = 0 .$$

Solving the equation we obtain

$$a_1^{(1)} = \frac{r}{\epsilon} \left(\frac{1}{12} \frac{\alpha}{G_N^2 m^3} - \frac{1}{24} \frac{1}{m} \right) . \quad (3.9)$$

The surviving terms of this sum (both of the logarithmic terms of $h_1^{(3)}$ and $\delta^{(1)}h_1^{(3)}$), with this choice of the coupling constant $a_1^{(1)}$, give the renormalized metric component at this order

$$h_1^{(3)\text{ren}}(r) = -\frac{8}{3} \frac{m\alpha G_N^2}{r^3} \ln cr + \frac{4}{3} \frac{m^3 G_N^3}{r^3} \ln cr .$$

Form this is possible to obtain $h_2^{(3)\text{ren}}(r) = -3h_1^{(3)\text{ren}}(r)$.

All the renormalized components are

$$\begin{cases} h_0^{(3)\text{ren}}(r) = -\frac{2m\alpha G_N^2}{r^3} - \frac{2m^3 G_N^3}{r^3} \\ h_1^{(3)\text{ren}}(r) = -\frac{8}{3} \frac{m\alpha G_N^2}{r^3} \ln cr + \frac{4}{3} \frac{m^3 G_N^3}{r^3} \ln cr \\ h_2^{(3)\text{ren}}(r) = 8 \frac{m\alpha G_N^2}{r^3} \ln cr - 4 \frac{m^3 G_N^3}{r^3} \ln cr . \end{cases}$$

These are in agreement with the 4.23, in particular the classical computations done for obtaining the 4.23 seem to predict perfectly the corrections given by the quantum field computations, guessing exactly the coefficients of the logarithmic term, and therefore giving information about the counterterms needed for the renormalization procedure.

3.1.2 Renormalised metric in five dimensions ($d = 4$)

The metric in $d = 4$ has ultraviolet divergences from one and two loop order. We will perform the renormalization procedure using respectively the corrections 3.5 and 3.6 given by the non-minimal coupling 3.1.

One-loop renormalization $d = 4$

At one loop order the metric components in general dimension are

$$\begin{cases} h_0^{(2)}(r) = \frac{8(d-2)^2}{(d-1)^2} m^2 G_N^2 \tilde{\rho}(r)^2 + \frac{2(d-2)}{d-1} \alpha G_N \tilde{\rho}(r)^2 \\ h_1^{(2)}(r) = -4 \frac{2d^2-9d+14}{(d-4)(d-1)^2} m^2 G_N^2 \tilde{\rho}(r)^2 - \frac{2(d-3)}{(d-4)(d-1)} \alpha G_N \tilde{\rho}(r)^2 \\ h_2^{(2)}(r) = \frac{4(d-2)^2(3d-2)}{(d-4)(d-1)^2} m^2 G_N^2 \tilde{\rho}(r)^2 + \frac{2(d-2)^2}{(d-4)(d-1)} \alpha G_N \tilde{\rho}(r)^2, \end{cases}$$

with the radial parameter defined by

$$\tilde{\rho}(r) \equiv \frac{\Gamma\left(\frac{d-2}{2}\right)}{\pi^{\frac{d-2}{2}}} \frac{1}{r^{d-2}} \stackrel{d=4}{=} \frac{\Gamma\left(\frac{d-2}{2}\right)}{\pi} \frac{1}{r^{d-2}}.$$

We can evaluate all the non divergent terms that are simple functions of d to simplify the expressions. These terms are not relevant in the dimensional regularization technique.

We obtain

$$\begin{cases} h_0^{(2)}(r) = \frac{4}{3} \frac{\alpha G_N}{\pi^2 r^4} + \frac{32}{9} \frac{m^2 G_N}{\pi^2 r^4} \\ h_1^{(2)}(r) = \frac{1}{d-4} \left(\Gamma\left(\frac{d-2}{2}\right) \right)^2 \left(-\frac{2}{3} \frac{\alpha G_N}{\pi^2} - \frac{40}{9} \frac{G_N^2 m^2}{\pi^2} \right) r^{-2(d-2)} \\ h_2^{(2)}(r) = \frac{1}{d-4} \left(\Gamma\left(\frac{d-2}{2}\right) \right)^2 \left(\frac{8}{3} \frac{\alpha G_N}{\pi^2} + \frac{160}{9} \frac{G_N^2 m^2}{\pi^2} \right) r^{-2(d-2)}. \end{cases}$$

Again the $h_0^{(2)}$ component is non divergent and no renormalization procedure has to be carried on. The $h_1^{(2)}$ and $h_2^{(2)}$ components are related by the identity $h_2^{(2)} = -4h_1^{(2)}$ and so we can focus only on one component, the $h_1^{(2)}$.

Introducing the dimensional regularization in this dimension we define $d = 4 + \epsilon$ and the radial dependence of the $h_1^{(2)}$ can be expressed as

$$r^{-2(d-2)} = r^{-2(2+\epsilon)} = r^{-4} e^{\ln(r^{-2\epsilon})} = r^{-4} e^{-2\epsilon \ln r} \simeq r^{-4} (1 - 2\epsilon \ln r),$$

and the Gamma term becomes

$$\left(\Gamma\left(\frac{d-2}{2}\right) \right)^2 = \left(\Gamma\left(1 + \frac{\epsilon}{2}\right) \right)^2 \simeq \left(1 - \frac{1}{2} \gamma \epsilon \right)^2 \simeq 1 - \gamma \epsilon.$$

Performing this substitution for d the metrics component becomes

$$\begin{aligned} h_1^{(2)}(r) &= \frac{1}{\epsilon} \left(-\frac{2}{3} \frac{\alpha G_N}{\pi^2} - \frac{40}{9} \frac{G_N^2 m^2}{\pi^2} \right) \left(\Gamma\left(\frac{d-2}{2}\right) \right)^2 r^{-2(2+\epsilon)} \simeq \\ &\simeq \frac{1}{\epsilon} \left(-\frac{2}{3} \frac{\alpha G_N}{\pi^2 r^4} - \frac{40}{9} \frac{G_N^2 m^2}{\pi^2 r^4} \right) (1 - \gamma \epsilon) (1 - 2\epsilon \ln r) \simeq \\ &\simeq \frac{1}{\epsilon} \left(-\frac{2}{3} \frac{\alpha G_N}{\pi^2 r^4} - \frac{40}{9} \frac{G_N^2 m^2}{\pi^2 r^4} \right) \left\{ 1 - 2\epsilon \left[\ln\left(\frac{\sqrt{\pi} r}{m G_N}\right) + \ln\left(e^{\frac{\gamma}{2}}\right) \right] \right\} \equiv \\ &\equiv \frac{1}{\epsilon} \left(-\frac{2}{3} \frac{\alpha G_N}{\pi^2 r^4} - \frac{40}{9} \frac{G_N^2 m^2}{\pi^2 r^4} \right) (1 - 2\epsilon \ln cr), \end{aligned}$$

where we regularised the argument of the logarithm and defined the constant $c \equiv \frac{\sqrt{\pi}e^{\gamma/2}}{mG_N}$.

In the limit $\epsilon \rightarrow 0$ the only divergent term is the first. We can absorb this divergence in a smart choice of the coupling constant $a_1^{(1)}$. At this order, the contribution arising from the non-minimal coupling 3.1 is given by the corrections 3.5 given by the tree-level diagram with the new non-minimal coupling vertex.

We have to express this contribution in terms of the ϵ as:

$$\begin{aligned} \delta^{(1)}h_1^{(2)}(r) &= \frac{16a_1^{(1)}\Gamma(\frac{d}{2})}{\pi^{\frac{d-2}{2}}} \left(\frac{(G_N m)^{\frac{1}{d-2}}}{r} \right)^d = \longleftarrow (d=4) \\ &= -16a_1^{(1)}\Gamma(\frac{d}{2})\frac{(G_N m)^2}{\pi}r^{-d} \simeq -16a_1^{(1)}\frac{(G_N m)^2}{\pi}r^{-4} - 16\epsilon a_1^{(1)}\frac{(G_N m)^2}{\pi}r^{-4} \ln cr . \end{aligned}$$

We can choose the coupling constant $a_1^{(1)}$ in order to delete the divergence in the first term of the $h_1^{(2)}$. The right choice of this constant is given solving the equation

$$-16a_1^{(1)}\frac{(G_N m)^2}{\pi}r^{-4} + \frac{1}{\epsilon} \left(-\frac{2}{3}\frac{\alpha G_N}{\pi^2 r^4} - \frac{40}{9}\frac{G_N^2 m^2}{\pi^2 r^4} \right) = 0 ,$$

that gives

$$a_1^{(1)} = \frac{1}{\epsilon} \left(\frac{1}{24}\frac{\alpha}{G_N m^2 \pi} + \frac{5}{18}\frac{1}{\pi} \right) . \quad (3.10)$$

The surviving terms with the substitution of the constant found above gives the renormalized metrics component:

$$\begin{cases} h_0^{(2)\text{ren}}(r) = \frac{4}{3}\frac{\alpha G_N}{\pi^2 r^4} + \frac{32}{9}\frac{m^2 G_N}{\pi^2 r^4} \\ h_1^{(2)\text{ren}}(r) = \frac{2}{3}\frac{\alpha G_N}{\pi^2 r^4} \ln cr + \frac{40}{9}\frac{m^2 G_N}{\pi^2 r^4} \ln cr \\ h_2^{(2)\text{ren}}(r) = -\frac{8}{3}\frac{\alpha G_N}{\pi^2 r^4} \ln cr - \frac{160}{9}\frac{m^2 G_N}{\pi^2 r^4} \ln cr . \end{cases}$$

These are in agreement with the components derived from the post-Minkowskian expansion at this loop order in $d=4$ (4.24).

Two-loop renormalization $d=4$

The two-loop terms of the metrics component in any dimension are given by 3.7. This time all the components diverge in $d=4$. Performing the same evaluation done above

$$\begin{cases} h_0^{(3)}(r) = \frac{1}{d-4} \left(-\frac{11}{9}\frac{m\alpha G_N^2}{\pi^3} - \frac{320}{27}\frac{m^3 G_N^3}{\pi^3} \right) r^{-3(d-2)} \\ h_1^{(3)}(r) = \frac{1}{d-4} \left(\frac{8}{9}\frac{m\alpha G_N^2}{\pi^3} + \frac{160}{27}\frac{m^3 G_N^3}{\pi^3} \right) r^{-3(d-2)} \\ h_2^{(3)}(r) = \frac{1}{d-4} \left(-\frac{16}{9}\frac{m\alpha G_N^2}{\pi^3} - \frac{320}{27}\frac{m^3 G_N^3}{\pi^3} \right) r^{-3(d-2)} . \end{cases}$$

We can focus only on the $h_1^{(3)}$ because the others are $h_0^{(3)} = h_2^{(3)} = -2h_1^{(3)}$. Introducing again $d=4+\epsilon$ and performing the radial expansion as done for the one-loop case, we found

$$\frac{1}{\epsilon} \left(\frac{8}{9}\frac{m\alpha G_N^2}{\pi^3 r^6} + \frac{160}{27}\frac{m^3 G_N^3}{\pi^3 r^6} \right) - \left(\frac{8}{9}\frac{m\alpha G_N^2}{\pi^3 r^6} + \frac{160}{9}\frac{m^3 G_N^3}{\pi^3 r^6} \right) \ln cr .$$

At two-loop order the corrections induced by the non-minimal coupling 3.1 are from both the one-loop diagrams with a coupling vertex and are 3.6.

The $h_1^{(3)}$ correction can be rewritten in $d = 4$ as

$$\begin{aligned}\delta^{(1)}h_1^{(3)}(r) &= -64a_1^{(1)}\frac{(d-2)\Gamma(\frac{d}{2})^2}{(d-1)\pi^{d-2}}\left(\frac{(G_N m)^{\frac{1}{d-2}}}{r}\right)^{2(d-1)} = \longleftarrow (d=4) \\ &= -\frac{64}{3}a_1^{(1)}\frac{\Gamma(\frac{d}{2})^2}{\pi^2}(G_N m)^3 r^{-2(d-1)} \simeq -\frac{64}{3}a_1^{(1)}\frac{G_N^3 m^3}{\pi^2 r^6} + \frac{128}{3}\epsilon\frac{G_N^3 m^3}{\pi^2 r^6} \ln cr .\end{aligned}$$

In this case we don't have the freedom to change the coupling constant because has been already fixed during the one loop computations (3.10). We see that this choice of the constant delete also the divergences of the two loop computations, in other words it satisfies the equation

$$\frac{1}{\epsilon}\left(\frac{8}{9}\frac{m\alpha G_N^2}{\pi^3 r^6} + \frac{160}{27}\frac{m^3 G_N^3}{\pi^3 r^6}\right) - \frac{64}{3}a_1^{(1)}\frac{G_N^3 m^3}{\pi^2 r^6} = 0 .$$

The surviving terms gives the renormalized metrics component and are:

$$\begin{cases} h_0^{(3)}(r)^{\text{ren}} = \frac{16}{9}\frac{m\alpha G_N^2}{\pi^3 r^6} \ln cr + \frac{320}{27}\frac{m^3 G_N^3}{\pi^3 r^6} \ln cr \\ h_1^{(3)}(r)^{\text{ren}} = -\frac{8}{9}\frac{m\alpha G_N^2}{\pi^3 r^6} \ln cr - \frac{160}{27}\frac{m^3 G_N^3}{\pi^3 r^6} \ln cr \\ h_2^{(3)}(r)^{\text{ren}} = \frac{16}{9}\frac{m\alpha G_N^2}{\pi^3 r^6} \ln cr + \frac{320}{27}\frac{m^3 G_N^3}{\pi^3 r^6} \ln cr . \end{cases}$$

As we will see in the next chapter these are in agreement with the components derived from the post-Minkowskian expansion at this loop order in $d = 4$ (4.24).

Chapter 4

Metrics in de Donder gauge

In this chapter we derive the post-Minkowskian expansion of the Reissner–Nordström–Tangherlini metric in de Donder gauge. We will compare the results in general dimension with the metric components given directly by the diagram computations of Chapter 2. For fixed dimensions ($d = 3$ and $d = 4$) we will obtain values in agreement with the renormalized quantities obtained in Chapter 3. We will see that the log terms arising from the renormalization process are perfectly expected by the post-Minkowskian expansion of the metric in de Donder gauge.

For the following treatment it is useful to derive a metric of the form

$$ds^2 = C(r)dt^2 - \frac{dr^2}{C(r)} - r^2 d\Omega_{d-1}^2 \quad (4.1)$$

in the de Donder gauge, where d is the number of spatial dimensions and $d\Omega_{d-1}^2 = d\theta_1^2 + \sin^2 \theta_1 d\theta_2^2 + \dots + \prod_{i=1}^{d-1} \sin^2 \theta_i d\theta_{d-1}^2$ the area of the unit $(d-1)$ -sphere. The metric 4.1 is expressed in polar coordinates, but it is useful to change the spatial part to Cartesian coordinates:

$$\left\{ \begin{array}{l} x_1 = r \cos \theta_1 \\ x_2 = r \sin \theta_1 \cos \theta_2 \\ \vdots \\ x_{d-1} = r \sin \theta_1 \sin \theta_2 \dots \sin \theta_{d-2} \cos \theta_{d-1} \\ x_d = r \sin \theta_1 \sin \theta_2 \dots \sin \theta_{d-2} \sin \theta_{d-1} . \end{array} \right. \quad (4.2)$$

In this system the metric reads

$$ds^2 = C(r)dt^2 - d\vec{x}^2 - \frac{1 - C(r)}{C(r)} \frac{(\vec{x} \cdot d\vec{x})^2}{r^2} . \quad (4.3)$$

We want to derive the de Donder gauge condition in Cartesian coordinates. The gauge freedom of the theory allows us to make a general coordinate transformation such that $(t, \vec{x}) \rightarrow (t, f(r)\vec{x})$ with $r = \sqrt{x^i x^i}$. This change of coordinates varies the

quantities in 4.3 as

$$\begin{aligned} dx^{i^2} &\rightarrow f^2 dx^{i^2} + \left(\frac{\vec{x} \cdot d\vec{x}}{r} \right)^2 (r^2 f'^2 + 2r f f') \\ \frac{(\vec{x} \cdot d\vec{x})^2}{r^2} &\rightarrow \frac{(\vec{x} \cdot d\vec{x})^2}{f^2 r^2} (f^2 + r f f')^2 \\ C(r) &\rightarrow C(f(r)r) . \end{aligned} \quad (4.4)$$

We thus define the functions $h_i(r)$ that characterise the metric in the following way

$$ds^2 \equiv h_0(r) dt^2 - h_1(r) d\vec{x}^2 - h_2(r) \frac{(\vec{x} \cdot d\vec{x})^2}{r^2} . \quad (4.5)$$

Substituting 4.4 into 4.3 and after some simplifications we obtain the $h_i(r)$ in these variables

$$\begin{cases} h_0(r) = C(f(r)r) \\ h_1(r) = f(r)^2 \\ h_2(r) = -f(r)^2 + \frac{(f + r f')^2}{C(f(r)r)} . \end{cases} \quad (4.6)$$

The metric components are now functions of $f(r)$ and the gauge freedom is explicit in the metric. We can fix the gauge in the metric fixing this function from the gauge condition. We will obtain the metric in de Donder gauge. The gauge equation represents a differential equation in the metrics component (as it will be proven) and substituting 4.6 we can solve this equation for $f(r)$.

The de Donder gauge is defined by the differential equation

$$\eta^{\mu\nu} (g_{\beta\nu,\mu} + g_{\mu\beta,\nu} - g_{\mu\nu,\beta}) = 0 . \quad (4.7)$$

where the metric tensor is given by 4.5 and has non-vanishing components:

$$\begin{cases} g_{00} = h_0(r) \\ g_{ij} = -h_1(r) \delta_{ij} - h_2(r) \frac{x^i x^j}{r^2} . \end{cases}$$

Observing that the components 4.8 are independent of time the gauge condition 4.7 becomes

$$-\partial_j g_{00} - 2\partial_i g_{ji} + \partial_j g_{ii} = 0 ,$$

where i and j are the indices only of the spatial coordinates. Substituting the components 4.8 and computing it for $j = 1$ we have

$$\begin{aligned} \partial_1 h_0(r) + \partial_1 (dh_1(r) + h_2(r)) &= 2\partial_i \left(h_1(r) \delta_{i1} + \frac{x_i x_1}{r^2} h_2(r) \right) \Bigg|_{\frac{\partial}{\partial x^1} = \frac{\partial r}{\partial x^1} \frac{\partial}{\partial r} = \frac{x^1}{r} \frac{\partial}{\partial r}} \\ \frac{x^1}{r} \partial_r (h_0(r) + (d-2)h_1(r) + h_2(r)) &= 2\partial_i \left(\frac{x_i x_1}{r^2} h_2(r) \right) \\ &= 2 \frac{x_i x_1}{r^2} \partial_i h_2(r) + 2h_2(r) \partial_i \left(\frac{x_i x_1}{r^2} \right) \\ \frac{x^1}{r} \partial_r (h_0(r) + (d-2)h_1(r) - h_2(r)) &= 2h_2(r) \left[\partial_1 \left(\frac{x_1^2}{r^2} \right) + \partial_2 \left(\frac{x_1 x_2}{r^2} \right) + \dots \right] \\ &= 2h_2(r) \frac{x_1}{r^2} (d-1) , \end{aligned}$$

and finally we have the gauge condition expressed in terms of the $h_i(r)$

$$r \frac{d}{dr} \left(h_0(r) + (d-2)h_1(r) - h_2(r) \right) = 2(d-1)h_2(r) . \quad (4.8)$$

This is a differential equation and can be solved for the function $f(r)$ once the components 4.6 have been substituted. After finding the solution for $f(r)$ we can insert this in 4.6 to find the metric in this gauge. This will be done numerically for the Reissner–Nordström–Tangherlini metric in the last section of this chapter.

4.1 Schwarzschild-Tangherlini metric

The Schwarzschild-Tangherlini metric [28] is a generalisation in arbitrary dimensions of the Schwarzschild metric, the metric that describes the gravitational field generated by a spherical non-charged mass distribution. In spherical coordinates it reads as

$$ds^2 = \left(1 - \frac{\mu_m}{r^{d-2}} \right) dt^2 - \frac{dr^2}{1 - \frac{\mu_m}{r^{d-2}}} - r^2 d\Omega_{d-1}^2 . \quad (4.9)$$

The parameter μ_m is related to the mass and to the area of the $(d-1)$ -sphere by the law

$$\mu_m = \frac{16\pi G_N m}{(d-1)\Omega_{d-1}} \quad \text{con} \quad \Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma\left(\frac{d}{2}\right)} . \quad (4.10)$$

The metric 4.9 is of the form given in 4.1 with $C(r) = 1 - \frac{\mu_m}{r^{d-2}}$. We write the metric in the de Donder gauge using the equations 4.6, the $h_i(r)$ in this case are:

$$\begin{cases} h_0(r) = 1 - \frac{\mu_m}{f(r)^{d-2}r^{d-2}} \\ h_1(r) = f(r)^2 \\ h_2(r) = -f(r)^2 + \frac{(f + rf')^2}{1 - \frac{\mu_m}{f(r)^{d-2}r^{d-2}}} \end{cases} . \quad (4.11)$$

It is useful to perform another change of variables from r to x defined by

$$x \equiv \frac{\mu_m}{r^{d-2}} . \quad (4.12)$$

The equations 4.11 become:

$$\begin{cases} h_0(x) = 1 - \frac{x}{f(x)^{d-2}} \\ h_1(x) = f(x)^2 \\ h_2(x) = -f(x)^2 + \frac{(f + (2-d)xf'(x))^2}{1 - \frac{x}{f(x)^{d-2}}} \end{cases} , \quad (4.13)$$

where we used the following relation

$$r \frac{d}{dr} = r \frac{dx}{dr} \frac{d}{dx} = -(d-2)r \frac{\mu_m}{r^{d-1}} \frac{d}{dx} = (2-d)x \frac{d}{dx} .$$

This relation is also useful to derive the gauge condition 4.8 in the new variable. The de Donder gauge condition becomes

$$(2-d)x \frac{d}{dx} \left(h_0(x) + (d-2)h_1(x) - h_2(x) \right) = 2(d-1)h_2(x) . \quad (4.14)$$

Is possible to solve this equation expanding $f(x)$ as

$$f(x) = 1 + \sum_{n=1}^{\infty} a_n x^n \quad (4.15)$$

which is the post-Minkowskian expansion, where we include the asymptotic boundary condition $\lim_{x \rightarrow 0} f(x) = 1$.

In the next section we will see that this ansatz is not always true because in certain dimensions some of the a_n diverge. This divergence makes this expansion no longer possible and we will need to change this ansatz including logs at the divergent terms in the way described in the last part of this chapter. The new ansatz will allow us to obtain finite metric components in every dimension and these values will be compared with the metric arising from scattering amplitudes for a perfect match. The presence of this log terms for the divergent dimensions is expected in the quantum computations by the renormalization procedure of Chapter 3.

4.2 Reissner–Nordström–Tangherlini metric

The Reissner–Nordström metric is the metric of a spherically symmetric charged mass distribution in four dimensions. The generalisation in arbitrary dimensional space-time was given by Tangherlini [28] in the same article of the uncharged case. In spherical coordinates it has the expression

$$ds^2 = \left(1 - \frac{\mu_m}{r^{d-2}} + \frac{\mu_q}{r^{2(d-2)}} \right) dt^2 - \frac{dr^2}{1 - \frac{\mu}{r^{d-2}} + \frac{\mu_q}{r^{2(d-2)}}} - r^2 d\Omega_{d-1}^2 , \quad (4.16)$$

where

$$\mu_m = \frac{16\pi G_N m}{(d-1)\Omega_{d-1}} \quad \text{and} \quad \mu_q = \frac{2G_N}{(d-2)(d-1)} Q^2 . \quad (4.17)$$

The metric above is equivalent to the metric 4.1 with $C(r) = 1 - \frac{\mu}{r^{d-2}} + \frac{\mu_q}{r^{2(d-2)}}$. As we did for the Schwarzschild–Tangherlini metric we perform the change of variable to x defined by 4.12. The gauge condition is equal to the previous case 4.14 and the $h_i(x)$ become:

$$\begin{cases} h_0(x) = 1 - \frac{x}{f(x)^{d-2}} \left(1 - \frac{\mu_q}{\mu_m} \frac{x}{f^{d-2}} \right) \\ h_1(x) = f(x)^2 \\ h_2(x) = -f(x)^2 + \frac{(f + (2-d)xf'(x))^2}{1 - \frac{x}{f(x)^{d-2}} \left(1 - \frac{\mu_q}{\mu_m} \frac{x}{f(x)^{d-2}} \right)} . \end{cases} \quad (4.18)$$

After finding the $f(x)$ from the gauge condition 4.14 we will substitute it inside 4.18 in order to obtain the Reissner–Nordström–Tangherlini metric components in the de Donder gauge. These components will be used as a check for the quantum computations of the previous chapters.

4.3 Numerical Computations of Metrics in de Donder gauge

In this section we will describe the numerical computation done using *Wolfram Mathematica* in order to solve the equations 4.18 in the de Donder gauge (4.14). We summarise the de Donder gauge condition and expression for the metrics components:

$$(2-d)x \frac{d}{dx} (h_0(x) + (d-2)h_1(x) - h_2(x)) = 2(d-1)h_2(x) \quad (4.19)$$

$$\begin{cases} h_0(x) = 1 - \frac{x}{f(x)^{d-2}} \left(1 - \frac{\mu_q}{\mu_m} \frac{x}{f(x)^{d-2}} \right) \\ h_1(x) = f(x)^2 \\ h_2(x) = -f(x)^2 + \frac{(f + (2-d)xf'(x))^2}{1 - \frac{x}{f(x)^{d-2}} \left(1 - \frac{\mu_q}{\mu_m} \frac{x}{f(x)^{d-2}} \right)} \end{cases} \quad (4.20)$$

Substituting the metric components inside the gauge condition we obtain a differential equation for $f(x)$ that it is possible to solve with the initial condition $f(0) = 1$. All the computations are done in the Reissner–Nordström–Tangherlini case because the Schwarzschild–Tangherlini one can be extracted from it by focusing on the terms that do not depend on the electric charge.

As said, it is possible to make an ansatz on $f(x)$ of the form

$$f(x) \simeq 1 + \sum_{i=1}^n a_i x^i, \quad (4.21)$$

where n is the expansion order, in the limit $n \rightarrow \infty$ the expansion is exact. The gauge condition fixes all the n coefficients a_i (from the equivalence of two n -th order polynomials) and so it determines the expression of $f(x)$. This process is reported the script 4.1.

```

n := 3
f[x_] := 1 + Sum[a[i] x^i, {i, 1, n}] + O[x]^(n+1);
Print["f(x) = ", TraditionalForm[f[x]]]

h0[x_] = 1 - x / (f[x]^(d-2) (1 - (q^2 x) / (mu^2 f[x]^(d-2)))) + O[x]^(n+1);
h1[x_] = f[x]^2 + O[x]^(n+1);
h2[x_] = -f[x]^2 + ((f[x] - (d-2) x f'[x])^2) / (1 - x / (f[x]^(d-2) (1 - (q^2 x) / (mu^2 f[x]^(d-2)))) + O[x]^(n+1);

gauge = 2 (d-1) h2[x] == (2-d) x Dt[h0[x] + (d-2) h1[x] - h2[x], x];
expansion = LogicalExpand[gauge];
vars := List[]
Do[AppendTo[vars, ai], {i, n}]
solution := Solve[expansion, vars]
x[r_] := (4 (d-2) / (d-1)) G m p[r] + O[G]^(n+1)
(* p[r_] := (Gamma[d/2-1] Pi^(1-d/2) / r^(d-2)) *)
mu := (8 G m Gamma[d/2] Pi^(1-d/2)) / (d-1)
q := Sqrt[(2 Pi^(1-d) G) / ((d-2) (d-1))] Gamma[d/2] e
e := Sqrt[4 Pi alpha]

```

Script 4.1. Script for the computation of $f(x)$ in generic dimension with $n = 3$.

The solution for general dimension up to $O(G^4)$ is found to be:

$$\begin{aligned}
h_0(r) &= 1 + \left(\frac{2(d-2)\alpha\tilde{\rho}(r)^2}{d-1} - \frac{4(d-2)m\tilde{\rho}(r)}{d-1} \right) G + \\
&\quad + \left(\frac{8(d-2)^2 m^2 \tilde{\rho}(r)^2}{(d-1)^2} - \frac{4(d-2)^2 (3d-11) m \alpha \tilde{\rho}(r)^3}{(d-4)(d-1)^2} \right) G^2 \\
&\quad + \frac{8(7-3d)(d-2)^3 m^3 \tilde{\rho}(r)^3 G^3}{(d-4)(d-1)^3} + O(G^4) \\
h_1(r) &= 1 + \left(\frac{4m\tilde{\rho}(r)}{d-1} - \frac{2(d-3)\alpha\tilde{\rho}(r)^2}{(d-4)(d-1)} \right) G + \\
&\quad + \left(\frac{8(3d^3 - 25d^2 + 69d - 65) m \alpha \tilde{\rho}(r)^3}{3(d-4)(d-3)(d-1)^2} + \frac{4(-2d^2 + 9d - 14) m^2 \tilde{\rho}(r)^2}{(d-4)(d-1)^2} \right) G^2 + \\
&\quad + \frac{8(7d^4 - 63d^3 + 214d^2 - 334d + 212) m^3 \tilde{\rho}(r)^3 G^3}{3(d-4)(d-3)(d-1)^3} + O(G^4) \\
h_2(r) &= \frac{2(d-2)^2 \alpha \tilde{\rho}(r)^2 G}{(d-4)(d-1)} + \\
&\quad + \left(\frac{4(d-2)^2 (-3d^3 + 19d^2 - 33d + 17) m \alpha \tilde{\rho}(r)^3}{(d-4)(d-3)(d-1)^3} + \frac{4(d-2)^2 (3d-2) m^2 \tilde{\rho}(r)^2}{(d-4)(d-1)^2} \right) G^2 + \\
&\quad + \frac{8(d-2)^2 (-2d^3 + 13d^2 - 25d + 10) m^3 \tilde{\rho}(r)^3 G^3}{(d-4)(d-3)(d-1)^3} + O(G^4),
\end{aligned} \tag{4.22}$$

where we have defined

$$\tilde{\rho}(r) \equiv \frac{\Gamma(\frac{d}{2} - 1) \pi^{1-d/2}}{r^{d-2}}$$

as the new radial parameter without the dependence on m , α and G_N . The metric is manifestly divergent in $d = 3$ and $d = 4$ as expected from the general analysis of equation 2.10. In these dimensions, the ansatz 4.21 is no longer right because the coefficients go to infinity. This ansatz has to be changed in the following way. Every time there is a divergent coefficient we have to add to the term at the coefficient's order all the terms with a power of logarithm permitted by the order of the polynomial. Every new term includes a new coefficient in the expansion, all of these coefficients can be evaluated by the gauge condition, from the equivalence of the coefficients of different polynomial plus logs powers. All the new coefficients will be fixed apart from the first divergent one. The procedure explained above is contained in the script 4.2 for the case $d = 3$ and in the script 4.3 for $d = 4$. For example in the first case the first divergent coefficient is at the third power of x , so the new ansatz on $f(x)$ is

$$f(x)_{d=3} = 1 + a_1 x + a_2 x^2 + (a_{3,2} \log(x)^2 + a_{3,1} \log(x) + a_3) x^3 + O(x^4) .$$

For $d = 4$ the first divergent coefficient is at second order in x and the new ansatz is

$$f(x)_{d=4} = 1 + a_1 x + (a_{2,1} \log(x) + a_2) x^2 + (a_{3,2} \log(x)^2 + a_{3,1} \log(x) + a_3) x^3 + O(x^4) .$$

The results obtained for the case $d = 3$ are:

$$\begin{aligned} h_0(r) &= 1 + \left(\frac{\alpha}{r^2} - \frac{2m}{r} \right) G + \left(\frac{2m^2}{r^2} - \frac{2m\alpha}{r^3} \right) G^2 + \frac{2m^3 G^3}{r^3} + O(G^4) \\ h_1(r) &= 1 + \frac{2mG}{r} + \left(\frac{5m^2}{r^2} + \frac{8\alpha \log\left(\frac{2Gm}{cr}\right) m}{3r^3} \right) G^2 + \frac{4m^3 \left(3 - \log\left(\frac{2Gm}{cr}\right) \right) G^3}{3r^3} + O(G^4) \\ h_2(r) &= -\frac{\alpha G}{r^2} + \left(\frac{2m\alpha \left(-12 \log\left(\frac{2Gm}{cr}\right) - 7 \right)}{3r^3} - \frac{7m^2}{r^2} \right) G^2 + \frac{2m^3 \left(6 \log\left(\frac{2Gm}{cr}\right) - 19 \right) G^3}{3r^3} + O(G^4) . \end{aligned} \tag{4.23}$$

The results obtained for the case $d = 4$ are:

$$\begin{aligned}
h_0(r) &= 1 + \left(\frac{4\alpha}{3\pi^2 r^4} - \frac{8m}{3\pi r^2} \right) G + \left(\frac{32m^2}{9\pi^2 r^4} - \frac{8m \left(\log \left(\frac{8Gm}{3c\pi r^2} \right) \alpha + 4\alpha \right)}{9\pi^3 r^6} \right) G^2 + \\
&\quad + \frac{32m^3 \left(-5 \log \left(\frac{8Gm}{3c\pi r^2} \right) - 3 \right) G^3}{27\pi^3 r^6} + O(G^4) \\
h_1(r) &= 1 + \left(\frac{4m}{3\pi r^2} - \frac{\alpha \log \left(\frac{8Gm}{3c\pi r^2} \right)}{3\pi^2 r^4} \right) G + \\
&\quad + \left(\frac{4 \left(1 - 5 \log \left(\frac{8Gm}{3c\pi r^2} \right) \right) m^2}{9\pi^2 r^4} + \frac{4 \left(3 \log \left(\frac{8Gm}{3c\pi r^2} \right) \alpha + 11\alpha \right) m}{27\pi^3 r^6} \right) G^2 + \\
&\quad + \frac{16m^3 \left(15 \log \left(\frac{8Gm}{3c\pi r^2} \right) + 4 \right) G^3}{81\pi^3 r^6} + O(G^4) \\
h_2(r) &= \frac{2 \left(2\alpha \log \left(\frac{8Gm}{3c\pi r^2} \right) - \alpha \right) G}{3\pi^2 r^4} + \left(\frac{40m^2 \left(2 \log \left(\frac{8Gm}{3c\pi r^2} \right) + 1 \right)}{9\pi^2 r^4} - \frac{8m \left(\log \left(\frac{8Gm}{3c\pi r^2} \right) \alpha + 13\alpha \right)}{9\pi^3 r^6} \right) G^2 + \\
&\quad + \frac{32m^3 \left(-5 \log \left(\frac{8Gm}{3c\pi r^2} \right) - 4 \right) G^3}{27\pi^3 r^6} + O(G^4) .
\end{aligned} \tag{4.24}$$

These metric components can be compared with the ones arising from the quantum computations giving us a confirmation of the validity of the analysis performed in the first chapters. The presence of these new log terms is expected by the quantum theory and it is due to the necessity of a renormalization of the components at the order in which these terms appear.

This expansion can be done at every n , but increasing this value takes exponentially long times of computations. The simulations have been carried out up to $n = 8$ included.

These expressions (in particular for large values of n) can be useful in guessing the type of diagrams that appear at each order by focusing on the power of the coupling constants. Also the prediction of the needed counterterms can be useful for many-loops predictions. In particular at large n powers of the log terms higher than one appear, which are due to the multi-loop diagram contribution of the first non-minimal coupling vertex (e.g. at two loops we have 3.6). These can be difficult to compute but can be extracted directly from the metric components avoiding loop computations.


```

In[*]:= Clear[d]
divergent := List[]
flag = 0;
Do[If[Limit[solution[[All, i, 2]][[1]], d -> dLimit] === Indeterminate || flag == 1,
    AppendTo[divergent, i];
    flag = 1], {i, n}]
Print["Divergent coefficient : " divergent]
{3 Divergent coefficient :}

Change ansatz on f(x) including logarithms to all the divergent coefficients.

In[*]:= Clear[a]
Clear[c]
If[divergent != {},
    f[x_] := 1 + Sum[a_i x^i, {i, 1, n}] + Sum[Sum[x^(i) a_{i,j} Log[x]^j, {j, 1, i-1}], {i, divergent}] + O[x]^(n+1)
Print["f(x) = ", TraditionalForm[f[x]]]
SetAttributes[c, Constant]

f(x) = 1 + a_1 x + a_2 x^2 + x^3 (a_{3,2} log^2(x) + a_{3,1} log(x) + a_3) + O(x^4)

Compute the new coefficients of f(x)

In[*]:= d := dLimit
h_0[x_] = 1 - x / (f[x]^(d-2)) (1 - (q^2 x) / (mu^2 f[x]^(d-2))) + O[x]^(n+1);
h_1[x_] = f[x]^2 + O[x]^(n+1);
h_2[x_] = -f[x]^2 + (f[x] - (d-2) x f'[x])^2 / (1 - x / (f[x]^(d-2)) (1 - (q^2 x) / (mu^2 f[x]^(d-2)))) + O[x]^(n+1);
gauge = 2 (d-1) h_2[x] - (2-d) x Dt[h_0[x] + (d-2) h_1[x] - h_2[x], x];
If[divergent != {},
    expansion := List[];
    Do[AppendTo[expansion, Coefficient[Normal[2 (d-1) h_2[x]], x^i] ==
        Coefficient[Normal[(2-d) x Dt[h_0[x] + (d-2) h_1[x] - h_2[x], x]], x^i]], {i, 1, n}];
    Do[
        Do[AppendTo[expansion, Coefficient[Normal[2 (d-1) h_2[x]], Log[x]^j x^i] ==
            Coefficient[Normal[(2-d) x Dt[h_0[x] + (d-2) h_1[x] - h_2[x], x]], Log[x]^j x^i]], {j, 1, i-1}],
        {i, divergent}];
    vars := List[];
    Do[If[i >= First[divergent], Do[AppendTo[vars, a_{i,j}], {j, i-1}];
        If[i != First[divergent], AppendTo[vars, a_i], {i, n}];
    solution = Solve[expansion, vars]]

Out[*]:= {{a_1 -> 1/2, a_2 -> 1/2, a_{3,1} -> (8 q^2 - mu^2) / (12 mu^2), a_{3,2} -> 0}}

In[*]:= Print["f(x) = ", f[x] /. solution[[1]]]
If[divergent != {},
    def = Coefficient[Normal[f[x] /. solution[[1]], x^(First[divergent])] /
        Coefficient[Normal[f[x] /. solution[[1]], Log[x] x^(First[divergent])] == Log[x / c];
    Print["Change of variable defined by: ", TraditionalForm[def]];
    s = Solve[def, a_First[divergent]];
    Print["f(x) = ", f[x] /. solution[[1]] /. s[[1]]]

f(x) = 1 + x/2 + x^2/2 + (8 q^2 Log[x] - mu^2 Log[x] + 12 mu^2 a_3) x^3 / (12 mu^2) + O[x]^4

Change of variable defined by: (12 a_3 mu^2 + 8 q^2 log(x) - mu^2 log(x)) / (8 q^2 - mu^2) = log(x/c)

f(x) = 1 + x/2 + x^2/2 + (8 q^2 Log[x/c] - mu^2 Log[x/c]) x^3 / (12 mu^2) + O[x]^4

```

Script 4.2. Script for the computation of $f(x)$ in $d = 3$ with $n = 3$.

Change ansatz on $f(x)$ including logarithms to all the divergent coefficients.

```
In[ ]:= Clear[a]
Clear[c]
If[divergent != {},
  f[x_] := 1 + Sum[a[i] x^i, {i, 1, n}] + Sum[Sum[x^(i) a[i,j] Log[x]^j, {j, 1, i-1}], {i, divergent}] + O[x]^(n+1)
Print["f(x) = ", TraditionalForm[f[x]]]
SetAttributes[c, Constant]

f(x) = 1 + a1 x + x^2 (a2,1 log(x) + a2) + x^3 (a3,2 log^2(x) + a3,1 log(x) + a3) + O(x^4)
```

Compute the new coefficients of $f(x)$

```
In[ ]:= d := dLimit

h0[x_] = 1 - x / (f[x]^(d-2) (1 - (q^2 x) / (mu^2 f[x]^(d-2)))) + O[x]^(n+1);
h1[x_] = f[x]^2 + O[x]^(n+1);
h2[x_] = -f[x]^2 + (f[x] - (d-2) x f'[x])^2 / (1 - x / (f[x]^(d-2) (1 - (q^2 x) / (mu^2 f[x]^(d-2))))) + O[x]^(n+1);

gauge = 2 (d-1) h2[x] == (2-d) x Dt[h0[x] + (d-2) h1[x] - h2[x], x];
If[divergent != {},
  expansion := List[];
  Do[AppendTo[expansion, Coefficient[Normal[2 (d-1) h2[x]], x^i] ==
    Coefficient[Normal[(2-d) x Dt[h0[x] + (d-2) h1[x] - h2[x], x], x^i]], {i, 1, n}];
  Do[
    Do[AppendTo[expansion, Coefficient[Normal[2 (d-1) h2[x]], Log[x]^j x^i] ==
      Coefficient[Normal[(2-d) x Dt[h0[x] + (d-2) h1[x] - h2[x], x], Log[x]^j x^i]], {j, 1, i-1}],
    {i, divergent}];
  vars := List[];
  Do[If[i >= First[divergent], Do[AppendTo[vars, a[i,j]], {j, i-1}]];
  If[i != First[divergent], AppendTo[vars, a[i]], {i, n}];
  solution = Solve[expansion, vars]
```

$$\text{Out[]:= } \left\{ \left\{ a_1 \rightarrow \frac{1}{4}, a_{2,1} \rightarrow \frac{-4 q^2 - 5 \mu^2}{32 \mu^2}, a_{3,1} \rightarrow \frac{3 \times (4 q^2 + 5 \mu^2)}{128 \mu^2}, a_{3,2} \rightarrow 0, a_3 \rightarrow \frac{11 q^2 + \mu^2 - 36 \mu^2 a_2}{48 \mu^2} \right\} \right\}$$

```
In[ ]:= Print["f(x) = ", f[x] /. solution[[1]]]
If[divergent != {},
  def = Coefficient[Normal[f[x] /. solution[[1]], x^(First[divergent])] /
    Coefficient[Normal[f[x] /. solution[[1]], Log[x] x^(First[divergent])] == Log[x/c];
  Print["Change of variable defined by: ", TraditionalForm[def]];
  s = Solve[def, a[First[divergent]]];
  Print["f(x) = ", f[x] /. solution[[1]] /. s[[1]]]

f(x) = 1 + x/4 + ( (-4 q^2 Log[x] - 5 mu^2 Log[x] + 32 mu^2 a2) x^2 ) / (32 mu^2) + ( (88 q^2 + 8 mu^2 + 36 q^2 Log[x] + 45 mu^2 Log[x] - 288 mu^2 a2) x^3 ) / (384 mu^2) + O[x]^4
```

Change of variable defined by: $\frac{32 a_2 \mu^2 - 4 q^2 \log(x) - 5 \mu^2 \log(x)}{-5 \mu^2 - 4 q^2} = \log\left(\frac{x}{c}\right)$

$$f(x) = 1 + \frac{x}{4} + \frac{\left(-4 q^2 \log\left[\frac{x}{c}\right] - 5 \mu^2 \log\left[\frac{x}{c}\right]\right) x^2}{32 \mu^2} + \frac{\left(88 q^2 + 8 \mu^2 + 36 q^2 \log\left[\frac{x}{c}\right] + 45 \mu^2 \log\left[\frac{x}{c}\right]\right) x^3}{384 \mu^2} + O[x]^4$$

Script 4.3. Script for the computation of $f(x)$ in $d = 4$ with $n = 3$.

Conclusions

It has been shown for the first time how to reconstruct the Reissner–Nordström–Tangherlini metric from scattering amplitudes and the renormalized components for fixed dimensions (in four and five dimensions) up to two-loop order. This work extends the work [18] in which this metric has been computed in four dimensions at one-loop order. In the conclusions of [17] the authors pointed out that such confirmation should be computed.

We have shown how the integration over the time component of loop momenta cuts the scalar line of the diagrams to tree-like structures in the static limit [13]. These terms generate the long-distance classical contributions [11, 16] from the rules of unitary cuts [14]. The EFT methods helped us to construct the theory and separate the contributions from light and heavy fields and also to renormalize the theory. It is then possible to obtain the static classical contributions to the metric as described above, at each loop computing the contributions from only tree-like structures of massless internal particles. As prescribed from the EFT methods at each loop order we had to add in the lagrangian all the interacting terms that are invariant under a coordinate transformation. A smart choice of the coupling constants permits to renormalize the metric components and give the finite contributions to each loop order.

Fortunately the problem of the different basis of the *in* and *out* states, expected by the quantization over a general background geometry [29], is not present in our computations since we did a perturbative expansion around flat space-time and the external particles are well defined.

A possible extension of this work could be computing the general dimension Kerr metric. This could be a more important computation from the astrophysical point of view in order to try to describe black holes with these metrics because it is known that the charged case is not the right model for real black holes. The Kerr metric can be obtained by adding a spinorial field. This leads to new propagators and vertices given in literature [18], that has to be included in the construction of the diagrams. Other interesting results can be found by computing the corrections to the propagation of the fields due to the coupling with gravity, (for example to the photon propagation) and trying to match the non-minimal coupling constant that should be in principle the same.

Scattering amplitudes have an analytic dependence on the dimension and give the opportunity to study exotic metrics (such as black holes) in general dimension. These methods give also an alternative to computing classical quantities, such as the gravitational waves. This opens the doors to a new physics of computation of gravitational process that can be more efficient for certain situations [13, 30, 31].

Appendix A

Faddeev-Popov ghosts

A theory with a gauge invariance has bad behaviours and divergences if we try to apply the standard path integral formalism. In the definition of the generating functional, there is the integration over all the possible configurations of the field, including the ones that are equivalent up to a gauge transformation. Thus we integrate over an infinite set of copies of the same physic situation and then obtain divergences. Therefore we need to fix this independence and evaluate the integrand once for all the different gauge orbits. This gauge fixing method is called Faddeev-Popov [10] method and has been introduced in 1967.

Taking the abelian gauge theory with the transformation rule

$$A_\mu \rightarrow A_\mu^{(\Omega)} = A_\mu + \partial_\mu \Omega \quad (\text{A.1})$$

and the gauge fixing condition namely $f(A_\mu) \equiv F(x)$. This method starts from the identity

$$1 = \int \delta\Omega \, \delta\left(f(A_\mu^{(\Omega)}) - F(x)\right) \Delta(A) \quad \text{with} \quad \Delta(A) \equiv \det\left(\frac{\delta f(A^{(\Omega)}(x))}{\delta\Omega(x)}\right),$$

where we have defined the Faddeev-Popov determinant $\Delta(A)$, which in general is invariant in Ω .

Inserting this identity inside the partition function perform the right way of imposing a gauge fixing.

This becomes

$$Z(J_\mu) = \frac{1}{Z(0)} \int \delta\Omega \delta A_\mu \, \delta\left(f(A_\mu^{(\Omega)}) - F(x)\right) \Delta(A) e^{iS}.$$

The above expression does not depend on F , we can therefore insert the Gaussian integral unit in a functional expression

$$1 = N(\xi) \int \delta F e^{-\frac{i}{2\xi} \int d^4x F(x)^2},$$

where ξ is the gauge parameter and $N(\xi)$ the normalization constant. This substitution yield to

$$Z(J_\mu) = \frac{N(\xi)}{Z(0)} \int \delta\Omega \delta A_\mu \delta F \, \delta\left(f(A_\mu^{(\Omega)}) - F(x)\right) \Delta(A) e^{iS - \frac{i}{2\xi} \int d^4x F(x)^2},$$

integrating over Ω and F

$$Z(J_\mu) = \frac{1}{Z(0)} \int \delta A_\mu \Delta(A) e^{iS - \frac{i}{2\xi} \int d^4x f(A_\mu)^2},$$

and we obtain a well defined partition function and the usual gauge fixing term called before $-\frac{1}{2}C_\mu^2$. The reason why $\Delta(A)$ is called determinant is because this term can be seen as an integral over an artificial fermion fields c . The study of the Grassmann variables and in particular the Gaussian integrals in these variables gives

$$\Delta(A) = \det \left(\frac{\delta f}{\delta \Omega} \right) = \int \delta c \delta \bar{c} e^{i \int d^4x \bar{c} \frac{\delta f}{\delta \Omega} c}. \quad (\text{A.2})$$

This field is the so called ghosts field and it does not correspond to any observable particle because any physical asymptotic state is impossible and these particles appear only inside the diagrams.

Appendix B

Sum over the permutations

Taking a vector $(x_1, x_2, \dots, x_l)$ we want to demonstrate the identity

$$\sum_{\sigma \in \mathcal{G}} \prod_{i=1}^{l+1} \frac{1}{\sum_{j=1}^i x_{\sigma(j)}} = \prod_{i=1}^{l+1} \frac{1}{x_i} , \quad (\text{B.1})$$

where $\sum_{\sigma \in \mathcal{G}}$ is the sum over the group $\mathcal{G}$ of all possible permutations of $l+1$ elements. This demonstration has been given by Levy and Sucher [32] in 1969.

Defining

$$\Sigma(l) \equiv \sum_{\sigma \in \mathcal{G}} \prod_{i=1}^{l+1} \frac{1}{\sum_{j=1}^i x_{\sigma(j)}} = \Sigma_l(x) \Big|_{x=0} ,$$

with $\Sigma_l(x)$ defined by

$$\begin{aligned} \Sigma_l(x) &\equiv \sum_{\sigma \in \mathcal{G}} \prod_{i=1}^{l+1} \frac{1}{\sum_{j=1}^i x_{\sigma(j)} + x} \\ &= \sum_{\sigma \in \mathcal{G}} \frac{1}{x_{\sigma(1)} + x} \times \frac{1}{x_{\sigma(1)} + x_{\sigma(2)} + x} \times \dots \times \frac{1}{x_{\sigma(1)} + \dots + x_{\sigma(l)} + x} . \end{aligned}$$

Adding $\frac{1}{x}$ on both sides and using the representation $\frac{1}{x} = \int_0^\infty e^{-\alpha x} d\alpha$ with $\text{Re}(x) > 0$

$$\frac{1}{x} \Sigma_l(x) = \sum_{\sigma \in \mathcal{G}} \int_0^\infty d\alpha_0 \dots \int_0^\infty d\alpha_l e^{-\alpha_0 x} e^{-\alpha_1 (x_{\sigma(1)} + x)} \dots e^{-\alpha_l (x_{\sigma(1)} + \dots + x_{\sigma(l)} + x)} .$$

Defining $\beta \equiv \alpha_0 + \dots + \alpha_l$ and

$$\begin{aligned} \beta_1 &= \alpha_1 + \dots + \alpha_l \\ \beta_2 &= \alpha_2 + \dots + \alpha_l \\ &\vdots \\ \beta_l &= \alpha_l , \end{aligned}$$

it is possible to isolate the terms $x_{\sigma(i)}$ as

$$\frac{1}{x} \Sigma_l(x) = \sum_{\sigma \in \mathcal{G}} \int_{\beta \geq \beta_1 \geq \dots \geq \beta_l \geq 0} d\beta d\beta_1 \dots d\beta_l e^{-\beta x} e^{-\beta_1 x_{\sigma(1)}} \dots e^{-\beta_l x_{\sigma(l)}} .$$

It is possible to make the permutation group act on the part of the variables β introducing the inverse permutation $\pi(j)$ defined by $\beta_{\pi(j)} \equiv \beta_{\sigma^{-1}(j)}$, so that

$$e^{-\beta_1 x_{\sigma(1)}} \dots e^{-\beta_l x_{\sigma(l)}} = e^{-\beta_{\sigma^{-1}(1)} x_1} \dots e^{-\beta_{\sigma^{-1}(l)} x_l} ,$$

and the integral becomes

$$\frac{1}{x} \Sigma_l(x) = \sum_{\sigma \in \mathcal{G}} \int_{\beta \geq \beta_{\sigma^{-1}(1)} \geq \dots \geq \beta_{\sigma^{-1}(l)} \geq 0} d\beta d\beta_{\sigma^{-1}(1)} \dots d\beta_{\sigma^{-1}(l)} e^{-\beta x} e^{-\beta_{\sigma^{-1}(1)} x_1} \dots e^{-\beta_{\sigma^{-1}(l)} x_l} .$$

The integral condition has to hold for every permutation and so we can substitute the sum over all permutations with the condition that for every fixed k , $\beta \geq \beta_{\sigma^{-1}(k)} \equiv \beta_k$. It is possible therefore to split the integrals and integrate over the β_k as

$$\begin{aligned} \frac{1}{x} \Sigma_l(x) &= \int_0^\infty d\beta e^{-\beta x} \int_0^\beta d\beta_1 \dots \int_0^\beta d\beta_l e^{-\beta_1 x_1} \dots e^{-\beta_l x_l} \\ \Sigma_l(x) &= x \int_0^\infty e^{-\beta x} \prod_{i=1}^l \left(\frac{1 - e^{-\beta x_i}}{x_i} \right) \\ &= \int_0^\infty d\beta e^{-\beta x} \frac{\partial}{\partial \beta} \prod_{i=1}^l \left(\frac{1 - e^{-\beta x_i}}{x_i} \right) . \end{aligned} \quad \begin{array}{l} \left. \begin{array}{l} \text{Integrating over } \beta_k \\ \text{Integrating by parts} \end{array} \right\}$$

Now it is immediate to prove B.1, computing the right side and using the previous result noticing that for $x = 0$ the integral simplifies with the derivative giving

$$\Sigma_l(0) = \prod_{i=1}^l \left(\frac{1 - e^{-\beta x_i}}{x_i} \right) \Big|_{\beta=0}^{\beta=\infty} = \prod_{i=1}^l \frac{1}{x_i} ,$$

which is the B.1.

Appendix C

Master Integral

C.1 Master Integral Fourier transform

In order to compute the Fourier transform of the Master Integral the following identities will be useful.

First we compute the Fourier transform

$$F(\alpha, r) \equiv \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} \frac{e^{i\vec{x} \cdot \vec{q}}}{|q|^\alpha} \stackrel{?}{=} \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma\left(\frac{d-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)} \left(\frac{2}{r}\right)^{d-\alpha}. \quad (\text{C.1})$$

In order to solve this integral, we use the proprieties of the Bessel functions and the Hankel transform.

Taking the expression of the Fourier anti-transformation of a general function $f(\mathbf{q})$

$$F(\mathbf{r}) = \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} f(\mathbf{q}) e^{i\vec{x} \cdot \vec{q}}.$$

Rewrite in hyper-spherical coordinates

$$(q_1, q_2, \dots, q_d) \rightarrow (q, \theta_1, \dots, \theta_{d-1}) \quad \text{with} \quad \begin{aligned} q &\in [0, \infty) \\ \theta_{i < d-1} &\in [0, \pi) \\ \theta_{d-1} &\in [0, 2\pi) \end{aligned}$$

(both q and x) the decomposition of the d -dimensional plane wave in hyper-spherical harmonics $Y_{l,m}$ and Bessel functions $J_\nu(qr)$ is [33]

$$e^{i\vec{q} \cdot \vec{x}} = (2\pi)^{\frac{d}{2}} (qr)^{1-\frac{d}{2}} \sum_{l=0}^{\infty} i^l J_{\frac{d}{2}-1+l}(qr) \sum_m Y_{l,m}(\Omega_r) Y_{l,m}^*(\Omega_q),$$

where we summarise all the angle variables in the vector Ω for q and r .
Substituting in the integral

$$F(\mathbf{r}) = (2\pi)^{-\frac{d}{2}} r^{1-\frac{d}{2}} \sum_{l=0}^{\infty} i^l \sum_m Y_{l,m}(\Omega_r) \int_0^\infty dq q^{\frac{d}{2}} J_{\frac{d}{2}-1+l}(qr) \int d\Omega_q f(\mathbf{q}) Y_{l,m}^*(\Omega_q).$$

Expanding also $f(\mathbf{q})$ and $F(\mathbf{r})$ in hyper-spherical coordinates the $Y_{l,m}$ simplifies and we have the expression for the coefficients of the expansion of the form

$$F_{l,m}(r) = (2\pi)^{-\frac{d}{2}} r^{1-\frac{d}{2}} i^l \int_0^\infty q dq q^{\frac{d}{2}-1} f_{l,m}(q) J_{\frac{d}{2}-1+l}(qr) ,$$

we express it in this form (in particular the q in the integrand) in order to explicit the Hankel transform.

In the hypothesis of radial symmetry ($f(\mathbf{q}) = f(q)$) we have the identity

$$F(r) = (2\pi)^{-\frac{d}{2}} r^{1-\frac{d}{2}} \int_0^\infty q dq q^{\frac{d}{2}-1} f(q) J_{\frac{d}{2}-1}(qr) ,$$

which is the Hankel transform (in reality it is the anti-transform) of the function $q^{\frac{d}{2}-1} f(q)$ of order $\frac{d}{2} - 1$.

In particular, the ν th order Hankel transform of the function $g(q)$ defined for $q > 0$ is

$$G_\nu(r) \equiv \mathcal{H}_\nu\{g(q)\} \equiv \int_0^\infty q dq g(q) J_\nu(qr) .$$

The integral we want to solve (C.1) is of the type described with $f(q) = q^{-\alpha}$. It takes therefore the expression

$$F(r, \alpha) = (2\pi)^{-\frac{d}{2}} r^{1-\frac{d}{2}} \mathcal{H}_{\frac{d}{2}-1}\{q^{\frac{d}{2}-\alpha-1}\} . \quad (\text{C.2})$$

The ν -order Hankel transform for $f(r) = r^{-\mu}$ is computed in [34] (Table 9.2 Item (2)). Inverting this identity we obtain the transformation for the general exponent μ

$$\mathcal{H}_\nu\{q^{\mu-2}\} = \frac{2^{\mu-1}}{r^\mu} \frac{\Gamma\left(\frac{\nu+\mu}{2}\right)}{\Gamma\left(\frac{\nu-\mu+2}{2}\right)} .$$

In our case we have $\mu = 1 + \frac{d}{2} - \alpha$ and $\nu = \frac{d}{2} - 1$, inserting inside C.2 we obtain

$$F(\alpha, r) = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma\left(\frac{d-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)} \left(\frac{2}{r}\right)^{d-\alpha} .$$

From this it is possible to compute more complicated Fourier anti-transforms.

Using the identity

$$\partial_i \partial_j (r^2)^\alpha = 2\alpha (r^2)^{\alpha-1} \left(\delta_{ij} + 2(\alpha-1) \frac{x_i x_j}{\vec{x}^2} \right) ,$$

we can obtain the new Fourier transformations deriving the $F(\alpha, r)$ both sides as

$$F_{ij}(\alpha, r) \equiv \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} \frac{q_i q_j}{q^{\alpha+2}} e^{i\vec{x} \cdot \vec{q}} = \frac{1}{\alpha} \left(\delta_{ij} + (\alpha-d) \frac{x_i x_j}{\vec{x}^2} \right) F(\alpha) .$$

In particular, we see that $F(0, r) = 0$ and so the only terms that will contribute are of the form

$$F_{ij}(0, r) = \left(\delta_{ij} - d \frac{x_i x_j}{\vec{x}^2} \right) \frac{\Gamma\left(\frac{d}{2}\right)}{2\pi^{\frac{d}{2}} r^d} .$$

From these identities it is straightforward to obtain the Master Integral transformations substituting the expression [17] (Equation (2.31))

$$J^{(l)}(q^2) = \frac{(q^2)^{\frac{l(d-2)}{2}}}{(4\pi)^{\frac{ld}{2}}} \frac{\Gamma(l+1 - \frac{ld}{2})\Gamma(\frac{d-2}{2})^{l+1}}{\Gamma\left(\frac{(l+1)(d-2)}{2}\right)}.$$

The results obtained are:

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q}\cdot\vec{x}} \frac{J^{(l)}(q^2)}{q^2} &= \left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{4\pi^{d/2} r^{d-2}} \right)^{l+1} \equiv \left(\frac{4\rho(r, d)}{\kappa^2} \right)^{l+1} \\ \int_{\mathbb{R}^d} \frac{d^d q}{(2\pi)^d} e^{i\vec{q}\cdot\vec{x}} \frac{q_\mu q_\nu}{q^4} J^{(l)}(q^2) &= \left(\frac{\Gamma\left(\frac{d-2}{2}\right)}{4\pi^{d/2} r^{d-2}} \right)^{l+1} \frac{1}{2-l(d-2)} \left(-\delta_{ij} + (l+1)(d-2) \frac{x_i x_j}{r^2} \right). \end{aligned}$$

C.2 Master Integral identities

Taking the following integral

$$I(q) \equiv \int \frac{d^d l}{(2\pi)^d} \frac{1}{l^2(l-q)^2}$$

we can construct the one-loop master integral as $J_1(q^2) = q^2 I(q)$.

Can be show that the integral

$$I_\mu(q) \equiv \int \frac{d^d l}{(2\pi)^d} \frac{l_\mu}{l^2(l-q)^2}$$

gives a contribution to the master integral performing the substitution $l + \frac{q}{2}$ as

$$I_\mu(q) = \int \frac{d^d l}{(2\pi)^d} \frac{(l + \frac{q}{2})_\mu}{(l + \frac{q}{2})^2(l - \frac{q}{2})^2} = \int \frac{d^d l}{(2\pi)^d} \frac{l_\mu}{(l + \frac{q}{2})^2(l - \frac{q}{2})^2} + \frac{1}{2} q_\mu I(q).$$

The first term of the last equation is null because the integrand is an odd function in l and the integral is performed over a symmetric domain.

Therefore this integral can be rewritten as

$$I_\mu(q) = \frac{1}{2} q_\mu I(q),$$

and the integrals appearing in the scattering computations gives the contribution to the master integral

$$\int \frac{d^d l}{(2\pi)^d} \frac{q \cdot l}{l^2(l-q)^2} = q_\mu I_\mu(q) = \frac{1}{2} J_1(q^2).$$

For the more tricky integral defined by

$$I_{\mu\nu}(q) \equiv \int \frac{d^d l}{(2\pi)^d} \frac{l_\mu l_\nu}{l^2(l-q)^2}$$

we perform the decomposition

$$I_{\mu\nu}(q) \equiv A q_\mu q_\nu I(q) + B \eta_{\mu\nu} q^2 I(q) .$$

We can then obtain the coefficients A and B computing the following projections:

$$\begin{cases} \eta^{\mu\nu} I_{\mu\nu} = \int \frac{d^d l}{(2\pi)^d} \frac{1}{(l-q)^2} = 0 \\ q^\mu q^\nu I_{\mu\nu} = \int \frac{d^d l}{(2\pi)^d} \frac{(q \cdot l)^2}{l^2 (l-q)^2} = \frac{1}{4} \int \frac{d^d l}{(2\pi)^d} \frac{((q-l)^2 - q^2 - l^2)^2}{l^2 (l-q)^2} = \frac{1}{4} q^4 I(q) . \end{cases}$$

From the decomposition of $I_{\mu\nu}$

$$\begin{cases} \eta^{\mu\nu} I_{\mu\nu} = A + (d+1)B \\ q^\mu q^\nu I_{\mu\nu} = (A+B)q^4 I(q) . \end{cases}$$

Solving for the coefficients we find $A = \frac{d+1}{d}$ and $B = -\frac{1}{4d}$, that gives

$$I_{\mu\nu} = -\frac{1}{4d} \left[q^2 \eta^{\mu\nu} - (d+1) q^\mu q^\nu \right] I(q) .$$

Lite-Red

For the two-loop integrals the previous algebraic methods can be performed on certain types of integrals, as it is shown in the scattering computations, but other types of integral can not be easily rewritten to explicit the contributions to the master integral. One of these is the following integral

$$\int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{q \cdot l_1 q \cdot l_2}{l_1^2 l_2^2 (l_1 + l_2)^2 (l_1 + l_2 - q)^2} ,$$

found in the two-loop diagram with the 3graviton vertex ($\textcircled{c}$). It is evident that this integral has contribution proportional to $J_2(q^2)$ but the denominator expression makes impossible to use symmetries or expansions of the scalar products.

In order to perform this integral we should derive the master integral both sides and obtain identities that can be used to re-express the integral [27].

The package of *Mathematica* called **LiteRed**¹ performs these computations. In the Script C.1 we loaded the package and obtain all the necessary identities from the master integral defined by all the independent momenta (the momenta of the two-loop diagram $\textcircled{c}$). In the second script (Script C.2) the actual computation to obtain C.5 is performed.

¹www.inp.nsk.su/~lee/programs/LiteRed/

Finding reduction 2LOOP

Out[*]= Finding reduction rules and saving

```

In[*]:= << LiteRed'; (*Loading the package*)
SetDirectory[NotebookDirectory[]];
(*Setting working directory*)
SetDim[d]; (*d stands for the dimensionality*)
Declare[{l1, l2, q}, Vector];
(*vector variables*)

In[*]:= NewBasis[v2, {l1 - q, l2 + l1, l1, l2, l1 + l2 - q}, {l1, l2}, Directory -> "v2 dir"]
(*Basis definition. A vector in the second argument is a shortcut for massless propagator, e.g. p-l shortcuts sp[p-l,p-l],
third option is the directory where all definitions for the basis are saved*)
GenerateIBP[v2] (*IBP generation. Can be also invoked by passing GenerateIBP->True option in NewBasis call*)
AnalyzeSectors[v2] (*Zero and simple sectors determination. Can be also invoked by passing AnalyzeSectors->True option in NewBasis call.*)
FindSymmetries[v2] (*Finding unique and mapped sectors. Can be also invoked by passing FindSymmetries->True option in NewBasis call.*)

In[*]:= SolvejSector /@ UniqueSectors[v2] (*Solving IBP identities for unique sectors.*)

In[*]:= DiskSave[v2] (*Saving to default directory.*)
Quit[] (*Quitting the kernel*)

```

Script C.1. Script for the computation of master integral identities.

Loading the basis and reducing

```

In[*]:= << LiteRed'; (*Loading the package*)
SetDirectory[NotebookDirectory[]];
(*Setting working directory*)
SetDim[d]; (*d stands for the dimensionality*)
Declare[{l1, l2, q}, Vector]; (*vector variables*)
<< "v2 dir/v2";

In[*]:= jnum := Toj[v2, j[v2, 0, 1, 1, 1, 1] × sp[l2, q] × sp[l1, q]]
IBPReduce[jnum] // Factor
Out[*]= 
$$\frac{(-7 + 2d) j[v2, 0, 0, 1, 1, 1] q \cdot q}{6 \times (-4 + d)}$$


```

Script C.2. Script for the computation of the integral C.5

Useful identities

The following identities are useful in the scattering computations, these have been computed analytically when possible and with LiteRed when necessary.

$$\int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{l_1 \cdot l_2}{l_1^2 l_2^2 (l_1 + l_2 - q)^2} = \frac{1}{6} J_2(q^2) \quad (C.3)$$

$$\int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{q \cdot l_1 l_1 \cdot l_2}{l_1^2 l_2^2 (l_1 + l_2)^2 (l_1 + l_2 - q)^2} = \frac{1}{6} J_2(q^2) \quad (C.4)$$

$$\int \frac{d^d l_1}{(2\pi)^d} \frac{d^d l_2}{(2\pi)^d} \frac{q \cdot l_1 q \cdot l_2}{l_1^2 l_2^2 (l_1 + l_2)^2 (l_1 + l_2 - q)^2} = \frac{2d - 7}{6(d - 4)} J_2(q^2) \quad (C.5)$$

Appendix D

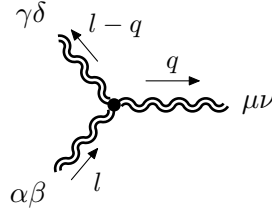
Three Graviton vertex

In this chapter we will obtain the general expression of the three graviton vertex contracted with two graviton propagators, namely

$$\tilde{\tau}_{\alpha\beta\gamma\delta}^{\mu\nu} \equiv P_{\alpha\beta}{}^{ab} P_{\gamma\delta}{}^{cd} \tau_{abcd}^{\mu\nu} , \quad (\text{D.1})$$

which will be useful for different amplitudes computations.

Taking the expression of the vertex given in [11]:



$$= \tau_{\alpha\beta\gamma\delta}^{\mu\nu}(l, q) =$$

$$\begin{aligned} &= \frac{i\kappa}{2} \left\{ P_{\alpha\beta\gamma\delta}^{(4)} \left[l^\mu l^\nu + (l - q)^\mu (l - q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] \right. \\ &\quad + 2q_\lambda q_\sigma \left[I_{\alpha\beta}{}^{\lambda\sigma} I_{\gamma\delta}{}^{\mu\nu} + I_{\gamma\delta}{}^{\lambda\sigma} I_{\alpha\beta}{}^{\mu\nu} - I_{\alpha\beta}{}^{\lambda\mu} I_{\gamma\delta}{}^{\sigma\nu} - I_{\alpha\beta}{}^{\sigma\nu} I_{\gamma\delta}{}^{\lambda\mu} \right] \\ &\quad + q_\lambda q^\mu \left(\eta_{\alpha\beta} I_{\gamma\delta}{}^{\lambda\nu} + \eta_{\gamma\delta} I_{\alpha\beta}{}^{\lambda\nu} \right) + q_\lambda q^\nu \left(\eta_{\alpha\beta} I_{\gamma\delta}{}^{\lambda\mu} + \eta_{\gamma\delta} I_{\alpha\beta}{}^{\lambda\mu} \right) \\ &\quad - q^2 \left(\eta_{\alpha\beta} I_{\gamma\delta}{}^{\mu\nu} + \eta_{\gamma\delta} I_{\alpha\beta}{}^{\mu\nu} \right) - \eta^{\mu\nu} q^\lambda q^\sigma \left(\eta_{\alpha\beta} I_{\gamma\delta}{}^{\lambda\sigma} + \eta_{\gamma\delta} I_{\alpha\beta}{}^{\lambda\sigma} \right) \\ &\quad + 2q^\lambda \left[I_{\alpha\beta}{}^{\sigma\lambda} I_{\gamma\delta}{}^{\sigma\nu} (l - q)^\mu + I_{\alpha\beta}{}^{\lambda\sigma} I_{\gamma\delta}{}^{\sigma\mu} (l - q)^\nu - I_{\gamma\delta}{}^{\lambda\sigma} I_{\alpha\beta}{}^{\sigma\nu} l^\mu - I_{\gamma\delta}{}^{\lambda\sigma} I_{\alpha\beta}{}^{\sigma\mu} l^\nu \right] \\ &\quad + q^2 \left(I_{\alpha\beta}{}^{\sigma\mu} I_{\gamma\delta}{}^{\sigma\nu} + I_{\alpha\beta}{}^{\sigma\nu} I_{\gamma\delta}{}^{\sigma\mu} \right) + \eta^{\mu\nu} q^\lambda q_\sigma \left(I_{\gamma\delta}{}^{\rho\sigma} I_{\alpha\beta}{}^{\lambda\rho} + I_{\alpha\beta}{}^{\rho\sigma} I_{\gamma\delta}{}^{\lambda\rho} \right) \\ &\quad + (l^2 + (l - q)^2) \left(I_{\alpha\beta}{}^{\sigma\mu} I_{\gamma\delta}{}^{\sigma\nu} + I_{\gamma\delta}{}^{\sigma\mu} I_{\alpha\beta}{}^{\sigma\nu} - \frac{1}{2} \eta^{\mu\nu} P_{\alpha\beta\gamma\delta}^{(4)} \right) \\ &\quad \left. - l^2 \eta_{\alpha\beta} I_{\gamma\delta}{}^{\mu\nu} - (l - q)^2 \eta_{\alpha\beta\gamma\delta} I_{\alpha\beta}{}^{\mu\nu} \right\} , \quad (\text{D.2}) \end{aligned}$$

where the tensor $I_{\alpha\beta\gamma\delta}$ is

$$I_{\mu\nu\alpha\beta} = \frac{1}{2} \eta_{\mu\alpha} \eta_{\nu\beta} + \frac{1}{2} \eta_{\mu\beta} \eta_{\nu\alpha} ,$$

we will compute the $\tilde{\tau}_{0000}^{\mu\nu}$ component given in [17] and then it will be found the general expression of this vertex and it's consistency comparing the two results.

Being the tensor part of the graviton propagator in the de Donder gauge in D dimensions

$$P_{\mu\nu\alpha\beta} \equiv \frac{1}{2}\eta_{\mu\alpha}\eta_{\nu\beta} + \frac{1}{2}\eta_{\mu\beta}\eta_{\nu\alpha} - \frac{1}{D-2}\eta_{\mu\nu}\eta_{\alpha\beta} ,$$

then we have

$$\begin{aligned} \tilde{\tau}_{0000}^{\mu\nu}(l, q) &\equiv P_{00}^{\alpha\beta} P_{00}^{\gamma\delta} \tau_{\alpha\beta\gamma\delta}^{\mu\nu} = \left(\delta_0^\alpha \delta_0^\beta - \frac{1}{D-2} \eta^{\alpha\beta} \right) \left(\delta_0^\gamma \delta_0^\delta - \frac{1}{D-2} \eta^{\gamma\delta} \right) \tau_{\alpha\beta\gamma\delta}^{\mu\nu} = \\ &= \underbrace{\delta_0^\alpha \delta_0^\beta \delta_0^\gamma \delta_0^\delta \tau_{\alpha\beta\gamma\delta}^{\mu\nu}}_{\text{(a)}} - \underbrace{\frac{1}{D-2} \delta_0^\alpha \delta_0^\beta \eta^{\gamma\delta} \tau_{\alpha\beta\gamma\delta}^{\mu\nu}}_{\text{(b)}} - \underbrace{\frac{1}{D-2} \delta_0^\gamma \delta_0^\delta \eta^{\alpha\beta} \tau_{\alpha\beta\gamma\delta}^{\mu\nu}}_{\text{(c)}} + \underbrace{\frac{1}{(D-2)^2} \eta^{\alpha\beta} \eta^{\gamma\delta} \tau_{\alpha\beta\gamma\delta}^{\mu\nu}}_{\text{(d)}} . \end{aligned}$$

The proprieties needed for the calculations are

$$P_{0000}^{(4)} = \frac{1}{2}, \quad I_{00}^{\mu\nu} = \delta_0^\mu \delta_0^\nu, \quad I_{00\lambda}^\mu I_{00}^{\lambda\nu} = \delta_0^\mu \delta_0^\nu .$$

From the expression of the vertex τ we have

$$\text{(a)} = \tau_{0000}^{\mu\nu} = -\frac{i\kappa}{2} \left\{ \frac{1}{2} \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] + (l^2 + (l-q)^2) \left(\delta_0^\mu \delta_0^\nu - \frac{1}{4} \eta^{\mu\nu} \right) \right\} .$$

For the second term will be useful the identity $\eta^{\gamma\delta} P_{00\gamma\delta}^{(4)} = \frac{2-D}{2}$,

$$\begin{aligned} \text{(b)} &= -\frac{1}{D-2} \eta^{\gamma\delta} \tau_{00\gamma\delta}^{\mu\nu} = \\ &= \frac{i\kappa}{2} \frac{1}{D-2} \left\{ \frac{2-D}{2} \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] + (4-D) q^2 \delta_0^\mu \delta_0^\nu + 2q^\mu q^\nu - 2q^2 \eta^{\mu\nu} + \right. \\ &\quad \left. + (l^2 + (l-q)^2) \left(\delta_0^\mu \delta_0^\nu - \frac{1}{4} \eta^{\mu\nu} \right) - l^2 \eta^{\mu\nu} - D(l-q)^2 \delta_0^\mu \delta_0^\nu \right\} . \end{aligned}$$

The third term is similar to the second because every part of the vertex τ is symmetric under the exchange of $\alpha\beta \leftrightarrow \gamma\delta$ less then the last row. Therefore the third term is

$$\begin{aligned} \text{(c)} &= -\frac{1}{D-2} \eta^{\alpha\beta} \tau_{00\alpha\beta}^{\mu\nu} = \\ &= \frac{i\kappa}{2} \frac{1}{D-2} \left\{ \frac{2-D}{2} \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] + (4-D) q^2 \delta_0^\mu \delta_0^\nu + 2q^\mu q^\nu - 2q^2 \eta^{\mu\nu} + \right. \\ &\quad \left. + (l^2 + (l-q)^2) \left(\delta_0^\mu \delta_0^\nu - \frac{1}{4} \eta^{\mu\nu} \right) - D l^2 \delta_0^\mu \delta_0^\nu - (l-q)^2 \eta^{\mu\nu} \right\} . \end{aligned}$$

For the last term is needed the identity $\eta^{\alpha\beta} \eta^{\gamma\delta} P_{\alpha\beta\gamma\delta}^{(4)} = \frac{D(2-D)}{2}$,

$$\begin{aligned} \text{(d)} &= \frac{1}{(D-2)^2} \eta^{\alpha\beta} \eta^{\gamma\delta} \tau_{\alpha\beta\gamma\delta}^{\mu\nu} = \\ &= -\frac{i\kappa}{2} \frac{1}{(D-2)^2} \left\{ \frac{D(2-D)}{2} \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] + (8-4D)(q^\mu q^\nu - q^2 \eta^{\mu\nu}) + \right. \\ &\quad \left. + \frac{1}{4} (D-2)^2 (l^2 + (l-q)^2) \eta^{\mu\nu} - D l^2 \eta^{\mu\nu} - D(l-q)^2 \eta^{\mu\nu} \right\} . \end{aligned}$$

Summing all the contributions computed above we found the expression [17]

$$\begin{aligned} \tilde{\tau}_{0000}^{\mu\nu}(l, q) &= \frac{i\kappa}{2} \frac{1}{D-2} \left[(D-3) \left(l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu + \frac{3}{2} \eta^{\mu\nu} q^2 \right) - \right. \\ &\quad \left. - 2(D-3) \left(\vec{l}^2 + (\vec{l} + \vec{q})^2 \right) \left(\delta_0^\mu \delta_0^\nu - \frac{\eta^{\mu\nu}}{4} \right) - 2(D-4) \vec{q}^2 \delta_0^\mu \delta_0^\nu \right] . \quad (\text{D.3}) \end{aligned}$$

This result will be useful as a consistency check for the general expression of $\tilde{\tau}$.

Using the following proprieties:

- $P_{\alpha\beta\gamma\delta}P_{(4)}^{\gamma\delta\mu\nu} = I^{\mu\nu}_{\alpha\beta},$
- $\eta^{\alpha\beta}P_{\alpha\beta\mu\nu} = -\frac{2}{D-2}\eta_{\mu\nu}$
- $P_{\alpha\beta\gamma\delta}I^{\gamma\delta}_{\mu\nu} = P_{\alpha\beta\mu\nu}$
- $P_{\alpha\beta\gamma\delta}P_{(4)}^{\gamma\delta\rho\sigma}P_{\rho\sigma\mu\nu} = P_{\alpha\beta\mu\nu} ,$

is possible to compute the general expression of the vertex. The result obtained is

$$\begin{aligned}
\tilde{\tau}_{\alpha\beta\gamma\delta}^{\mu\nu}(l, q) &\equiv P_{\alpha\beta}^{ab}P_{\gamma\delta}^{cd}\tau_{abcd}^{\mu\nu}(l, q) = \\
&= \frac{i\kappa}{2} \left\{ P_{\alpha\beta\gamma\delta} \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2}\eta^{\mu\nu} q^2 \right] \right. \\
&\quad + 2q_\lambda q_\sigma \left[P_{\alpha\beta}^{\lambda\sigma} P_{\gamma\delta}^{\mu\nu} + P_{\gamma\delta}^{\lambda\sigma} P_{\alpha\beta}^{\mu\nu} - P_{\alpha\beta}^{\lambda\mu} P_{\gamma\delta}^{\sigma\nu} - P_{\alpha\beta}^{\sigma\nu} P_{\gamma\delta}^{\lambda\mu} \right] \\
&\quad - \frac{2}{D-2} q_\lambda q^\mu \left(\eta_{\alpha\beta} P_{\gamma\delta}^{\lambda\nu} + \eta_{\gamma\delta} P_{\alpha\beta}^{\lambda\nu} \right) - \frac{2}{D-2} q_\lambda q^\nu \left(\eta_{\alpha\beta} P_{\gamma\delta}^{\lambda\mu} + \eta_{\gamma\delta} P_{\alpha\beta}^{\lambda\mu} \right) \\
&\quad + \frac{2}{D-2} q^2 \left(\eta_{\alpha\beta} P_{\gamma\delta}^{\mu\nu} + \eta_{\gamma\delta} P_{\alpha\beta}^{\mu\nu} \right) + \frac{2}{D-2} \eta^{\mu\nu} q^\lambda q^\sigma \left(\eta_{\alpha\beta} P_{\gamma\delta}^{\lambda\sigma} + \eta_{\gamma\delta} P_{\alpha\beta}^{\lambda\sigma} \right) \\
&\quad + 2q^\lambda \left[P_{\alpha\beta}^{\sigma\lambda} P_{\gamma\delta}^{\mu\nu} (l-q)^\mu + P_{\alpha\beta}^{\lambda\sigma} P_{\gamma\delta}^{\mu\nu} (l-q)^\nu - P_{\gamma\delta}^{\lambda\sigma} P_{\alpha\beta}^{\mu\nu} l^\mu - P_{\gamma\delta}^{\lambda\sigma} P_{\alpha\beta}^{\mu\nu} l^\nu \right] \\
&\quad + q^2 \left(P_{\alpha\beta}^{\sigma\mu} P_{\gamma\delta}^{\nu\sigma} + P_{\alpha\beta}^{\nu\sigma} P_{\gamma\delta}^{\sigma\mu} \right) + \eta^{\mu\nu} q^\lambda q_\sigma \left(P_{\gamma\delta}^{\rho\sigma} P_{\alpha\beta}^{\lambda\rho} + P_{\alpha\beta}^{\rho\sigma} P_{\gamma\delta}^{\lambda\rho} \right) \\
&\quad + (l^2 + (l-q)^2) \left(P_{\alpha\beta}^{\sigma\mu} P_{\gamma\delta}^{\nu\sigma} + P_{\alpha\beta}^{\nu\sigma} P_{\gamma\delta}^{\sigma\mu} - \frac{1}{2}\eta^{\mu\nu} P_{\alpha\beta\gamma\delta} \right) \\
&\quad \left. + \frac{2}{D-2} \left(l^2 \eta_{\alpha\beta} P_{\gamma\delta}^{\mu\nu} + (l-q)^2 \eta_{\gamma\delta} P_{\alpha\beta}^{\mu\nu} \right) \right\} . \tag{D.4}
\end{aligned}$$

It is possible to check the consistency of this expression setting to zero all the lower indices and compare with the vertex D.3. For this check the following proprieties have been used:

$$P_{0000} = \frac{D-3}{D-2} \quad \text{and} \quad P_{00\mu\lambda}P_{00\lambda\nu} = \frac{D-4}{D-2}\delta_{0\mu}\delta_{0\nu} + \frac{1}{(D-2)^2}\eta_{\mu\nu} ,$$

and the result obtained is in agree with D.3 [17], giving us the hope that all these long computations led us to a correct result.

The following components of the vertex D.4 are used in the computations:

$$\begin{aligned}
\tilde{\tau}_{00\gamma\delta}^{\mu\nu}(l, q) &= \frac{i\kappa}{2} \left\{ \left(\delta_\gamma^0 \delta_\delta^0 - \frac{1}{D-2}\eta_{\gamma\delta} \right) \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2}\eta^{\mu\nu} q^2 \right] \right. \\
&\quad + 2q_\lambda q_\sigma P_{\gamma\delta}^{\sigma\lambda} \delta_0^\mu \delta_0^\nu + \frac{4}{(D-2)^2} q^\mu q^\nu \eta_{\gamma\delta} + \frac{2}{D-2} q^2 \eta_{\gamma\delta} \delta_0^\mu \delta_0^\nu - \frac{4}{(D-2)^2} q^2 \eta^{\mu\nu} \eta_{\gamma\delta} \\
&\quad + \frac{2}{D-2} q^\lambda q^\mu P_{\gamma\delta}^{\lambda\nu} + \frac{2}{D-2} q^\lambda q^\nu P_{\gamma\delta}^{\lambda\mu} - 2q_\lambda l^\mu P_{\gamma\delta}^{\lambda\sigma} \delta_0^\nu - 2q_\lambda l^\nu P_{\gamma\delta}^{\lambda\sigma} \delta_0^\mu \\
&\quad + q^2 P_{\gamma\delta}^{\sigma\lambda} \delta_0^\mu + q^2 P_{\gamma\delta}^{\sigma\lambda} \delta_0^\nu - \frac{2}{D-2} q^2 P_{\gamma\delta}^{\mu\nu} - \frac{2}{D-2} q_\sigma q_\lambda \eta^{\mu\nu} P_{\gamma\delta}^{\sigma\lambda} \\
&\quad + (l^2 + (l-q)^2) \left[P_{\gamma\delta}^{\sigma\mu} \delta_0^\nu + P_{\gamma\delta}^{\sigma\mu} \delta_0^\mu - \frac{2}{D-2} P_{\gamma\delta}^{\mu\nu} - \frac{1}{2}\delta_\gamma^0 \delta_\delta^0 \eta^{\mu\nu} + \frac{1}{2(D-2)} \eta_{\gamma\delta} \eta^{\mu\nu} \right] \\
&\quad \left. + \frac{2}{D-2} l^2 P_{\gamma\delta}^{\mu\nu} + \frac{2}{D-2} (l-q)^2 \delta_0^\mu \delta_0^\nu \eta_{\gamma\delta} - \frac{2}{(D-2)^2} (l-q)^2 \eta^{\mu\nu} \eta_{\gamma\delta} \right\} . \tag{D.5}
\end{aligned}$$

$$\begin{aligned}
\tilde{\tau}_{00\gamma\delta}^{00}(l, q) &= \frac{i\kappa}{2} \left\{ \frac{1}{2(D-2)} \left[(D-6)q^2 + (3D-6)l^2 + (3D-10)(l-q)^2 \right] \delta_\gamma^0 \delta_\delta^0 \right. \\
&\quad + \frac{1}{2(D-2)^2} \left[-(D-2)q^2 - (3D-6)l^2 + (D-2)(l-q)^2 \right] \eta_{\gamma\delta} \\
&\quad \left. + \frac{2D-6}{D-2} q_\lambda q_\delta \right\} . \tag{D.6}
\end{aligned}$$

$$\begin{aligned}
\tilde{\tau}_{00\gamma\delta}^{ij}(l, q) = & \frac{i\kappa}{2} \left\{ \left(\delta_\gamma^0 \delta_\delta^0 - \frac{1}{D-2} \eta_{\gamma\delta} \right) \left[l^i l^j + (l-q)^i (l-q)^j + q^i q^j + \frac{3}{2} \delta^{ij} q^2 \right] + \frac{4}{(D-2)^2} q^i q^j \eta_{\gamma\delta} \right. \\
& + \frac{4}{(D-2)^2} q^2 \delta^{ij} \eta_{\gamma\delta} + \frac{2}{D-2} q^\lambda q^i P_{\gamma\delta\lambda}{}^j + \frac{2}{D-2} q^\lambda q^j P_{\gamma\delta\lambda}{}^i - \frac{2}{D-2} q^2 P_{\gamma\delta}{}^{ij} + \frac{2}{D-2} q_\sigma q_\lambda \delta^{ij} P_{\gamma\delta}{}^{\sigma\lambda} \\
& + (l^2 + (l-q)^2) \left[-\frac{2}{D-2} P_{\gamma\delta}{}^{ij} + \frac{1}{2} \delta_\gamma^0 \delta_\delta^0 \delta^{ij} - \frac{1}{2(D-2)} \eta_{\gamma\delta} \delta^{ij} \right] \\
& \left. + \frac{2}{D-2} l^2 P_{\gamma\delta}{}^{ij} + \frac{2}{(D-2)^2} (l-q)^2 \delta^{ij} \eta_{\gamma\delta} \right\} . \tag{D.7}
\end{aligned}$$

Appendix E

Feynman Rules

- Scalar propagator of mass m :

$$\text{---}\overset{q}{\text{---}} = \frac{1}{q^2 - m^2 + i\epsilon} .$$

- Photon propagator in the Feynman gauge [35]:

$$\mu \text{---}\overset{q}{\text{---}} \nu = -\frac{i\eta^{\mu\nu}}{q^2 + i\epsilon} .$$

- Graviton propagator derived in the first chapters: with $P_{\alpha\beta\mu\nu}$ defined by

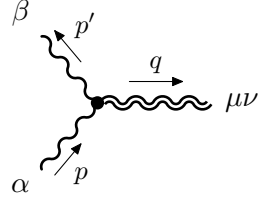
$$\alpha\beta \text{---}\overset{q}{\text{---}} \mu\nu = i\frac{P_{\alpha\beta\mu\nu}}{q^2 + i\epsilon} .$$

$$P_{\alpha\beta\mu\nu} \equiv \frac{1}{2} \left(\eta^{\mu\alpha} \eta^{\nu\beta} + \eta^{\mu\beta} \eta^{\nu\alpha} - \frac{2}{D-2} \eta^{\mu\nu} \eta^{\alpha\beta} \right) . \quad (\text{E.1})$$

- The 2scalar-1graviton vertex is given by:

$$\begin{aligned} \text{Diagram: } & \text{Two incoming lines with momenta } p \text{ and } p' \text{ meet at a vertex, from which a wavy line with momentum } q \text{ and indices } \mu\nu \text{ emerges.} \\ & = \tau_{\mu\nu}(p, p', m) = \quad (\text{E.2}) \\ & = -\frac{i\kappa}{2} \left[(p_\mu p'_\nu + p_\nu p'_\mu) - \eta_{\mu\nu} (p \cdot p' - m^2) \right] . \end{aligned}$$

- The two photons one graviton vertex has the expression:

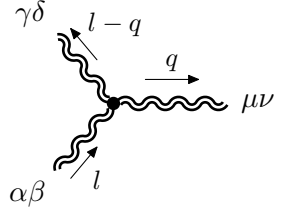


$$= \tau_{\mu\nu\alpha\beta}(p, p') =$$

$$= i\kappa \left\{ P_{\mu\nu\alpha\beta}^{(4)}(p \cdot p') + \frac{1}{2} \left[\frac{1}{2} \eta_{\mu\nu} (p_\alpha p'_\beta + p_\beta p'_\alpha) + \eta_{\alpha\beta} (p_\mu p'_\nu + p_\nu p'_\mu) \right. \right. \quad (E.3)$$

$$\left. \left. - \frac{1}{2} \left(\eta_{\beta\nu} (p_\mu p'_\alpha + p_\alpha p'_\mu) + \eta_{\alpha\nu} (p_\mu p'_\beta + p_\beta p'_\mu) + \eta_{\alpha\mu} (p_\nu p'_\beta + p_\beta p'_\nu) + \eta_{\beta\mu} (p_\nu p'_\alpha + p_\alpha p'_\nu) \right) \right] \right\}.$$

- The 3graviton vertex given by Donoghue [11] has the expression:



$$= \tau_{\alpha\beta\gamma\delta}^{\mu\nu}(l, q) =$$

$$= \frac{i\kappa}{2} \left\{ P_{\alpha\beta\gamma\delta}^{(4)} \left[l^\mu l^\nu + (l-q)^\mu (l-q)^\nu + q^\mu q^\nu - \frac{3}{2} \eta^{\mu\nu} q^2 \right] \right.$$

$$+ 2q_\lambda q_\sigma \left[I_{\alpha\beta}^{\lambda\sigma} I_{\gamma\delta}^{\mu\nu} + I_{\gamma\delta}^{\lambda\sigma} I_{\alpha\beta}^{\mu\nu} - I_{\alpha\beta}^{\lambda\mu} I_{\gamma\delta}^{\sigma\nu} - I_{\alpha\beta}^{\sigma\nu} I_{\gamma\delta}^{\lambda\mu} \right]$$

$$+ q_\lambda q^\mu \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\lambda\nu} + \eta_{\gamma\delta} I_{\alpha\beta}^{\lambda\nu} \right) + q_\lambda q^\nu \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\lambda\mu} + \eta_{\gamma\delta} I_{\alpha\beta}^{\lambda\mu} \right)$$

$$- q^2 \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\mu\nu} + \eta_{\gamma\delta} I_{\alpha\beta}^{\mu\nu} \right) - \eta^{\mu\nu} q^\lambda q^\sigma \left(\eta_{\alpha\beta} I_{\gamma\delta}^{\lambda\sigma} + \eta_{\gamma\delta} I_{\alpha\beta}^{\lambda\sigma} \right)$$

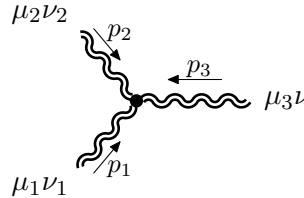
$$+ 2q^\lambda \left[I_{\alpha\beta}^{\sigma\lambda} I_{\gamma\delta}^{\mu\nu} (l-q)^\mu + I_{\alpha\beta}^{\lambda\sigma} I_{\gamma\delta}^{\mu\nu} (l-q)^\nu - I_{\gamma\delta}^{\lambda\sigma} I_{\alpha\beta}^{\mu\nu} l^\mu - I_{\gamma\delta}^{\lambda\sigma} I_{\alpha\beta}^{\mu\nu} l^\nu \right]$$

$$+ q^2 \left(I_{\alpha\beta}^{\sigma\mu} I_{\gamma\delta}^{\nu\sigma} + I_{\alpha\beta}^{\nu\sigma} I_{\gamma\delta}^{\mu\sigma} \right) + \eta^{\mu\nu} q^\lambda q_\sigma \left(I_{\gamma\delta}^{\rho\sigma} I_{\alpha\beta}^{\lambda\rho} + I_{\alpha\beta}^{\rho\sigma} I_{\gamma\delta}^{\lambda\rho} \right)$$

$$+ (l^2 + (l-q)^2) \left(I_{\alpha\beta}^{\sigma\mu} I_{\gamma\delta}^{\nu\sigma} + I_{\alpha\beta}^{\sigma\nu} I_{\gamma\delta}^{\mu\sigma} - \frac{1}{2} \eta^{\mu\nu} P_{\alpha\beta\gamma\delta}^{(4)} \right)$$

$$\left. - l^2 \eta_{\alpha\beta} I_{\gamma\delta}^{\mu\nu} - (l-q)^2 \eta_{\alpha\beta\gamma\delta} I_{\alpha\beta}^{\mu\nu} \right\}. \quad (E.4)$$

- The 3graviton vertex derived by DeWitt [5] is:



$$= \tau^{\mu_1\nu_1\mu_2\nu_2\mu_3\nu_3}(p_1, p_2, p_3) =$$

$$= \frac{i\kappa}{2} \text{Sym} \left[-\frac{1}{4} P_3(p \cdot p' \eta^{\mu_1\nu_1} \eta^{\mu_2\nu_2} \eta^{\mu_3\nu_3}) - \frac{1}{4} P_6(p^{\mu_2} p'^{\nu_2} \eta^{\mu_1\nu_1} \eta^{\mu_3\nu_3}) + \frac{1}{4} P_3(p \cdot p' \eta^{\mu_1\mu_2} \eta^{\nu_1\nu_2} \eta^{\mu_3\nu_3}) \right.$$

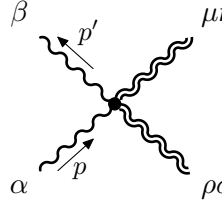
$$+ \frac{1}{2} P_6(p \cdot p' \eta^{\mu_1\nu_1} \eta^{\mu_2\mu_3} \eta^{\nu_2\nu_3}) + P_3(p^{\mu_2} p'^{\nu_3} \eta^{\mu_1\nu_1} \eta^{\nu_2\mu_3}) - \frac{1}{2} P_3(p^{\nu_2} p'^{\mu_1} \eta^{\nu_1\mu_2} \eta^{\mu_3\nu_3})$$

$$+ \frac{1}{2} P_3(p^{\mu_3} p'^{\nu_3} \eta^{\mu_1\mu_2} \eta^{\nu_1\nu_2}) + \frac{1}{2} P_6(p^{\mu_3} p'^{\nu_3} \eta^{\mu_1\mu_2} \eta^{\nu_1\nu_2}) + P_6(p^{\mu_2} p'^{\nu_3} \eta^{\nu_2\mu_1} \eta^{\nu_1\mu_2})$$

$$\left. + P_3(p^{\mu_2} p'^{\mu_1} \eta^{\nu_2\mu_3} \eta^{\nu_3\nu_1}) - P_3(p \cdot p' \eta^{\nu_1\mu_2} \eta^{\nu_2\mu_3} \eta^{\nu_3\mu_1}) \right].$$

The "Sym" indicates that a symmetrization has to be performed on each index pair. The "P" indicates that a summation has to be performed on all the distinct permutation of the momentum-index triplets (the pedices gives the number of terms)¹.

- The 2photons-2graviton vertex is given in [36] for generic massive spin-1 particle. Can be also found in [37].



$$= \tau_{\rho\sigma,\alpha,\beta}^{\mu\nu}(p, p') =$$

$$\begin{aligned}
&= -i \frac{\kappa^2}{4} \left\{ [p_\beta p'_\alpha - \eta_{\alpha\beta} (p \cdot p' - m^2)] (\eta_{\mu\rho} \eta_{\nu\sigma} + \eta_{\mu\sigma} \eta_{\nu\rho} - \eta_{\mu\nu} \eta_{\rho\sigma}) \right. \\
&\quad + \eta_{\mu\rho} [\eta_{\alpha\beta} (p_\nu p'_\sigma + p_\sigma p'_\nu) - \eta_{\alpha\nu} p_\beta p'_\sigma - \eta_{\beta\nu} p_\sigma p'_\alpha \\
&\quad - \eta_{\beta\sigma} p_\nu p'_\alpha - \eta_{\alpha\sigma} p_\beta p'_\nu + (p \cdot p' - m^2) (\eta_{\alpha\nu} \eta_{\beta\sigma} + \eta_{\alpha\sigma} \eta_{\beta\nu})] \\
&\quad + \eta_{\mu\sigma} [\eta_{\alpha\beta} (p_\nu p'_\rho + p_\rho p'_\nu) - \eta_{\alpha\nu} p_\beta p'_\rho - \eta_{\beta\nu} p_\rho p'_\alpha \\
&\quad - \eta_{\beta\rho} p_\nu p'_\alpha - \eta_{\alpha\rho} p_\beta p'_\nu + (p \cdot p' - m^2) \eta_{\alpha\nu} \eta_{\beta\rho} + \eta_{\alpha\rho} \eta_{\beta\nu}] \\
&\quad + \eta_{\nu\rho} [\eta_{\alpha\beta} (p_\mu p'_\sigma + p_\sigma p'_\mu) - \eta_{\alpha\mu} p_\beta p'_\sigma - \eta_{\beta\mu} p_\sigma p'_\alpha \\
&\quad - \eta_{\beta\sigma} p_\mu p'_\alpha - \eta_{\alpha\sigma} p_\beta p'_\mu + (p \cdot p' - m^2) (\eta_{\alpha\mu} \eta_{\beta\sigma} + \eta_{\alpha\sigma} \eta_{\beta\mu})] \\
&\quad + \eta_{\nu\sigma} [\eta_{\alpha\beta} (p_\mu p'_\rho + p_\rho p'_\mu) - \eta_{\alpha\mu} p_\beta p'_\rho - \eta_{\beta\mu} p_\rho p'_\alpha \\
&\quad - \eta_{\beta\rho} p_\mu p'_\alpha - \eta_{\alpha\rho} p_\beta p'_\mu + (p \cdot p' - m^2) (\eta_{\alpha\mu} \eta_{\beta\rho} + \eta_{\alpha\rho} \eta_{\beta\mu})] \\
&\quad - \eta_{\mu\nu} [\eta_{\alpha\beta} (p_\rho p'_\sigma + p_\sigma p'_\rho) - \eta_{\alpha\rho} p_\beta p'_\sigma - \eta_{\beta\rho} p_\sigma p'_\alpha \\
&\quad - \eta_{\beta\sigma} p_\rho p'_\alpha - \eta_{\alpha\sigma} p_\beta p'_\rho + (p \cdot p' - m^2) (\eta_{\alpha\rho} \eta_{\beta\sigma} + \eta_{\beta\rho} \eta_{\alpha\sigma})] \\
&\quad - \eta_{\rho\sigma} [\eta_{\alpha\beta} (p_\mu p'_\nu + p_\nu p'_\mu) - \eta_{\alpha\mu} p_\beta p'_\nu - \eta_{\beta\mu} p_\nu p'_\alpha \\
&\quad - \eta_{\beta\nu} p_\mu p'_\alpha - \eta_{\alpha\nu} p_\beta p'_\mu + (p \cdot p' - m^2) (\eta_{\alpha\mu} \eta_{\beta\nu} + \eta_{\beta\mu} \eta_{\alpha\nu})] \\
&\quad + (\eta_{\alpha\rho} p_\mu - \eta_{\alpha\mu} p_\rho) (\eta_{\beta\sigma} p'_\nu - \eta_{\beta\nu} p'_\sigma) + (\eta_{\alpha\sigma} p_\nu - \eta_{\alpha\nu} p_\sigma) (\eta_{\beta\rho} p'_\mu - \eta_{\beta\mu} p'_\rho) \\
&\quad \left. + (\eta_{\alpha\sigma} p_\mu - \eta_{\alpha\mu} p_\sigma) (\eta_{\beta\rho} p'_\nu - \eta_{\beta\nu} p'_\sigma) + (\eta_{\alpha\rho} p_\nu - \eta_{\alpha\nu} p_\rho) (\eta_{\beta\sigma} p'_\mu - \eta_{\beta\mu} p'_\sigma) \right\} .
\end{aligned}
\tag{E.5}$$

¹For example:

$$\text{Sym}[P_3(p^{\mu_2} p'^{\mu_1} \eta^{\nu_2 \mu_3} \eta^{\nu_3 \nu_1})] = p_1^{\mu_2} p_2^{\mu_1} \eta^{\nu_2 \mu_3} \eta^{\nu_3 \nu_1} + p_1^{\mu_3} p_3^{\mu_1} \eta^{\nu_3 \mu_2} \eta^{\nu_2 \nu_1} + p_2^{\mu_3} p_3^{\mu_2} \eta^{\nu_3 \mu_1} \eta^{\nu_1 \nu_2} .$$

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